

Regeneration Perspectives in Fluoroquinolone-Associated Disability (FQAD)

A Clinical-Translational Working Model — Version 74

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Regeneration perspectives in Fluoroquinolone-Associated Disability (FQAD): A clinical-translational working model

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Author: S. F.

Type: Position paper / clinical-translational working model (not peer-reviewed)

Note: This document is a working model based on published evidence, biological plausibility and clinical observation. It does not replace medical diagnosis or therapy. The evidence classification ([GREEN]/[YELLOW]/[RED]) consistently indicates the level of certainty of each statement.

S. F., March 2026

Index Terms: Fluoroquinolones, Mitochondria, Disability, Biomarkers, Translational Medicine, Multisystem Toxicity, Regeneration, Senescence, Post-Exertional Malaise, Mitochondrial Base Editing, Epigenetic Reprogramming, Therapeutic Ketosis, Endophenotypes, Pharmacogenomics, Molecular Docking, Chelation, Tissue Destabilization, Ion-Trapping, MMP Cascade, CYP1A2, Enantiomer, Cross-Vulnerability, C-F Bond Stability, Fluorine Enabler, Cumulative Damage, Enrofloxacin Residues, Dietary Triggers, Reaction Cascade, Phototoxicity, Lipid Peroxidation, Nemonoxacin, Multiple Chemical Sensitivity, DAMPs, Regulatory T-Cells, SOD2, Glutathione S-Transferase, Butyrate, HDAC Inhibition, Low-Dose Naltrexone, Specialized Pro-resolving mediators

Abstract

Background: Fluoroquinolone-Associated Disability (FQAD) is a clinically recognized but mechanistically incompletely understood multisystem lesion following fluoroquinolone exposure. Validated biomarkers, subtyping and causal therapies are lacking. **Aim:** Systematic presentation of a staged regeneration model with a clear separation between clinically acceptable care, translational hypotheses and speculative future technologies. **Method:** Narrative literature analysis of current evidence (as of March 2026). Databases: PubMed, PMC, Google Scholar, ClinicalTrials.gov, regulatory sources (FDA, EMA, BfArM). Period: 2002–2026, with a focus on post-2017 evidence. Evidence weighting: human studies > animal models > in vitro data > expert inference. In-house quantitative model calculations (e.g. chelation stoichiometry, OH• hit distribution) are marked as such and do not represent clinical measurements. If there are contradictory findings, the more conservative interpretation is preferred. **Key messages:** (1) FQAD should be understood primarily as a clinical multiorgan umbrella syndrome with several possible endophenotypes, not as a uniform pathway disease. (2) Partial regeneration up to extensive functional restoration is in principle conceivable from a cell biological perspective, but disease models, biomarkers and intervention studies are currently lacking. (3) The most realistic approach is a staged model consisting of damage stabilization, organ phenotyping, symptom- and organ-related therapy and careful re-conditioning. **Implications:** This document describes hypotheses and research directions — not a therapy recommended for use, even off-label.

Reading guide (updated for 3-pillar structure)

For patients and caregivers: Start with the safety information and the physician handout. Then part 1 (origin, [GREEN] sections) for the basic understanding. Part 3 (Healing, [GREEN] Damage Stabilization) for information relevant to immediate action.

For doctors: Physician handout (1 page), then part 2 (diagnosis, [GREEN] routine diagnostics). Part 1 [GREEN]/[YELLOW] for mechanistic background. Part 3 [GREEN]/[YELLOW] for therapy options.

For researchers: The entire document; especially part 1 [YELLOW]/[RED] (pathomechanics), part 2 [RED] (research diagnostics), part 3 [RED] (curative horizon).

Evidence classification and traffic light system

This document uses the following markings throughout: | Icon | Level of evidence | Significance for practice | —|—|—| | [GREEN]**Routine**| Well supported (primary literature + regulatory recognition) or close to guidelines | Generally medically acceptable; can be considered in standard

care | | [YELLOW]**Individual**| Plausible (mechanistically justified, but FQAD-specific not systematically investigated) | Individualized medical consideration after risk-benefit assessment | | [RED]**Research only**| Hypothetical or speculative | Purely experimental; not for self-application; only in the context of controlled studies |

Guiding Section: All drugs and procedures mentioned in this document that are not explicitly approved for FQAD are considered experimental. They should only be considered as part of controlled studies or after careful medical risk-benefit assessment. Substances from the curative horizon (Section XIV) have the same status: *Currently no clinical recommendation for FQAD; research only.*

Clinical handout: FQAD brief overview for the treating physician (1 page)

What is FQAD? Fluoroquinolone-Associated Disability — regulatory recognized risk of long-lasting, disabling and potentially irreversible side effects following fluoroquinolone exposure by FDA (2016/2018) and EMA (2018) [9], [10]. Since 2025/2026, first ICD codes: Germany: Y57.90! (since Jan. 2026), USA: T36.AX5 (since Oct. 2025). No validated biomarker, no approved causal therapy. FQAD is currently a diagnosis of exclusion; The aim of the translational research (AIFM1/IDH2 fingerprint, lactate-pyruvate ratio) is to make FQAD an objectively positive diagnosis in the medium term.

Immediate contraindications for FQAD in particular: - No other fluoroquinolones - No concurrent corticosteroids (officially recognized high risk factor for tendon rupture) - No NSAIDs promptly (GABA escalation risk, package insert warning) - No forced stress training (risk of PEM/rhabdomyolysis if deconditioned)

Basic diagnostics (level 1, [GREEN] routine): Blood count, CRP, B12/folate, HbA1c, TSH, ferritin, CK, creatinine/eGFR (if underweight: cystatin C), liver values, vitamin D, magnesium (whole blood), zinc, selenium, ECG. Orthostasis test for dysautonomic symptoms.

Key laboratory values for referral to specialist center: - Lactate-pyruvate quotient (note collection protocol: undamaged, immediately on ice, L+P together) - Skin biopsy (intraepidermal nerve fiber density/IENF) for mainly small fiber neuropathy - Calprotectin + tryptase for GI symptoms and especially MCAS

Crash protocol for acute decompensation: Absolute stimulus protection, no diagnostic stress tests, short-term H1/H2 blockade if mast cells are suspected (cetirizine + famotidine), electrolyte balancing, no graded exercise. Monitoring until restabilization.

Core Endophenotypes: Musculoskeletal-tendinous | Neuropathic-dysautonomous | PEM/ME-CFS-like | Neuropsychiatric | GI-immunological (incl. MCAS/MCS) | Multisystem. Therapy and diagnostics depend on the dominant endophenotype.

I. Safety notice

This document is a research and position paper. It does not represent a treatment protocol, does not contain any therapy recommendations and does not replace medical advice.

The off-label use of the substances discussed in this paper (including dasatinib, urolithin A, elamipretide) is associated with considerable, sometimes serious, risks; Safety and efficacy data for FQAD are lacking [1]–[4]. Every therapeutic attempt must be carried out under medical supervision and ideally as part of clinical studies.

Central nervous system drugs (SSRIs, SNRIs, gabapentinoids) can be poorly tolerated in some of those affected by FQAD; Clinically, paradoxical reactions are frequently reported [5]. If use is necessary, titration should be carried out carefully and individually. Direct FQAD-specific safety data is missing.

II. What this model explicitly does not claim

-No proven unit mechanics.FQAD is not shown to be fully explained by a single mitochondrial cascade. **-Not a validated biomarker.**No test today reliably says who is actively disease-driven, which tissue dominates or who would respond to which therapy [1], [6]. **-No approved causal therapy.**There is no approved or evidence-based recommended medication for FQAD. **-No prediction of course.**It is unclear which patients spontaneously improve, become chronic or worsen. **-Not equated with genetic mitochondrial diseases.**FQAD is an acquired, toxic injury - not analogous to monogenetic diseases [7], [8]. **-No off-label self-therapy.**Substances from the curative horizon (Section XIV) are not suitable for self-administration. **-No automatic transferability**of mitochondrial medicine licenses to FQAD. **-Do not equate biomarker improvement with clinical cure.**Whether normalized laboratory values actually mean functional gain is not clear in FQAD. **-Not “fluoride poisoning”.**FQAD is not the same as fluorosis. The covalently bound fluorine atom is not released in the body in toxic relevant quantities (see D4).

PART 1: PATHOGENESIS – What breaks and why?

From active ingredients to tissue damage: regulation, pathomechanics and individual vulnerability

1.1 Regulatory and epidemiological classification ([GREEN] Established)

Regulatory classification

The term FQAD was coined by the FDA in safety communications; The risk of long-lasting, disabling and potentially irreversible side effects is recognized by regulators [9]. EMA and BfArM have given comparable assessments [10]. There is no ICD code, no validated biomarker and no approved causal therapy.

Epidemiology

The true frequency spectrum of severe long-term courses is unclear; Reliable population-based incidence data are lacking [11]. The courses range from complete recovery within weeks to years of severe disability. Factors for favorable or unfavorable outcomes are largely unknown.

Prescription numbers and estimated side effect burden ([GREEN] Established):

The AOK Scientific Institute (WIdO, 2019) [52] has extrapolated the additional risks of fluoroquinolones compared to other antibiotics based on study results: Of 3.3 million FQ-treated patients in Germany (2018), an estimated >40,000 patients suffer from additional side effects (nervous system, aorta, tendon rupture), with an estimated ~140 additional deaths/year that would not have occurred with alternative antibiotics. Extrapolated to the EU (~42 million FQ patients/year, based on EMA estimate of 350 million daily doses/year [53]), this corresponds to ~360,000 additional side effects and ~1,260 additional deaths per year. Despite BfArM application restrictions (2019) [56], FQ were the fourth most heavily prescribed antibiotic group in Germany in 2019 with ~17 million defined daily doses.

Organ-specific incidence rates from registry data and studies:

Side effect	Incidence (all patients)	Estimated cases EU / year	Source
Epigenetic tissue damage	~10%	~4,200,000	Register data
Tendon damage	0.5-1%	210,000-420,000	AkdÄ
Tendon ruptures	0.08-0.2%	30,000-84,000	Register data

Side effect	Incidence (all patients)	Estimated cases EU / year	Source
Acute liver damage	0.64%	~250,000 (of which ~50,000 permanent, ~2,500 lethal)	Register data
Aortic aneurysms	0.009-0.3%	3,600-120,000	FDA
Peripheral neuropathy	0.02-0.1%	10,000-60,000	Register data
Acute kidney injury	~0.06%	~27,500	Register data
Retinal detachment	~0.04%	~16,000	Register data
Severe ventricular arrhythmia	0.002-0.047%	840-19,740	Register data
Stevens-Johnson syndrome	~0.0008%	~346 (of which 5-10% are fatal: ~17-34 deaths / year)	Register data
Gastrointestinal perforation	~0.0005%	~228 (of which 10% are fatal: ~23 deaths / year)	Register data

Pharmacokinetics of Ciprofloxacin ([GREEN] Established):

Maximum serum concentration is reached 1 hour after oral administration. Half-life: ~4 hours. Elimination: 40-50% unchanged in urine, 25-40% metabolized. Important: Of the identified metabolites, some have a weaker effect, but others have a stronger effect than the parent substance ciprofloxacin itself - so the toxicity does not end with the elimination of the parent substance. Toxicological data: LDLo (Human, oral): 5,714 µg/kg; LD50 (mouse, oral): 5,000 mg/kg.

Increased risk in seniors (≥65 years): In the EU, an estimated ~13.4 million seniors/year receive fluoroquinolones (double the prescription rate). Large cohort studies show for seniors receiving FQ therapy: tendon ruptures 2.1% (~268,000/year EU), aortic aneurysms 1.1% (~147,000/year EU), retinal detachments 0.2% (~27,000/year EU).

EMA pharmacovigilance data (2000-2017) [54]: Ciprofloxacin: 22,353 total cases, 1,161 deaths (most common: toxic epidermal necrolysis, fever, multi-organ failure). Levofloxacin: 30,170 total cases, 1,532 deaths (most common: pneumonia, multi-organ failure, sepsis). For both substances, tendonitis/tendon rupture are among the most commonly reported side effects - although the package insert classifies them as “very rare”. **FDA FQAD data (1997-2015) [55]:** FDA documented 178 FQAD cases with a median duration of disabling adverse events of 61.2 weeks (14 months); the longest documented duration was 9 years at the time of reporting. The proportion of disability reports among all serious side effect reports for oral antibiotics was dramatically higher for FQ than for all other antibiotics: ofloxacin 31.1%, ciprofloxacin 29.3%, levofloxacin 26.9% - compared to, for example, amoxicillin 6.0% or azithromycin 1.2%.

Concentration-dependent cytotoxicity of ciprofloxacin ([GREEN] Established):

Concentration	Cytotoxic effect
>1 mg / L	Necrosis of human chondrocytes (cartilage cells)
>2.5 mg / L	DNA fragmentation and apoptosis of human T cells
>3 mg / L	Mitochondria damage and apoptosis of human tendon cells
>5 mg / L	Proliferation inhibition of human fibroblasts (cytostatic)
>10 mg / L	Mitochondrial dysfunction and ultrastructural mito-damage in mammalian cells
>20 mg / L	Mitochondrial DNA (mtDNA) damage, growth inhibition
>25 mg / L	Cell cycle disruption, genomic mutation, mtDNA loss, aneuploidy

Tissue concentrations following a single 500 mg dose of ciprofloxacin (oral, 70 kg):

Organ / tissue	Concentration (mg / L)	Exceeding the toxicity threshold
Serum / Muscle	~3	≥ Tendon cell apoptosis (3 mg / L)
bones	1.5-6	Chondrocyte necrosis from 1 mg / L
muscle	6-12	≥ Fibroblast inhibition (5 mg / L)
Prostate / intestinal	6-9	≥ Fibroblast inhibition (5 mg / L)
Granulocytes	18-21	≥ mtDNA damage (20 mg / L)
lungs	6-30	Up to the aneuploidy threshold
bile	3-30	Up to the aneuploidy threshold
kidney	30-150	Far above all toxicity thresholds

This comparison shows: Even a single standard therapeutic dose reaches concentrations in several organs that trigger cytotoxic and genotoxic effects in vitro - especially in the kidney (30-150 mg/L vs. genotoxicity threshold 25 mg/L) and granulocytes (18-21 mg/L vs. mtDNA damage threshold 20 mg/L). In the umu test, ciprofloxacin showed a stronger genotoxic effect than the chemotherapy drugs etoposide, doxorubicin, cisplatin, fluorouracil, dacarbazine, bleomycin and mitomycin C.

Market withdrawals of fluoroquinolones due to serious side effects:

year	Active ingredient	Reason for return
1992	Temafloxacin	Liver, kidney toxicity, HUS, anaphylaxis
1993	Lomefloxacin	Phototoxicity, liver toxicity, CNS ADR
1998	Rosoxacin	Marketing decision
1999	Trovafloxacin	Liver failure, liver toxicity
1999	Grepafloxacin	Cardiotoxicity
2000	Pefloxacin	Phototoxicity, tendinitis
2001	Clinafloxacin	Phototoxicity, sugar metabolism
2001	Sparfloxacin	Phototoxicity, CNS ADR
2004	Fleroxacin	Phototoxicity, CNS ADR
2004	Gatifloxacin	sugar metabolism
2016	Enoxacin	Interactions

12 fluoroquinolones have been withdrawn from the market since 1992 due to serious side effects. The remaining systemic FQs in Germany (ciprofloxacin, levofloxacin, moxifloxacin, norfloxacin, ofloxacin, delafloxacin) all carry the EMA warnings from 2019.

Economic burden due to FQ-associated damage in Germany ([YELLOW] modeling)

Level of evidence:[YELLOW] The individual cost components are based on reliable external sources (WiDO [52], Kuula et al. [71], Pasternak/BMJ, Morales/JAMA Neurol.). The total is a separate extrapolation based on these sources with considerable uncertainties (± 30 –50%), as prospective German register data are missing. All calculation steps are disclosed below.

Calculation basis: ~4 million FQ therapies/year (WiDO [52]) $\approx$ 3 million patients. Empirical cost ratio: €440/therapy (Kuula et al. [71], Finland 2008–2019) + 20% correction for DE personnel costs/morbidity = ~€528/therapy. Incidences of side effects from WiDO/AOK [52]: tendon

damage 0.6%, liver damage 0.6%, psychiatric NW 0.5%; Aortic aneurysms 0.05% (Pasternak et al., BMJ 2018); peripheral neuropathy 0.05% (Morales et al., JAMA Neurol. 2019); FQAD 0.1% $\pm$ 0.05% (FDA Briefing 2015 [55]).

Damage type	Annual cases (DE)	Calculation	Ø Costs / case (€)	Total cost (€)
Tendon tears / tendonitis	~24,000	4 million $\times$ 0.6% [52]	~6,500 (Rehab + AU, Kuula model [71])	~156 million
Liver / kidney damage	~21,000	4 million $\times$ (0.6% – overlap) [52]	~10,000 (inpatient + aftercare)	~210 million
Aortic aneurysms	~5,000	4 million $\times$ ~0.13% (Pasternak BMJ projection)	~40,000 (OP + Intensive)	~200 million
Psychiatric NW	~30,000	4 million $\times$ ~0.5% (Arabyat et al.)	~3,000 (outpatient + medication)	~90 million
FQAD (chronic, $\geq$ 2 organ systems)	~3,000	4 million $\times$ 0.1% – overlap with individual categories [55]	~30,000 (long-term care + disability)	~90 million
Total direct costs				≈0.75 billion €
Indirect costs		Lost work (Ø 20 days $\times$ BW productivity [71]) ~0.3 billion; Early retirement / care (~20,000 people $\times$ €25,000 / a) ~0.5 billion; Tax / social losses (~15% of productivity losses) ~0.15 billion		≈1.0 billion €
Total burden 2025 (extrapolation)				≈€3.5-4.5 billion / year

Disclosed model calculation - derivation of €3.5-4.5 billion/year:

Path 1: Bottom-Up (sum of damage categories)

Direkte Kosten:

Tendons:	24.000 cases $\times$ 6.500 €	=	156 million €
Liver/kidney:	21.000 cases $\times$ 10.000 €	=	210 million €
Aorta:	5.000 cases $\times$ 40.000 €	=	200 million €
Psychiatry:	30.000 cases $\times$ 3.000 €	=	90 million €
FQAD chronic:	3.000 cases $\times$ 30.000 €	=	90 million €

Total direct: ≈ 746 million € ≈ 0,75 billion €

Indirekte Kosten:

Sick leave:	~83.000 cases $\times$ Ø 20 AU-Tage $\times$ 180 €/Tag (BW-Produktivität)	
	= 83.000 $\times$ 3.600 €	≈ 299 million € ≈ 0,3 billion €
Early retirement/care:	~20.000 persons $\times$ 25.000 €/a	≈ 500 million € ≈ 0,5 billion €
Tax/social losses:	~15% of productivity losses	≈ 150 million € ≈ 0,15 billion €

Total indirect: ≈ 949 million € ≈ 1,0 billion €

Bottom-Up total: 0,75 + 1,0 = 1,75 billion € (lower bound, excluding dark figure)

Path 2: Top-Down (Kuula model [71])

4 Mio. therapies $\times$ 528 €/Therapie (Kuula 440 € + 20% DE-Korrektur)
= 2,11 billion € (direct + indirect base costs per prescription year)

Path 3: Synthesis of both paths + long-term costs

Bottom-Up akut: 1,75 billion € (Pfad 1)

Top-Down Kuula: 2,11 billion € (Pfad 2)
 + Langzeitkosten chronifizierter legacy cases (FQAD + Pflege aus Vorjahren):
 Estimated 150,000–450,000 permanent injuries × Ø 5.000–10.000 €/a
 Care/disability = 0,75–4,5 billion € (extremely wide range)
 Conservative: ~€1.5 billion (midpoint at 250,000 × €6,000/yr)

Total range:

Lower bound: 1.75B (Bottom-Up) + 1.5B (legacy) ≈ €3.25B → rounded €3.5B

Upper bound: 2.11B (Top-Down) + 2.5B (legacy) ≈ €4.6B → rounded €4.5B

Result: ≈ €3.5 – 4.5 billion/year (2025)

Sensitivity check of the main drivers:

Parameters	Conservative	Medium	Upper limit	Impact on overall
FQAD chronicity rate	0.05%	0.1%	0.3%	±0.5 billion €
Ø Costs / Aortic Aneurysm	€25,000	€40,000	€60,000	±0.1 billion €
Number of old cases (cumulative)	150,000	250,000	450,000	±1.5 billion €
Kuula-DE correction	+10%	+20%	+30%	±0.2 billion €

→**Largest uncertainty factor:** The cumulative number of old cases (permanent damage from previous years). Without a prospective FQAD registry, this number cannot be reliably limited.

With stable prescription rates, the linear projection (2026–2030) results in ~€20 billion cumulatively, with ~70% attributable to indirect effects. Cave: This projection does not take into account the downward trend in prescriptions since 2019 (–33.9% [57]).

Epidemiological comparison: FQ permanent damage vs. widespread neurological diseases

Calculation method for prevalence estimate: With ~4 million FQ therapies/year since widespread market introduction (~1995) and a conservative rate of permanent damage of 0.1–0.3% (based on WiDO-NW rates [52] × estimated chronicity rate 15–30%), this results in a cumulative prevalence of ~25 years 150,000–450,000 people with permanent FQ-related restrictions. The range is considerable because neither mortality nor spontaneous remission are systematically recorded in this cohort. The lower limit (150,000) assumes a high level of spontaneous remission, the upper limit (450,000) assumes predominantly chronicity. | disease | Valued those affected (DE, 2025) | Source | |—|—|—| | **Parkinson's** | ~420,000 | German Parkinson Society 2024 [72] | | **FQ-associated permanent damage** | **150,000–450,000** | Extrapolation: WiDO incidences [52] × cumulative prescriptions × chronicity assumption (own model calculation, not externally validated) | | **Multiple Sclerosis** | ~250,000 | DMSG / RKI 2024 [73] | | **FQAD (Multisystem form)** | **90,000–180,000** | Extrapolation: FDA FQAD rate 0.1% [55] × cumulative prescriptions (own model calculation, not externally validated) |

According to this projection, the number of people permanently damaged by fluoroquinolones is in the same order of magnitude as the Parkinson's prevalence and overlaps with the MS prevalence - but without systematic recording in disease statistics, without an ICD code and without specialized care structures. **These estimates are model calculations with high uncertainty and require validation by prospective registry data.** Epidemiological “invisibility” is a central health care problem, regardless of the exact number.

Population-based incidence trends — Achilles tendon ruptures and FQ prescriptions:

Several European countries show a temporal parallel between increasing FQ use and Achilles tendon ruptures: Finland [58]: Incidence of non-traumatic Achilles tendon ruptures increased from 2.1 (95% CI 0.3–7.7) per 100,000 person-years (1979) to 21.5 (95% CI 14.6–30.6) in 2011 — a

>10-fold increase, coinciding with the broad FQ rollout. Funen County (Denmark) [59]: increase from 22 to 32 per 100,000 inhabitants (1991–2002). Malmö (Sweden) [60]: 10 cases/year (1950–1973) → 38 cases/year (1987–1991). These ecological correlations do not prove causation, but are consistent with known FQ tendon toxicity and the concomitant increase in FQ prescription numbers in the 1990s.

Structure-activity relationships on the FQ molecule: The side effects of fluoroquinolones are not randomly distributed, but correlate with specific positions on the quinolone backbone [61]: Position C-6 (fluorine atom): responsible for genotoxicity and CNS toxicity. FQ are significantly more clastogenic (chromosome-breaking) than non-fluorinated quinolones. Position R7: controls GABA binding (→ seizures, anxiety) and theophylline interaction. Position C-8: controls phototoxicity (C-8F > C-8Cl > N-8 > C-8H > C-8OMe). Position R1: controls theophylline interaction and genetic toxicity. Position R5 (metal binding site): controls chelation with divalent cations (Mg²⁺, Zn²⁺, Fe²⁺) and interaction with antacids/milk.

Global dimension: Estimated proportion of the world population with at least one FQ treatment ([YELLOW] model estimate)

Level of evidence:[YELLOW] No direct epidemiological study or global registry on lifetime FQ use available. The following estimate is a data-based model derivation from DDD consumption data [74], [75], historical trends and age-weighted cumulative calculation. Bandwidth: ±10 percentage points.

Approximately **40-50% of people alive today** (approximately 3.2-4 billion out of ~8.1 billion) are estimated to have been treated with fluoroquinolones at least once.

Derivation:

Parameters	Value	Source
Global AB total consumption (2018)	~14.3 DDD / 1,000 population / day	Browne et al., Lancet Planet. Health 2021 [74]
FQ share in AB consumption	~1.0-1.8 DDD / 1,000 / day (weighted globally; up to 3.0 in Southeast Asia / MENA)	Browne et al. [74], Klein et al. [75]
Typical FQ course	7-10 days ≈ 7-8 DDD / treatment	Specialist information
Annual FQ exposure rate (current)	~4-7% of the world population (~320-570 million courses / year)	Calculation from DDD data
Historical average (1985-2025)	~3.5-5%/year (increase since market launch, in LMICs sometimes >1.8-fold)	Klein et al. [75], Linder et al. [76]

Cumulation “at least once” (lifetime):

Model formula: $1 - (1 - p)^t$, where p = annual exposure probability, t = exposure years.

Age group	Share of world population	Years of exposure	Cumulative risk	Result
<18 years	~20-25%	FQ is almost never prescribed (children’s contraindication)	~5-10% (indirect, e.g. LMICs without restriction)	Hardly exposed
18-65 years	~60%	15-45 years × 3.5-5%/year	$1 - (1 - 0.045)^{30} \approx 75\%$	Mostly exposed
>65 years	~15-20%	30-45+ years × 5-7%/year (double prescription rate for seniors)	>85%	Almost completely exposed
Weighted average				~40-50% (≈3.2-4.0 billion people)

Correction factors: Repeat prescriptions slightly reduce the number of unique people. HICs:

previously higher consumption, declining since FDA/EMA warnings (2013/2016/2019). LMICs: often higher today, but lower availability in the 1980s/90s. Deceased people and children bring the total number down.

Classification: The estimate of ~45% (approx. 3.6 billion people with at least one lifetime exposure) underlines the global dimension of the problem: Even with a conservative rate of severe permanent damage of 0.1% (1:1,000), there would be ~3.6 million potentially permanently injured people worldwide - without any systematic care structures, registry data or an ICD code existing for this population. Without the FDA/EMA warnings and the pediatric contraindication, the exposure rate would likely be significantly higher (55–65%).

Available alternatives: Conventional AB + phages + nemonoxacin → almost 100% replaceable ([GREEN] Established / [YELLOW])

A systematic indication analysis [63] shows that the majority of current FQ prescriptions in Germany can be replaced by conventional antibiotics, bacteriophage therapy or nemonoxacin (non-fluoroquinolone), without increasing mortality.

Indication	Share of FQ regulations	Conventional alternatives	Nemonoxacin (non-FQ quinolone) [69]	FQ replaceable?
Uncomplicated UTI	30% (~1.25 million / year)	Fosfomycin trometamol (single dose, cure rate 91%, E.coli resistance <1%), nitrofurantoin (93% cure), pivmecillinam	Not necessary — conventional AB sufficient	Yes — FQ ban (Category A) [63]
Acute bronchitis / sinusitis	~25% (~1.04 million / year)	Mostly viral → no AB necessary; bacterial: amoxicillin ± clavulanic acid, doxycycline	Not necessary — conventional AB sufficient	Yes — FQ ban (Category A)
Bacterial Gastroenteritis	~8% (~333,000 / year)	Mostly self-limiting; if indicated: azithromycin, rifaximin	Not necessary — conventional AB sufficient	Yes — FQ ban (Category A)
Ambulatory Pneumonia (CAP)	~10%	amoxicillin + macrolide; Ceftriaxone during hospitalization	Yes — Phase III: non-inferior vs. levofloxacin, fewer CNS / musculoskeletal ADRs [69]. Approved in Taiwan / China	Yes - completely replaceable with nemonoxacin
Complicated UTI / Pyelonephritis	~5%	Ceftriaxone i.v. → oral sequence; on MDR: Carbapenems	Yes — quinolone tissue permeability without fluorine toxicity. Residual chelation risk remains	Yes - can be replaced with nemonoxacin + phages
Chronic Prostatitis	~5%	Oral fosfomycin (new evidence), cotrimoxazole in sensitivity	Yes — Prostate penetration preserved (quinolone backbone), without phototoxicity / ion trapping	Yes - can be replaced with nemonoxacin (FQ gold standard replaced)

Bacteriophage therapy as a new FQ alternative ([YELLOW] clinical evidence growing):

Bacteriophages — viruses that specifically infect and lyse bacteria without harming human cells — are experiencing a renaissance as a solution to the antibiotic resistance problem [63], [64]. Clinical experience: Phage Therapy Unit Wroclaw (Poland): >3,000 patients since 2005; Eliava Institute Tbilisi (Georgia): >100,000 patients since the 1930s; national phage programs in Belgium and France since 2019.

Indication	Phage success rate	Special feature
Uncomplicated UTI (MDR)	85-95%	Recurrence rate 12% vs. 42% with antibiotics (12 months)

Indication	Phage success rate	Special feature
Pyelonephritis / Urosepsis (MDR)	75-88% eradication	Synergistic with carbapenems (dose reduction ~50%)
Chronic wound infections / osteomyelitis	60-85%	Biofilm penetration - advantage over conventional AB
Nosocomial pneumonia (VAP)	70-80%	For MDR-Pseudomonas / Acinetobacter
GI infections (MDR salmonella)	~90% eradication	Randomized Trial in Bangladesh (Pediatric)
Chronic prostatitis	70-85% clinical improvement	Tissue penetration comparable to FQ

Approval status (2025): EU: Compassionate use according to Helsinki Declaration Art. 37; Belgium: extemporaneous preparation approved (2019); Poland: national reimbursement (since 2021); Germany: Compassionate use possible, no structured reimbursement. **Key message for regulators:** With the three-way combination of conventional alternatives + bacteriophage therapy + nemonoxacin (non-fluoroquinolone), **almost 100% of the current FQ regulations** can be replaced in the future. Already today ~83% are covered by conventional AB + phages [63]. The ~17% previously considered “unreplaceable” (CAP, complicated UTI, prostatitis) can be addressed by nemonoxacin, which offers quinolone tissue penetration without the fluorine-mediated toxicity enhancement [69]. Remaining exceptions: MDR tuberculosis and febrile neutropenia in high-risk oncology patients, where FQ remains as a reserve agent pending full approval of nemonoxacin in the EU.

Nemonoxacin (Taigexyn) — the fluorine-free quinolone as a structural FQ replacement ([YELLOW] approved in Asia)

Nemonoxacin is the pharmacological counter-evidence for the enabling role of fluorine (see 1.8): a non-fluoroquinolone in which the fluorine atom at C-6 has been intentionally removed. It retains the antibacterial effectiveness of the quinolone backbone (topoisomerase inhibition), but eliminates the toxicity enhancers specifically mediated by the fluorine atom:

Property	Fluoroquinolones (Cipro / Levo)	Nemonoxacin	Clinical consequence
Phototoxicity	Yes (C-F cleavage by UV)	Virtually eliminated	No UV-induced radical fire
Lipophilia / Ion Trapping	High (fluorine increases membrane permeability)	Significantly reduced	Less intracellular accumulation
Metabolic stability	Very high (C-F resistant to liver degradation)	Reduced → faster degradation	Shorter tissue exposure
Chelation Risk	Yes (quinolone backbone)	Yes - residual risk remains	Mg ²⁺ /Zn ²⁺ chelation still possible
AIFM1 / IDH2 binding	Proven [2]	Not examined	Open research question
Approval	Worldwide	Taiwan (2014), China (oral + i.v.)	EU / USA: Phase III studies ongoing

Range of indications (approved/in studies): Community-acquired pneumonia (CAP) - here nemonoxacin shows non-inferiority to levofloxacin in phase III studies with a significantly lower side effect profile [69]. Additionally: skin/soft tissue infections (approved in China). **Importance for FQ restriction:** Nemonoxacin could, in the long term, address the remaining ~17% of FQ indications in which specific tissue penetration and the quinolone mode of action (topoisomerase inhibition) are currently considered non-substitutable - particularly outpatient pneumonia and complicated UTIs. The combination of conventional AB + phages + nemonoxacin could in the future enable a complete FQ replacement.

1.2 Core Evidence: Molecular Off-Targets ([GREEN] Established)

Core evidence on fluoroquinolone toxicity

Level of evidence:[GREEN] Well supported. Human cell chemoproteomic data [2] confirmed by independent mtDNA study [3]. Pending clinical translation.

The chemoproteomic study by Reinhardt et al. (TU Munich, 2025) [2] provides the most precise molecular mapping to date: Ciprofloxacin and levofloxacin bind directly to AIFM1 and IDH2, selectively disrupt complexes I and IV of the ETC, and the resulting dysfunction activates a rescue mechanism via IDH2, which is further enhanced by simultaneous FQ inhibition. Toxicity does not occur acutely but develops over prolonged treatment duration and cellular accumulation [2], [3].

1.3 The role of the fluorine atom: enabler, not poison ([GREEN]/[YELLOW])

D4. The role of the fluorine atom: enabler, not poison

It is often discussed in patient forums whether FQAD is “fluoride poisoning.” This hypothesis is chemically untenable, but the fluorine atom plays a crucial indirect role as an enabler of toxicity.

What is secured:

The fluorine atom at position C6 of the quinolone ring system is covalently bound via one of the strongest bonds in organic chemistry (C-F bond energy ~450–525 kJ/mol). Under physiological conditions (37°C, pH 7.4) no significant metabolic defluorination occurs (<0.1%). Ciprofloxacin is almost completely excreted in its fluorinated form. Direct fluoride poisoning (as with inorganic sodium fluoride) is physically and chemically impossible. Fluorosis (teeth, bones/osteosclerosis) and FQAD (soft tissue, nerves) have fundamentally different symptoms and different biochemical reaction pathways [51].

Three indirect roles of the fluorine atom in causing damage:

role	Mechanism	Consequence for FQAD
1. Lipophilicity enhancer	Fluorine at C6 significantly increases the lipid solubility of the molecule	Enables penetration of cell membranes and deep penetration into tendons, cartilage, CNS and mitochondria - the physical prerequisite for ion trapping (see D2)
2. Liver protective shield	The strong electron pull of the fluorine (inductive effect) electronically deactivates the quinolone ring and makes it difficult for CYP enzymes to attack	The molecule escapes liver breakdown and remains in the blood in its highly toxic 3D original form. Only this intact molecule can penetrate into enzymatic binding pockets (AIFM1, IDH2, topoisomerase) (see D3, metabolite detox). With slow metabolizers, this effect increases cumulatively.
3. Chelation enhancer	The chelation (Mg ²⁺ /Ca ²⁺ robbery) takes place on the oxygen atoms of the keto / carboxyl group at C3 / C4 - not on the fluorine itself. But the electron pull of the fluorine changes the charge distribution on this “claw” and can strengthen the bond to metal ions.	Fluorine is not the claw, but it sharpens it. The chelating ability of the entire molecule is influenced by the electronic modification of the ring [51].

The central answer: Fluorine → ROS → tissue damage

The causal pathway from the fluorine atom to mitochondrial ROS production in tenocytes (tendon cells) proceeds via a three-step detour: (1) fluorine protects the molecule from CYP degradation in the liver → the intact molecule remains in the blood; (2) Fluorine increases lipophilicity → the molecule penetrates deep into tendon tissue and accumulates in mitochondria via ion trapping; (3) there the intact, fluorinated molecule blocks the ETC via AIFM1/IDH2 [2] → electron congestion → superoxide → Fenton reaction → hydroxyl radicals (ROS) → mtDNA destruction → MMP escalation [4], [34].

The fluorine atom does not cause the damage directly as a toxic ion, but is the architectural

reason why the entire molecule (a) survives liver breakdown, (b) penetrates deep enough into the tissue and (c) remains precise enough to block the mitochondrial enzymes. Without the fluorine, the older, non-fluorinated quinolones (e.g. nalidixic acid) would be significantly less permeable to tissue and less potent - and would therefore be less likely to cause serious long-term damage.

1.4 Clinically discussed damage axes ([GREEN]/[YELLOW]/[RED] mixed)

Damage axes discussed clinically and mechanistically

FQAD is currently better understood as a clinical umbrella syndrome with several possible endophenotypes [1], [6]. The following axes are supported to varying degrees:

Well supported (primary literature + regulatory recognition): 1. Mitochondrial dysfunction caused by oxidative stress, Top2 β inhibition and direct ETC off-target effects (AIFM1, IDH2) [1]-[3] 2. Collagen damage due to upregulation of matrix metalloproteinases [13] 3. Peripheral nervous system due to oxidative stress, mitochondrial dysfunction or direct neurotoxicity [5], [14] 4. Neuropsychiatric effects due to GABA receptor modulation and NMDA receptor overactivation [5]

Plausible (mechanistically justified, but not systematically investigated for FQAD): 5. Dysautonomia and small fiber neuropathy, partly with autoimmune components (GPCR autoantibodies) [15], [16] 6. Endocrine dysregulation (HPA axis) during chronic stress [17]

Hypothetical (biologically conceivable, but FQAD-specific has hardly been investigated): 7. Gastrointestinal-immunological axis caused by antibiotic dysbiosis, barrier disruption and mast cell activation [18]-[20]. The prevalence of clearly defined MCAS trajectories within FQAD is not known; these considerations generate hypotheses. 8. Multiple Chemical Sensitivity (MCS) as a result of the MCAS/GABA double blockade: immunological overreaction (mast cells) + neuronal misinterpretation of chemical stimuli (GABA filter failure) could explain the chemical intolerance frequently reported in FQAD. FQAD-specific not examined.

1.5 Chelation and structural tissue destabilization ([YELLOW] Plausible)

D2. Chelation and structural tissue destabilization

Level of evidence:[YELLOW] Plausible. Chelating property pharmacokinetically established (interaction with antacids/Ca/Fe/Mg). Hypothesis of independent tissue damage is biologically plausible, but has not been proven as the main FQAD-specific mechanism in isolated human studies [51].

In addition to mitochondrial toxicity, fluoroquinolones have a second, often underestimated damage mechanism whose role, however, must be reinterpreted: they are potent chelating agents with high affinity for divalent cations, especially iron (Fe²⁺/Fe³⁺), calcium (Ca²⁺) and magnesium (Mg²⁺) [4], [13], [51].

The tetracycline paradox — why pure ion robbery cannot be the main cause ([GREEN] structural comparison):

Tetracyclines are also potent metal chelators and form stable, long-lived metal complexes — Friedman (2026) [79] showed via DFT/ORCA quantum chemistry calculations that the iron-tetracycline binding energy is even *higher* than the iron-ciprofloxacin binding energy. If pure ion removal were the main cause of FQAD, tetracyclines would have to trigger a comparable or more severe syndrome. They demonstrably do not (apart from aesthetic effects such as tooth discoloration via Ca²⁺ binding). This structure-activity comparison — a toxicological gold standard — proves: if two substance classes share the same nonspecific mechanism (chelation) but only one triggers a specific syndrome, the causality must lie in the *structural peculiarities* of the one class. In fluoroquinolones these are the quinoline/pyridine core and the piperazine ring (see molecular mimicry below) [2], [79].

The reinterpretation: metal binding as a weapon, not a robbery:

The consequence is a fundamental reweighting. Iron binding is less a “loss” for the cell than something that makes the FQ molecule *more dangerous*: in-silico docking simulations [77], [79] show that the iron-FQ complex (Fe-FQ, ~1:1 stoichiometry) has a *significantly higher* binding

energy to pyridine-ring-dependent enzymes than free ciprofloxacin. At the same time, iron binding causes the molecule to lose its lipophilicity → it can no longer cross membranes → it becomes trapped long-term in the cell compartments (ion trapping). The metal is thus not the “victim” of a robbery, but is used by the FQ molecule as a *chemical anchor* to (a) increase binding strength to essential enzymes and (b) lodge permanently in tissue. This directly links the chelation mechanism to off-target mimicry (see molecular mimicry section) and long-term persistence (see 1.11). The direct “ion robbery” from structural proteins is secondary; what matters is the off-target inhibition *enhanced* by metal binding. **Mechanism:** Biological tissues — tendons, cartilage, bones, cell membranes — are structurally stabilized by calcium bridges between proteins and matrix components. Fluoroquinolones can actively deprive these tissues of stabilizing ions (comparable to a strong complexing agent such as EDTA). This disintegration goes beyond a pure acid or radical effect - it directly attacks the physical statics of the tissue [51].

Evidence from in vitro experiments: Ciprofloxacin hydrochloride (the salt of a strong acid and a weak base) produces an aggressive, unbuffered environment at high local concentrations. In tissue experiments (500 mg ciprofloxacin tablet in a small amount of liquid, 8 hours exposure time) this leads to measurable dissolution of biological structures that cannot be explained by the pH value alone - control experiments with orange juice (pH ~3.5, organic acids, buffered) show no tissue decomposition at a comparable pH. The resulting clouding is not only due to insoluble excipients (cellulose, titanium dioxide, approx. 60-80% of the tablet weight), but also to real tissue dissolution due to ion removal. The crucial additional mechanism is chelation: Ciprofloxacin actively removes the structure-stabilizing cations from the tissue, which explains the differences to the pH-controlled orange juice experiment [51].

Consequences for FQAD damage Saxony:

Quantitative model calculation: chelation capacity in experiments vs. in vivo

A simplified but chemically correct example calculation for the in vitro scenario (500 mg ciprofloxacin tablet in 100 ml water, contact with muscle tissue):

Parameters	Value
Ciprofloxacin amount	500mg; molar mass ~331 g / mol → ~1.5 mmol active ingredient
Concentration in glass (100 ml)	15 mmol / L (~15,000 µmol / L)
Mg ²⁺ content muscle tissue	~10 mmol / kg (freely available)
Complex stability constant (K)	~10 ⁴ (log K ≈ 4.0) — strong bond

Law of mass action: At K = 10,000 and a molar excess of ciprofloxacin (15 mmol/L) compared to magnesium in the tissue (~10 mmol/L), complex formation occurs almost completely. Mathematically, ciprofloxacin binds ~100% of the freely available magnesium in the direct contact area - only traces (µmol range) of free Mg²⁺ remain. The electrostatic bridges between muscle proteins (actin, myosin) and collagen break down → denaturation → the meat loses its structural strength in hours [51].

Critical Reality Check — In-vitro vs. In-vivo:

Parameters	In the glass (experiment)	In the body (after 500 mg orally)
Ciprofloxacin concentration	~15,000 µmol / L	~7.5 µmol / L (Plasma Cmax)
concentration factor	Reference	~2,000 times lower
Tissue accumulation (tendon / cartilage)	—	Up to 10 times higher than plasma (~75 µmol / L) — still ~200 times lower than in glass

The conclusion: In the body, the drug can never remove all the magnesium from the tissue at once (otherwise a single tablet would be immediately lethal). The stoichiometry shows that for every ciprofloxacin molecule in the tendon there are at least ~133 magnesium ions. Even with immediate binding, in the absolute extreme case the drug only consumes ~0.75% of total

tissue magnesium. A direct structural collapse due to mass ion theft - as in a glass of water - is physically and mathematically impossible in the human body.

Then why does the tendon still tear? — Precision instead of mass

The drug does not bind structural magnesium indiscriminately, but rather targets the most sensitive parts of the cell: the enzymes. Many vital enzymes (topoisomerases for DNA repair, TIMP brake systems, mitochondrial dehydrogenases) have exactly a single magnesium ion in their active center. The ciprofloxacin does not have to suck the entire cell dry - it is enough if it sits in this enzymatic keyhole and blocks this one specific ion. The comparison: Chelation effect = match (makes the cell sick), MMP enzymes = dynamite (tear down the structure).

Ion trapping — why the drug gets stuck in cells: Ciprofloxacin is extremely lipophilic and penetrates cell membranes effortlessly. There are acidic compartments inside the cell (lysosomes, pH ~4.5). There the FQ molecule changes its electrical charge (protonation of the weak base) and can no longer penetrate the membrane - the ion trap snaps shut. While the plasma concentration has been falling for a long time, the drug accumulates intracellularly and reaches concentrations there that are many times the plasma level. This lysosomal storage could explain why symptoms persist or even increase for weeks to months after discontinuation [4], [51].

Exception kidney - crystalluria: Ciprofloxacin can actually physically crystallize in the urine at certain pH (sharp-edged needles), which mechanically injure the kidney epithelium. This is the only location in the body where physical aggregation of the active ingredient is clinically relevant.

The chronological three-phase domino effect:

Tissue damage in the body is a strict temporal chain reaction:

phase	Mechanism	Result
Phase 1: Chemical Overexploitation	FQ accumulates via ion trapping in tendon / cartilage; continuously removes Mg ²⁺ from enzymes and matrix proteins	Microscopically, collagen fibers lose their tension and fray; Enzymes (topoisomerases, TIMPs) lose their active center → DNA breaks, repair stops
Phase 2: Biological Alarm	Damaged collagen is recorded as an “injury”; At the same time, Mg ²⁺ deficiency in mitochondria generates massive oxidative stress (ROS)	Double alarm: defective collagen + oxidative stress = cellular distress signal
Phase 3: MMP Escalation	Cells release massive amounts of MMPs (MMP-2, MMP-9, MMP-13) - biological mission: “Clean up the broken collagen”	Since FQ continues to sit in the tissue via ion trapping, the alarm signal remains permanently on “red” → MMPs then break down not only damaged but also healthy tissue at an increased rate → chronic destruction over months [13]

The sponge analogy: The drug does not hit like a hammer, but rather acts like a sponge that sits deep in the cells for weeks and sucks the ions out of the building substance drop by drop. On a microscopic level, the thick collagen fibers lose their internal tension because the stabilizing cross-links are missing - they begin to fray. The tendon becomes “rotten” and loses its tensile strength without the tissue immediately visibly disintegrating as in a glass of water. It is a slow, cumulative structural impairment over weeks to months over weeks to months.

Critical limit of chemistry: Even if this “piece-by-piece” robbery were to continue for years, the ion withdrawal alone is not enough to cause entire muscle bundles or tendons to collapse completely. To do this, thousands of ion bridges would have to be exploded at the same time - the amount of medication in the body is stoichiometrically too low for this (see model calculation above: ~0.75%). Without the MMP enzymes, the tissue would develop microscopic cracks, but would not tear or shrink over a large area. Phase 3 (the MMP escalation) is therefore not an optional ingredient, but the mandatory prerequisite for the clinically visible structural damage.

Without the previous FQ-induced ion depletion and cell stress (phase 1+2), the MMPs would never be activated to this destructive extent. The drug provides the toxic trigger; the body carries out the destruction itself.

What breaks immediately WITHOUT MMPs (immediate functional damage):

Even without MMP activation, ion withdrawal causes direct damage on a functional level - the lack of individual ions in critical enzymes is enough for this:

-DNA Destruction: Topoisomerases often have a single Mg^{2+} in the active site; When blocked, the DNA breaks when read → cell death (apoptosis), without any MMPs [1], [3] **-Nerve short circuit:** Nerve pathways function via ion channels (Ca^{2+} , K^+ , Mg^{2+}); if FQ disrupts channel regulation, nerves fire uncontrollably or die → burning sensation, neuropathy as a direct chelation/channel effect [5] **-TIMP inhibition:** Zinc chelation by FQ additionally weakens the body's own MMP brakes (see TIMP section below) [13]

1.Local accumulation: Fluoroquinolones accumulate up to 10-fold higher in tendons and cartilage than in plasma, cumulatively after repeated administration over days [13], [51]. **2.The enzyme trap:** Even a small but constant Mg^{2+} withdrawal at the tissue interface can lower the activation threshold of the matrix metalloproteinases (MMPs). The MMPs then carry out the actual collagen destruction over weeks to months - the chelation effect is the igniter, the enzymes are the radical amplifier [13].

The calculation proves the destructive power in the water glass quantitatively; For FQAD pathology in the body, it is not the absolute chelation capacity that is crucial, but rather the targeted, cumulative disruption of ion-dependent cell processes in vulnerable tissue locations.

Damage axis	Chelation mechanism	Clinical manifestation
Tendons (tendinopathy / rupture)	Withdrawal of Ca^{2+}/Mg^{2+} from collagen fibers → loss of cross-linking and tear strength	Tendon becomes "rotten"; Even normal stress can trigger a rupture [13]
Cartilage (accelerated osteoarthritis)	Destabilization of the cartilage matrix by ion robbery in addition to MMP activation and chondrocyte apoptosis	Increased enzymatic + structural cartilage degradation
Nervous system (neuropathy)	Disruption of ion homeostasis on nerve cell membranes; Ca^{2+} channel dysregulation	Nerve burning, hypersensitivity, small fiber damage [5]
Bone (risk of osteoporosis)	Long-term withdrawal of Ca^{2+} and Mg^{2+} from bone tissue	Particularly relevant for immobile patients with already increased degradation
Mucous membranes	Local necrosis with high drug concentration at the tissue interface; Breaking through the mucosal barrier	Gastrointestinal erosions, local tissue damage [51]

Magnesium as a protective factor - extended reason: Since ciprofloxacin, as a chelating agent, actively "searches" for ions, if the magnesium level is low, the drug will obtain its binding partners directly from the tissue (bones, tendons, cartilage, nerve membranes). Sufficient magnesium supply is therefore not just a general cofactor, but rather a concrete protection against the ion-robbing effect of the active ingredient - the supply of magnesium before and during each fluoroquinolone administration is biologically based and is mentioned in specialist information as an indication of interaction [4], [51]. **Evidence status:** The chelating property of fluoroquinolones is pharmacokinetically established (hence the known interaction with antacids, calcium, iron, magnesium). The hypothesis that this mechanism makes an independent contribution to tissue damage beyond the reduction in bioavailability is biologically plausible, but has not been proven as the main FQAD-specific mechanism in isolated human studies [51].

Chelate interactions: bioavailability reduction through simultaneous ion supply

The chelating affinity of fluoroquinolones is so strong that when taken orally with certain ion-containing substances, it reduces the absorption of the active ingredient in the gastrointestinal tract by up to 90%. These pharmacokinetically established data provide quantitative evidence of the strength of the chelation effect (values refer primarily to ciprofloxacin; minimally weaker for levofloxacin/moxifloxacin, but highly clinically relevant):

Substance class	Examples	Reduction of bioavailability
Antacids ($\text{Al}^{3+}/\text{Mg}^{2+}$)	Maaloxan, Talcid, Rennie, Gaviscon	up to 85-90%
Iron preparations ($\text{Fe}^{2+}/\text{Fe}^{3+}$)	Ferrosanol, iron tablets	up to 80-90%
Zinc supplements	Zinc tablets, zinc histidine	approx. 50-70%
Calcium preparations (high doses)	Calcium effervescent tablets (500-1000 mg)	approx. 40-60%
Milk and dairy products	Cow's milk (from ~300 ml), yoghurt, quark	approx. 30-50% (up to 70% if consumed at high levels)
Multivitamins	A-Z tablets with mineral mix	approx. 30-50%
Magnesium supplements	Magnesium citrate, oxide	approx. 30-50%

Time window: These extreme reductions only occur when they are present in the stomach. After 2 hours in the body, the blockage effect decreases drastically. Specialist information therefore recommends: Take FQ at least 2 hours before or 4-6 hours after substances containing ions.

Double reading of this data: The same chelation affinity that blocks absorption in the intestine acts as an ion robber of enzymes and collagen after absorption in the tissue (see D5, front 0). The table thus indirectly quantifies the strength of the chelation effect that drives tissue damage after systemic distribution.

Emergency Note: In the event of an acute accidental overdose, strong binding antacids or iron preparations physically represent the strongest blockage of absorption. However, they never replace immediate contact with a poison control center or emergency medical care. **Scientific classification of the in vitro experiment:**

The experiment did not provide toxicological research with any new insights into the mechanism of action. Chelation, MMP upregulation and collagen damage caused by fluoroquinolones have been demonstrated in cell culture studies and animal experiments for decades [13], [51]. A toxicologist would classify the experiment as “non-physiological”: the flesh decays due to the sheer concentration of the active ingredient (brute chemical violence), while tissue destruction in the body occurs via the biological domino effect (mitochondria blockade → oxidative stress → MMP activation). The glass shows the correct end result (rotten tissue), but via a different route than in the living organism.

The true value is didactic: the experiment makes the invisible visible. It's physical evidence of the molecule's raw reactivity, helping sufferers contextualize the class of compounds to skeptical doctors — not a scientific breakthrough, but a powerful visual aid.

Why the stomach survives the pill — three protective mechanisms:

The obvious question as to whether the tablet “breaks down” the stomach in the same way as the meat in the glass is legitimate, but the answer is reassuring:

1. Mucous membrane armor: The stomach is built to handle extremely harsh chemicals. Stomach acid (pH 1.5-3) is far more corrosive than ciprofloxacin. A thick layer of mucosa with basic bicarbonate protects the epithelium - the tablet hits this protective armor, not naked tissue. **2. Dilution and transit:** Ingestion with 200-300 ml of water, dilution with gastric juices, mixing through peristalsis and transit into the small intestine within 1-2 hours - the hour-long point extreme concentration of the glass experiment never occurs. **3. Enrichment occurs elsewhere:** FQs accumulate not in the stomach, but in tendons, cartilage, macrophages and CNS [13]. The tissue damage there is not “chemical melting from outside”, but rather the gradual biological domino effect (Mito blockade → Mg^{2+} withdrawal → MMP activation over weeks to months).

What the fluoroquinolone actually destroys in the GI tract is not the stomach wall, but the microbiome - with the consequences (dysbiosis → barrier disorder → MCAS), which are described in detail in the GI-immunological axis (organ axis 3).

TIMPs — the natural brakes against MMP destruction: In order to inhibit the out-of-control

matrix metalloproteinases (MMPs), the body produces natural antagonists: TIMPs (Tissue Inhibitors of Metalloproteinases). These proteins act as brake blocks around the catalytic centers of the MMPs and stop the enzymatic breakdown of collagen. For TIMP synthesis, the body absolutely needs zinc (Zn^{2+}) as a cofactor - since fluoroquinolones also chelate zinc, an FQ-induced zinc deficiency can further weaken the body's own MMP brake. As a cofactor of glutathione peroxidase, selenium is indirectly involved in protection against oxidative tissue degradation [13], [34].

Consequence for supplementation: Zinc and selenium status should be tested in a laboratory and, if there is a deficiency, specifically compensated for - no blind high-dose supplementation, but diagnostics → compensation → monitoring (analogous to the magnesium strategy).

The Supplementation Paradox: The Same Ions as Saviors and Radical Amplifiers

The logic “the drug robs magnesium/zinc/calcium/iron, so you have to give them in large quantities” is convincing, but it is a biochemical minefield when FQAD is present. What prevents absorption in the stomach as a chelate blocker (see table above) can either start the healing process inside the damaged cell or act as a radical amplifier:

ion	Therapeutic potential	Risk at FQAD	Recommendation
Magnesium (Mg^{2+})	Wichtigster Baustein: füllt den zellulären „Mörtel“ in Enzymen und Kollagen wieder auf; As an NMDA antagonist, blocks the toxic Ca^{2+} influx [4], [51]	<i>Transport problem:</i> Damaged cells often have blocked or lost ion channels → Mg^{2+} floats in the blood, but has difficulty getting into the starved cell. Requires months of patience and bioavailable forms (glycinate, malate, threonate)	Priority 1. Laboratory-guided (whole blood analysis), dose up slowly, prefer bioavailable forms. No “a lot helps a lot”
Zinc (Zn^{2+}) + Manganese (Mn^{2+})	“Ignition key” for SOD (superoxide dismutase, the most important cellular fire extinguisher) and TIMPs (MMP brakes) [13], [34]	In case of overdose: copper depletion, GI irritation, paradoxical immunosuppression	Laboratory-guided testing and correcting if there is a deficiency. Not as high-dose monotherapy
Iron ($\text{Fe}^{2+}/\text{Fe}^{3+}$)	Essential for severe anemia; Blocks FQ absorption most strongly in the stomach (80-90%)	Absolute radical amplifier in active oxidative stress. Free iron catalyzes the Fenton reaction: $\text{Fe}^{2+} + \text{H}_2\text{O}_2 \rightarrow \text{OH}^\bullet$ (hydroxyl radicals). As long as the mitochondrial fire smolders, any unnecessary iron intake significantly increases radical production [4], [34]	Only in cases of laboratory-confirmed anemia and under medical supervision. Handle restrictively in the acute phase of FQAD. Check ferritin and transferrin saturation
Calcium (Ca^{2+})	Bone protection in the event of immobility; Collagen cofactor	Neurological risk: FQ block GABA receptors and overstimulate NMDA receptors → uncontrolled Ca^{2+} influx into nerve cells (excitotoxicity) → nerve death, burning, cramps. High doses of Ca^{2+} can increase this tsunami [5]	Do not reduce (risk of osteoporosis, see calcium / Mg section organ axis 3), but do not supplement in high doses either. Moderate dietary intake; Mg^{2+} as an NMDA opponent has priority

Tiered supplementation protocol (clinically pragmatic, not FQAD validated):

1. Acute phase (damage stabilization): Top up magnesium (priority 1) + check zinc/selenium and compensate if there is a deficiency. Iron and calcium restrictive (only in laboratory-confirmed deficiency). No blind polysupplementation. **2. Stabilization phase:** Further optimize magnesium and zinc in the laboratory. Iron only after the acute oxidative stress has subsided (clinically: no more active crashes, more stable exercise tolerance) and only if anemia is proven. **3. Re-conditioning:** Calcium and Vitamin D/K2 for bone protection, especially in cases of immobility. Always accompanied by sufficient magnesium as an NMDA antagonist.

The basic rule remains: diagnostics → targeted compensation → monitoring. “A lot helps a lot” doesn't work with FQAD — the system is so unstable that precision wins over dosage.

The Glycine Paradox: Why the Carrier May Be More Toxic Than the Mineral

Level of evidence:[YELLOW] Plausible. Glycine as an NMDA co-agonist is neurochemically proven; The specific hypersensitivity in FQAD has been clinically reported but not systematically investigated. A common clinical conundrum: FQAD sufferers report massive fatigue after as little as 20 mg of magnesium glycinate — a dose that is pharmacologically almost negligible (~2-3 mg elemental Mg, ~17 mg glycine). The standard answer, “Take more magnesium,” makes the problem worse. The explanation lies not in the magnesium, but in the carrier molecule:

Problem	Mechanism	Consequence
Glycine = NMDA co-agonist	Glycine is an obligate co-agonist on the GluN1 subunit of the NMDA receptor. If the nervous system is intact, this is harmless because GABA functions as a counterbalance [5]	With FQ-blocked GABA receptors, the brake is missing → just 17 mg of glycine can trigger a neuro-inflammatory cascade: NMDA overstimulation → excitotoxicity → the broken mitochondria cannot supply the energy → system crashes → leaden exhaustion
Reaction is qualitative, not quantitative	The problem is not “too little magnesium”, but rather the intolerance of the carrier substance when the nervous system is damaged More magnesium glycinate = more glycine = more NMDA stimulation = worse fatigue. The assumption that it is a simple deficiency that can be remedied by higher dosage is untenable	

MCAS-safe forms of magnesium (glycine-free):

shape	Why safer with FQAD / MCAS	Cave
Magnesium malate	Malate goes directly into the citrate cycle → supports ATP production instead of stimulating NMDA	For citrate sensitivity (Aspergillus fermentation) as an alternative to Mg citrate
Magnesium Taurate	Taurine has a GABAergic (calming) effect instead of excitatory → counterbalance to FQ-GABA blockade	For very severe MCAS, test individually
Magnesium oxide (low dose)	Simplest form, no bioactive ligand; low bioavailability = slow, gentle absorption	May cause diarrhea at higher doses
Transdermal MgCl₂	Completely bypasses the GI tract and nervous system (see foot bath protocol)	Dilute brine, only soles of feet (see paradoxical weakness reaction)

Molecular Mimicry: Functional Vitamin B6 Blockade as a Bridge Between Nervous System and Mast Cells ([YELLOW]→[GREEN] increasingly supported)

Evidence level: [YELLOW]→[GREEN] The structural similarity of the fluoroquinolone core to the pyridine ring (vitamin B6/pyridoxal, B3/NAD⁺) is chemically established. **Upgrade 2026:** In-silico docking simulations [77] show that the ciprofloxacin quinoline core (free or as an Fe-FQ complex) docks exactly where the pyridine ring of pyridoxal and NADH naturally dock — at the active pocket of the enzyme PdxK (pyridoxal kinase) and at AIFM1. Friedman (2026) [79] and Reinhardt et al. (2025) [2] independently support off-target binding as the primary mechanism. The functional consequence (competitive inhibition rather than true deficiency) is thus far better supported than in v73, though controlled FQAD-specific human studies are still lacking.

Fluoroquinolones possess a quinoline core with a **fused pyridine ring** that acts as a bioisostere (structural doppelganger) of the pyridine ring — the ring structure found in vitamin B6 (pyridoxal), vitamin B3 (niacin), and in NAD⁺/NADH [77], [79]. Through this “doppelganger problem” (molecular mimicry), FQ molecules can occupy the binding sites (lock-and-key principle) of

pyridine-ring-dependent enzymes and competitively displace the real cofactors — even though standard lab values show normal or even elevated B6 levels in the blood. This is therefore not a classic deficiency but a **functional blockade at the receptor or enzyme pocket** [77].

Amplification by the Fe-FQ complex: The docking data [77], [79] show that iron binding *enhances* this competitive inhibition — the Fe-FQ complex binds with higher energy to the pyridine-dependent enzyme pockets than free ciprofloxacin. This links the metal binding described in 1.5 directly to mimicry: chelation is the anchor, mimicry is the weapon.

Why this is catastrophic in FQAD — two systems fail simultaneously:

Blocked system	Mechanism	Clinical consequence
GABA production	Vitamin B6 (active form: P5P) is an obligate cofactor of glutamate decarboxylase (GAD), which converts glutamate (excitatory) to GABA (inhibitory). With B6 blockade, GABA synthesis fails [5], [77]	Nerves fire uncontrollably → dysautonomia, panic attacks, insomnia, muscle twitching. This additively amplifies the known direct GABA-A receptor blockade by FQ (see 1.4)
DAO production (histamine degradation)	B6 / P5P is the most important cofactor of diaminoxidase (DAO), which degrades histamine in the gut. With B6 blockade, DAO activity collapses [77]	Histamine accumulates uncontrollably → mast cell escalation (MCAS), immediate reactions to foods. This explains the molecular bridge between FQ exposure and the histamine intolerance frequently reported in FQAD

Additional receptor blockade: Nicotinic acetylcholine receptors (nACh):

Beyond the known GABA-A receptor blockade, fluoroquinolones also inhibit nicotinic acetylcholine receptors (nACh receptors) [77]. Acetylcholine is the most important neurotransmitter of the vagus nerve (parasympathetic system). nACh blockade explains at the molecular level the dysautonomia frequently seen in FQAD, particularly gastroparesis: without functioning acetylcholine receptors, there is no regulated intestinal peristalsis.

The kynurenine trap — why B6 supplementation can be paradoxically toxic:

In chronic B6 depletion, tryptophan (which should normally become calming serotonin) is diverted into the kynurenine pathway. This pathway produces neurotoxic and neuroinflammatory metabolites. If P5P is suddenly introduced into this state, the system can overreact — the so-called “paradoxical B6 intolerance” in FQAD [77].

Iron-fluoroquinolone coordination complexes — why the toxin remains “cemented into tissue”:

Iron can bind up to three ciprofloxacin molecules simultaneously, forming extremely stable coordination complexes (Fe-FQ₃ crystals). Normal hepatic detoxification enzymes (UGTs, ABCB transporters) cannot process these complexes [77]. This could explain why affected individuals still feel the substance is “still in the body” years after discontinuation — it is literally embedded in tissue as microcrystals. This mechanism amplifies the ion trapping effect (see 1.5) and provides an additional explanation for the extreme long-term persistence of symptoms.

AMPK suppression by ciprofloxacin — the missing switch for mitochondrial biogenesis ([YELLOW] Plausible):

Evidence level: [YELLOW] AMPK suppression by ciprofloxacin is demonstrated in a recent study (Zeng et al. 2024) [78]. The direct connection to PGC-1α blockade in FQAD is mechanistically compelling but clinically not validated.

Zeng et al. (2024) [78] showed that ciprofloxacin actively suppresses AMPK phosphorylation. AMPK is the primary cellular energy sensor and the obligate upstream activator of PGC-1α — the switch for mitochondrial biogenesis. If ciprofloxacin blocks AMPK, this mechanistically explains why PBMC lab results in FQAD patients show near-complete PGC-1α shutdown (see residual capacity calculation, Part 2): mitochondrial biogenesis is not only prevented by insufficient energy, but the build command itself is actively suppressed. The cell is stuck in a deadlock — too little ATP for repair, and the command to repair is simultaneously blocked.

Consequence for quercetin (see above): Quercetin enzymatically activates exactly this AMPK/SIRT1/PGC-1α signaling pathway, thereby directly addressing the ciprofloxacin-caused AMPK block [78].

Berberine warning ([RED] Contraindicated in FQAD):

Berberine is used in integrative medicine for insulin resistance and SIBO (small intestinal bacterial overgrowth). In FQAD, berberine is **absolutely contraindicated**: berberine possesses a chemical scaffold structurally extremely similar to the fluoroquinolone core structure. In a system sensitized to the FQ scaffold (particularly with MCAS), berberine can trigger the same cellular alarm as a mimicry substance and provoke a severe FQAD crash [77].

Extended molecular mimicry: The FQ molecule as a universal key ([YELLOW] Plausible)

Evidence level: [YELLOW] The structural similarities of the heterocyclic FQ ring structure to endogenous molecules are chemically established. The functional consequences of competitive displacement are mechanistically plausible and consistent with the clinical picture, but not isolated in controlled FQAD-specific studies [77].

Molecular mimicry is not limited to vitamin B6. The quinoline ring and nitrogen heterocycles of the FQ core structure occur in nature at numerous critical sites. The FQ molecule acts as a false universal key that fits into many different locks in the body:

Endogenous molecule	Structural similarity to FQ	Blocked function	Clinical consequence
Vitamin B6 (Pyridoxine)	Pyridine ring ≈ FQ heterocycle	GAD enzyme (glutamate→GABA) + DAO (histamine degradation)	GABA failure + MCAS escalation (see above) [77]
Vitamin B3 (Niacin / NAD+)	Pyridine ring ≈ FQ heterocycle	NAD+ metabolism → NAD+ is substrate for SIRT1 → SIRT1 activates PGC-1α	NAD+ depletion → SIRT1 fails → PGC-1α shutdown → no mitochondrial biogenesis. Additively amplifies AMPK suppression [78]
Kynurenic acid (endogenous 4-hydroxyquinoline)	Nearly identical: kynurenic acid IS the body's natural quinolone	Kynurenic acid calms NMDA receptors (natural glutamate antagonist). FQ displaces it and blocks instead of calming	Glutamate accumulates → NMDA overexcitation → neurotoxicity, anxiety, dysautonomia. Explains the paradoxical nerve damage from an “antibiotic”
Coenzyme Q10 (Ubiquinone)	Quinone ring structure related to quinolone	Q10 docking sites in the ETC (Complex I + III): FQ displaces endogenous Q10	Electron flow breaks → ATP stop + massive ROS. Explains why Q10 supplementation can be paradoxical in FQAD: more Q10 on an FQ-clogged road = more electron backup
Purines (Adenine, Guanine)	Purine double rings with nitrogen ≈ FQ scaffold	DNA intercalation: flat FQ molecule wedges between base pairs of mtDNA	Blocks reading and repair of mtDNA. Amplifies the Top2β mechanism (see 1.2)

The piperazine ring as a second off-target source — structure-activity evidence ([GREEN] supported by Duman 2025):

Besides the quinoline/pyridine core, most fluoroquinolones carry a **piperazine ring** as the “tail” of the molecule (position R7). Duman et al. (2025) [80] provided direct structure-activity evidence that this ring is also an independent pathological factor: fluoroquinolones with a **methy-lated piperazine ring** (ofloxacin, levofloxacin, gatifloxacin) show a **significantly increased risk of tendon rupture** compared to FQ with a non-methylated piperazine ring (ciprofloxacin, norfloxacin). The latter also increase the risk, but to a lesser degree.

This dose-response gradation along a single structural variable (methylation of the piperazine ring) is a strong argument: toxicity is mediated by a nitrogen-containing heterocycle and an off-target effect — not primarily by DNA damage or pure chelation. This supports the mimicry hypothesis independently of the pyridine argument and explains why different FQ are differently toxic (see also Cipro vs. Levo comparison, 1.8). Clinical consequence: where an FQ is

unavoidable and effective, an FQ with a non-methylated piperazine ring (e.g. ciprofloxacin) is tendentially preferable to the methylated agents (levofloxacin) with respect to tendon rupture.

Tonic water / quinine warning ([RED] Contraindicated in FQAD):

Quinine — the bitter compound in tonic water — is the chemical precursor from which fluoroquinolones were derived in the 1960s. Both share the same toxicological core: the **quinoline ring**. Quinine is thus even closer to the FQ core structure than berberine. Three reasons why tonic water is contraindicated in FQAD:

1. **Quinoline mimicry:** Traumatized mast cells scan the quinoline ring, recognize the “poison” pattern, and immediately trigger degranulation → crash, nerve pain.
2. **Carbonation + gastroparesis:** Gas cannot be transported away → stomach inflates → mechanical pressure on diaphragm and vagus nerve → Roemheld syndrome (tachycardia, blood pressure fluctuations, ATP consumption).
3. **Citrus flavoring:** Nearly all tonic waters contain citrus extracts — high-grade histamine liberators that force immediate mast cell degranulation in MCAS.

PQQ warning ([RED] Caution in FQAD):

PQQ (pyrroloquinoline quinone) is recommended as a “mito booster” for PGC-1 α activation. The name reveals the problem: **pyrroloquinoline quinone** contains exactly the quinoline ring. In FQAD patients whose mast cells are sensitized to this chemical scaffold, PQQ very frequently triggers massive crashes and nerve pain — despite the theoretically favorable mechanism of action.

Consequence for supplementation — the nitrogen-ring-free safe list:

Since the quinoline/pyridine ring is the central mimicry motif, substances **without nitrogen rings** are by definition not confusable with the FQ molecule and do not trigger a “doppelganger alarm” at mast cells:

Category	Safe substances	Why safe	MCAS trap to avoid
Pure elements (no ring structure possible)	Magnesium, zinc, selenium	Single atoms / ions → no confusion with FQ rings	Mg: no citrate (mold fermentation → MCAS); only malate, taurate, transdermal MgCl ₂ . Zinc: sulfate drops pure, no tablets with fillers
Structurally safe vitamins (no nitrogen ring)	Vitamin C (ascorbic acid), Vitamin D3 / K2	Simple sugar or steroid structure, no heterocycles	Vitamin C: only from tapioca (cassava), not corn-fermented, no citrus residues. D3 / K2: drops in MCT or olive oil, no orange oil as preservative
Linear fatty acids (no rings)	C8 MCT oil (caprylic acid), algae oil (EPA / DHA)	Straight carbon chains → structurally impossible to read as FQ doppelganger	C8 MCT: gastroparesis bypass (needs neither stomach acid nor bile), delivers ketones = clean energy without lactate
Free amino acids (no rings in EAAs)	Pure EAAs (leucine, isoleucine, valine, etc.)	Unbound, linear amino acids without toxic ring systems	No isolated tryptophan (→ kynurenine pathway). No flavoring, sweeteners, or fermentation products
Flavonoids (ring, but no nitrogen)	Quercetin (pure substance)	Flavonoid rings contain no nitrogen → not read as FQ doppelganger	Mast cell stabilizing rather than activating. Pure substance only, microdosing (see quercetin section above)

The toothpick method — introduction protocol for any new substance in FQAD/MCAS:

Even substances on the safe list must be introduced with extreme caution, as the system in alarm state evaluates every new input as a potential threat:

1. **Never start two substances simultaneously** — if a reaction occurs, it would be unclear which was the trigger.
2. **Toothpick dosing on day 1:** Moisten the tip of a toothpick, dip it into the powder → stir this microscopic amount into a glass of still water.

3. **Wait 48 hours and observe:** No nerve burning, no abdominal cramping, pulse stable → system has accepted the molecule as safe.
4. **Only then increase to knife-tip**, then to ¼ capsule, etc. Observe each increase for 48h again.

Safe B6 supplementation in FQAD ([YELLOW] Individualized, P5P only):

Rule	Rationale
Exclusively active form P5P (pyridoxal-5-phosphate) — never pyridoxine HCl	Inactive B6 (pyridoxine HCl) must be converted in the liver. With blocked enzymes, inactive B6 accumulates in the blood → high levels of inactive B6 are strongly neurotoxic and cause exactly the same nerve pain (neuropathies) as FQAD itself
Microdosing: 1-2 mg (one drop or crumb of powder)	Standard doses (20-100 mg) immediately overexcite pre-damaged nerves → burning, worsening
Rhythm: every 2nd-3rd day	Nervous system gives feedback: increasing burning, restlessness, or sleep worsening within 12-24h = stop signal
Form: Liquid drops or pure powder — no capsules with fillers	No microcrystalline cellulose, no TiO ₂ , no flavoring (MCAS triggers)

With 10+ years of FQAD/MCAS, even small amounts of certain foods can trigger acute fatigue, which is not “fatigue” but a measurable loss of ATP (metabolic PEM). The mechanism proceeds through three converging paths: 1. **Mast cell trigger → cytokine storm → ATP robbery:** Histamine liberators (avocado), salicylates (peppers, onions) or lectins (spelled/gluten) activate the unstable mast cells → massive release of inflammatory mediators → the immune system immediately withdraws all remaining energy (ATP) from the already weak mitochondria [19]. 2. **Sulfur Metabolism Overload:** Onions and cruciferous vegetables contain high amounts of sulfur. If there is a genetic or acquired weakness in sulfur metabolism (CBS or SUOX enzyme defects), the metabolism of sulfur compounds can produce toxic intermediates (sulfite) that put additional strain on the mitochondria. 3. **Salicylate intolerance as a third dimension:** In addition to histamine and oxalates, salicylate hypersensitivity is often present in FQAD/MCAS. Salicylates (found in peppers, tomatoes, spices, herbs) inhibit cyclooxygenase → leukotriene synthesis is redirected to the lipoxygenase pathway → additional mast cell activation via a non-histaminergic pathway.

The exhaustion is not a weakness, but a biochemical energy deficit: immune reaction + NMDA overstimulation + mitochondrial damage = the system literally runs out of power.

Quercetin as PGC-1α activator and mast cell stabilizer ([YELLOW] individualized)

Evidenzgrad:[YELLOW] PGC-1α-Aktivierung durch Quercetin ist in präklinischen Studien beschrieben; FQAD-specific data is missing. Quercetin is also a potent senolytic - in FQAD the dosage is therefore a balancing act between promoting biogenesis and the risk of senescence crash (see evaluation matrix, polyphenol trap).

Quercetin enzymatically activates the same signaling pathway as endurance training: the AMPK/SIRT1/PGC-1α axis → new formation of mitochondria (biogenesis). This directly addresses the PGC-1α blockade demonstrated by the PBMC laboratory (see mitochondrial residual capacity, part 2). At the same time, quercetin stabilizes mast cell membranes and inhibits histamine release — a dual benefit in FQAD/MCAS.

Problem	Solution	Justification
Standard capsules contain MCAS triggers	Mandatory pure substance: Ingredients list may only contain “quercetin” or “quercetin dihydrate”. No magnesium stearate, no titanium dioxide, no microcrystalline cellulose	Standard fillers are high-level MCAS triggers that negate gut benefit
Capsules lie in the paralyzed stomach	Open the capsule → sprinkle pure powder into still water or amino acid drink	Gastroparesis Bypass: Powder in solution passes through the stomach faster than solid capsules

Problem	Solution	Justification
Dosage: Senolytics vs. Biogenesis	Start: knife tip (□100 mg) for 3 days → if tolerated: ½ capsule → 1 capsule (250–500 mg). Do not start with the full dose	Quercetin is also a senolytic: Too high a dose kills masses of senescent cells → gigantic cell waste in an overloaded liver → week-long crash. Microdosing allows PGC-1α activation without a senescence avalanche
Natural sources are not enough	Top sources (capers 230 mg, lovage 170 mg) are not edible for MCAS / gastroparesis (histamine, fiber, acid). Only buckwheat (23 mg / 100g) is MCAS safe and easily digestible — but for therapeutic doses you would need to eat >2 kg / day	Pure substance as a “bypass” is safer than trying to achieve therapeutic doses through food

Omega-3 supplementation: TOTOX value and oxidation protection ([YELLOW])

Problem	Solution	Justification
Fish oil oxidized → mito-toxin + histamine	TOTOX value must be on the bottle and be <5. If it is not on it: do not buy	Oxidized fish oil produces lipid peroxides → direct mitochondrial poison + massive histamine release (nausea, racing heart). Fatal in FQAD
Fish oil = animal protein = MCAS risk	Liquid algae oil in a darkened glass bottle instead of fish oil capsules. Place in the refrigerator immediately after opening Algae oil: plant-based, histamine-free, provides direct EPA / DHA without fish protein. Dark glass protects against light oxidation	
Capsules for gastroparesis	1 tsp liquid oil → passes through the stomach almost effortlessly (no CCK reflex with small quantities)	Large soft gel capsules lie in the paralyzed stomach for hours
Vegetable ALA (linseed oil) is not enough	ALA → EPA / DHA conversion requires intact mitochondria + liver enzymes → massively restricted in FQAD (conversion rate estimated <5% instead of normal 5–15%)	You need the marine forms (EPA / DHA) directly - ALA from linseeds / walnuts is therapeutically inadequate

1.6 Ciprofloxacin vs. Levofloxacin ([YELLOW] Plausible)

D3. Ciprofloxacin vs. Levofloxacin: Structural differences and clinical consequences

Both molecules possess the same toxic fluoroquinolone double ring (the “toxicophoric core”) that is responsible for ETC blockade, chelation, and MMP escalation [2]. The Reinhardt study [2] shows that both fit into the binding pockets of AIFM1 and IDH2. The 3D differences mainly affect the side arms and determine the disposal and accumulation dynamics:

Property	Ciprofloxacin	Levofloxacin
Structural feature	Cyclopropyl group (flat, compact)	Additional oxazine ring; L-enantiomer (asymmetric, “sharpened”)
Main clearance	Dual: liver (CYP1A2) + kidney	Almost 100% kidney; Liver hardly plays a role
Plasma half-life	~4 h (with normal clearance)	~6–8 h (longer circulation time)
Primary eye of the needle	CYP1A2 capacity (liver genetics)	Creatinine clearance / eGFR (renal function)

Property	Ciprofloxacin	Levofloxacin
Ion trapping risk	Increased in slow detoxifiers (slow metabolizers). Increased in cases of impaired kidney function or dehydration; longer HWZ = more time for tissue accumulation	

Why Levofloxacin Often Causes More Severe Damage — Three Converging Factors:

1. Lack of “back door”: Ciprofloxacin has two pathways for degradation (liver + kidney); If one fails, the other can partially compensate. Levofloxacin only affects the kidneys — if you have even slightly impaired eGFR, dehydration or age, there is no Plan B. **2. Longer exposure time:** The half-life, which is almost twice as long, gives the molecule more time to penetrate tissue with poor blood circulation (tendons, cartilage, CNS) and to be stored there accumulatively via ion trapping. **3. Perfected 3D key:** As a pure L-enantiomer, Levofloxacin is the sharpened version - all ineffective molecular parts have been removed, which maximizes the binding affinity to human enzymes (AIFM1, IDH2, topoisomerase).

Metabolite detoxification — how the liver destroys the key:

When ciprofloxacin encounters CYP enzymes in the liver, the 3D shape of the molecule is changed (side chains broken off, sulfate/oxo groups stuck on → sulfociprofloxacin, oxo-ciprofloxacin). These metabolites have lost their perfect 3D fit for the enzymatic binding pockets — they no longer fit into the “vault rooms” of AIFM1 or topoisomerase II and can no longer chelate there. The metabolites are pharmacological “junk”: water-soluble, renally excretable, no longer capable of being used as weapons [51]. The FQAD problem is therefore not toxic liver degradation, but rather a lack of liver degradation: With genetic slow detoxifiers (slow metabolizers), the intact, highly toxic original molecule remains in the blood and has time to penetrate tissue via ion trapping.

Coffee sensitivity as a clinical CYP1A2 indicator:

Caffeine is broken down by CYP1A2 — the same enzyme that metabolizes ciprofloxacin (and which further inhibits ciprofloxacin). Patients who respond to a cup of coffee with heart palpitations that last for hours, insomnia, or extreme restlessness are most likely CYP1A2 slow detoxifiers. This coffee sensitivity is a simple, free clinical indication of increased risk of ciprofloxacin accumulation. This analogy does not apply to levofloxacin (no CYP1A2 dependence), where kidney function is the crucial bottleneck.

Related Drug Classes — Cross Vulnerability:

Drug class	Structural / functional analogy to FQ	Clinical Consequence for FQAD Risk Assessment
Malaria drugs (quinolines): Mefloquine, Chloroquine, Hydroxychloroquine	Same quinoline ring as chemical skeleton. Mefloquine: FDA black box warning for long-lasting, partly irreversible neurotoxicity	Patients with severe reactions to malaria prophylaxis / hydroxychloroquine often have a genetic vulnerability to this ring structure → extreme caution with FQ
Topoisomerase inhibitors (chemotherapy): Etoposide, Doxorubicin	Attack the same cellular target (topoisomerase II) as FQ — intentionally rather than “accidentally.” “Chemo-Brain” and neuropathy due to identical NAD ⁺ /ATP collapse	Patients after chemotherapy have no DNA repair regeneration reserve for new chemically induced DNA breaks
CYP Detox Twins: Codeine (CYP2D6), Antidepressants (CYP2D6), Isoniazid (NAT2)	Identical detoxification enzymes as for ciprofloxacin. Hypersensitivity to these substances = indication of slow metabolizer genetics	Extreme codeine sensitivity, increased SSRI effect or coffee sensitivity = clinical warning signal for increased risk of FQ accumulation

Underweight and creatinine trap: In patients with very low muscle mass (e.g. <50 kg), serum creatinine can life-threateningly overestimate kidney function. Creatinine is formed from muscle creatine - little muscle mass = little creatinine = falsely “perfect” eGFR values. The doctor does

not adjust the levofloxacin dose downward, and the toxic burden on the little remaining tissue is massively excessive. If you are underweight, kidney function should be determined using cystatin C (independent of muscle mass) instead of creatinine.

1.7 Full Reaction Cascade: 6 Fronts ([YELLOW] Plausible)

D5. Complete reaction cascade: From molecule to tissue damage

Level of evidence:[YELLOW] Individual mechanisms supported experimentally [1]–[4], [13], [34], [51]. The combination into a linear cascade with quantitative OH• hit distributions (~50/45/5%) is a pathobiochemically plausible working hypothesis; The percentages are estimates from competitive kinetics, not clinical measurements.

The following six fronts describe the chronological causal chain from chelation to the radical explosion to the final damage. Fronts 0-2 are triggers; Fronts 3-5 are the consequential damage caused by OH• radicals, whose distribution to cellular targets follows the competition kinetics. Evidence status: Individual mechanisms supported experimentally [1]–[4], [13], [34], [51]; the combination to form a linear cascade is a pathobiochemically plausible working hypothesis.

The triggers Front 0 — Enzymatic disarmament and loss of structure (chelation):

- (1) Intact, lipophilic FQ penetrates tendons/cartilage/intracellular space → (2) acts as a strong chelating agent → (3) local withdrawal of Mg^{2+} , Mn^{2+} , Zn^{2+} , Ca^{2+} → (4) cellular “fire extinguisher” (superoxide dismutase/SOD, Mn^{2+} -dependent) and DNA repair enzymes (polymerases, Mg^{2+} -dependent) lose their catalytic metal ion and fail → (5) Collagen fibrils lose stabilizing Mg^{2+} “mortar” (mechanical destabilization) → (6) FQ anchors itself permanently in the tissue through the bound ion (ion trapping, see D2) [4], [13], [51].

Front 1 - Metabolic short circuit (dark reaction in deep tissue): (1) Intact FQ blocks topoisomerase / ETC complexes I+IV [2] → (2) electron accumulation on the respiratory chain → (3) premature O_2 reduction → (4) superoxide ($O_2^{\bullet-}$) is formed → (5) SOD converts to H_2O_2 (but SOD is already weakened by front 0) → (6) Fenton reaction on free iron → mass emission of hydroxyl radicals (OH•) [4], [34].

Front 2 — UV phototoxicity (light reaction in skin/eyes):

Only relevant in exposed skin, not active in deep tissue: (1) FQ absorbs UVA photons → (2) molecule jumps to excited triplet state → (3) energy transfer to O_2 OR photolytic breakage of the C-F bond → (4) massive formation of singlet oxygen (1O_2), FQ radicals and local free fluoride (F^-). **Here - and only here - the drug actually decomposes itself and damages the tissue with its own debris. This is the only context in which true fluoride release occurs**— quantitatively small compared to the overall damage, but relevant locally in the skin for photodermatitis and photosensitivity.

The consequential damage: What the OH• radical hits the tissue As soon as OH• is created by front 1 and the protective shields by front 0 are missing, the competition kinetics takes effect - the radical reacts within nanoseconds with the next accessible biomolecule:

Front	Target	Estimated proportion of OH• hits	Mechanism	Result
3. Lipid peroxidation	Cell membranes	~50%	OH• snatches an H atom from fatty acid → lipid radical ($L^{\bullet}$) + O_2 → lipid peroxy radical ($LOO^{\bullet}$) → chain reaction (infinite loop) → membrane breakdown → highly toxic aldehydes (MDA, 4-HNE)	membrane integrity destroyed; 4-HNE migrates and damages distant cells
4. Protein oxidation	Structural proteins, enzymes	~45%	OH• attacks cysteine, methionine, tyrosine → protein radical → crosslinking (gluing) → protein carbonyl formation	Collagen and enzymes clump together; mechanical and biochemical function lost

Front	Target	Estimated proportion of OH• hits	Mechanism	Result
5. DNA damage	mtDNA, nuclear DNA	~1-5%	OH• (or diffused H ₂ O ₂) reaches DNA → ring oxidation on guanine → 8-oxo-dG + strand breaks → reading errors [34]	Lowest hit rate, but most severe long-term consequences: Mito function stop, apoptosis or senescence (see phases 3-5 of cascade F)

Drug degradation by radicals (OH• accidentally hits the FQ ring itself) is quantitatively negligible at <0.001% of hits - the drug does not “sacrifice itself” but remains intact in the tissue as an intact catalyst of destruction.

1.8 Nemonoxacin: The counter-evidence ([YELLOW] Plausible)

D6. Nemonoxacin: Evidence to the contrary from the missing fluorine atom

Nemonoxacin (Taigexyn) is a non-fluoroquinolone (des-fluoroquinolone): the developers intentionally omitted the fluorine atom at C-6. Sending this molecule through the causal cascade (D5) shows which fronts have been defused:

Front	Nemonoxacin status	Justification
0. Chelation	Partially received	The quinolone skeleton (3-carboxy-4-oxo structure) is still there → Mg ²⁺ /Zn ²⁺ bonding is still possible. But: without fluorine significantly lower lipophilicity → less deep tissue penetration → chelation effect more superficial / in the blood, where the body can compensate better
1. ETC blockade / dark reaction	Strongly weakened	Without fluorine: faster liver degradation (no CYP resistance), shorter circulation time → less ion trapping → lower mitochondrial accumulation → significantly reduced electron accumulation and ROS production
2. Phototoxicity	Virtually eliminated	Where there is no C-F, photolytic C-F cleavage cannot take place. Nemonoxacin shows virtually no phototoxicity in clinical studies
3-5. Consequential damage	Proportionally reduced	Less ROS formation (front 1 weakened) + lower tissue accumulation → less lipid peroxidation, protein oxidation and DNA damage

Conclusion: The developers of Nemonoxacin have a precise understanding of FQAD pathobiochemistry. By removing the fluorine atom, they have stripped the weapon of its armor (metabolic stability), its metabolic stability (lipophilicity) and its phototoxic potential (phototoxicity). A residual risk of tendon problems remains (the quinolone skeleton continues to chelate), but the catastrophic chain reaction - in which intact FQ triggers an eternal radical fire deep in the mitochondria - is toxicologically almost impossible. Nemonoxacin is the pharmacological counter-evidence for the enabling role of fluorine (see D4).

1.9 Pathobiochemical Cascade Model — Dual-Track with Mimicry Bridge ([YELLOW]/[RED])

Hypothetical pathobiochemical cascade model — dual track: off-target mimicry + chelation anchor

Level of Evidence:[RED] Hypothetical in its synthesis. **Conceptual update (v74):** the structural track is no longer presented as an isolated magnesium-deficiency process. Following Friedman (2026) [79] and Reinhardt et al. (2025) [2], the **molecular mimicry of the Fe-FQ complex forms the bridge** between the two tracks: the same metal binding that drives tissue accumulation (ion trapping) simultaneously increases binding energy to the pyridine-dependent off-target enzymes. Chelation and enzymatic toxicity are thus not two separate mechanisms but two faces of the same molecule.

The mimicry bridge — how the two tracks connect:

The older picture of two parallel, separate damage pathways (DNA/mitochondria vs. chelation/matrix) is outdated. The connecting mechanism is the Fe-FQ complex:

1. **Chelation (anchor):** FQ binds iron → loses lipophilicity → becomes trapped in tissue (ion trapping). The direct ion robbery from structural proteins is *not* the main damage (see tetracycline paradox, 1.5).
2. **Mimicry (weapon):** the Fe-FQ complex binds with increased energy to pyridine-ring-dependent enzymes (PdxK, AIFM1, IDH2) and receptors (nACh) — exactly where pyridoxal/NADH dock [77], [79]. This explains both the mitochondrial crisis (ETC) and the B6/B3/GABA failures.
3. **Convergence:** both effects emanate from the same accumulated molecule. Tissue accumulation (the “structure” track) and enzymatic functional failure (the “cell” track) are causally coupled, not parallel.

Therapeutic window — why the phase is decisive:

The intervention logic must be based on the temporal phase of the cascade: **-Phase 1-2 (Months 0-6):** ROS fire active. Priority: Mg^{2+} replenishment, radical protection, antioxidant support. Iron restrictive (Fenton). No forced training. **-Phase 3-4 (Year 1-4):** MMP escalation and tissue remodeling. Priority: Structural rehabilitation (careful eccentric training for tendinopathy), TIMP support (zinc), senescence monitoring. Pure antioxidant supplementation alone brings little functional benefit. **-Phase 5 (Year 4+):** SASP lockdown. The oxidative burn is often extinguished, but senescent cells dominate. Here, structural conversion work and - in the research context - senescence clearance come into focus. Antioxidants are then more supportive than causal.

This temporal allocation prevents chronic patients from demanding treatments that can only be justified pathobiochemically in the acute phase.

Evidence status: The following model integrates known individual mechanisms (Top2 β inhibition, Fenton reaction, PARP activation, PGC-1 α dysfunction, senescence) with the chelation/ion trapping mechanism (see Section D2) to form a coherent cascade hypothesis. The individual building blocks are experimentally supported [1]-[4], [13], [34], [51]; The combination into a linear process model with quantitative weights is a pathobiochemically plausible but not clinically validated working hypothesis.

Temporal Sequence — Dual Track (hypothetical)

phase	Period	Cellular Track (DNA / Mitochondria)	Structural track (chelate / matrix)	Clinical endpoint
1. Double toxic trigger	Day 1 – Week 4	FQ inhibits human topoisomerase II → immediate double-strand breaks of nuclear DNA [1], [3]. Parallel: ETC blockade via AIFM1 / IDH2 binding [2]	FQ accumulates in tendons / cartilage via ion trapping; acts as a sponge and continuously removes Mg^{2+} from matrix proteins / enzymes → Collagen loses microscopic tension and frays [13], [51]	Often still with few symptoms; the course for destruction is set in parallel
2. Oxidative fire + cellular alarm	Month 1-6	Blocked respiratory chain accumulates electrons → Fenton reaction → hydroxyl radicals physically damage mtDNA [4], [34]. PARP enzymes consume massive amounts of NAD ⁺ /ATP during nuclear DNA repair → energy collapse accelerates	Frayed collagen is registered as an “injury.” Mg^{2+} deficiency in mitochondria creates additional oxidative stress. Double alarm: defective collagen + ROS = cellular distress signal	First noticeable symptoms: fatigue, burning nerves, tendon pulling, diffuse pain

phase	Period	Cellular Track (DNA / Mitochondria)	Structural track (chelate / matrix)	Clinical endpoint
3. MMP escalation	Month 6 - Year 2	Guardian protein p53 detects irreparable DNA damage → senescence. Cell suffocates energetically → anaerobic emergency power (glycolysis). Possible PGC-1 α dysfunction blocks mitochondrial remodeling (placeholder, see evidence status)	In response to phase 2 distress signal: massive amounts of MMPs (MMP-2, -9, -13) released → Order: “clean up broken collagen”. Since FQ continues to persist in the tissue via ion trapping, the alarm remains on “red” → MMPs also eat healthy tissue without a brake [13]	Phase of the most severe chronic pain, tendon tears, deep exhaustion, PEM
4. Catabolism + production stop	Year 2-4	To maintain residual energy, the body burns its own muscle protein (gluconeogenesis) → ammonia, BCAA waste [4], [35]	DNA-damaged senescent fibroblasts completely stop producing collagen. MMPs continue to break down, but no replenishment → visible tissue degeneration (tendons, cartilage, fascia) [13]	Sarcopenia, accelerated osteoarthritis, increasing immobility
5. SASP lock-down	Year 4+	Accumulated senescent cells do not die but permanently release SASP factors → chronic inflammation meets metabolic inflexibility [34]	Structural final stage: Tissue is enzymatically decomposed and is no longer renewed; chronic joint / tendon degeneration, microcirculation disturbed	Chronic FQAD syndrome; Final state without external intervention

The key message: Chelation is not primarily an “ion robbery” that directly decomposes the structure (the tetracycline paradox refutes this, see 1.5), but the **anchor** that lodges the FQ molecule as an Fe-FQ complex in tissue and *amplifies* its **mimicry binding** to pyridine-dependent enzymes. The DNA/ETC damage simultaneously deprives the cell of any energy to defend itself or build new material. Both tracks originate from the same accumulated molecule and converge in phase 3 at MMP escalation — the point of clinically visible structural damage.

Four chain reactions and their valued contribution

Chain	Mechanism	Estimated share	Evidence status
1. Toxic Trigger (Dual)	Accumulation → protein binding (AIFM1 / IDH2) → ETC blockade + ion trapping → chelate attachment	~10%	Experimentally supported [2], [51]
2. Oxidative “hit” + structural weakening	Fenton reaction → mtDNA destruction + Mg ²⁺ withdrawal → collagen fraying	~25%	Mechanistically supported [4], [13], [34]
3. Biogenesis Disruption + MMP Escalation	Possible PGC-1 α dysfunction → lack of new mitostructure → anaerobic emergency power + MMP permanent activation due to persistent ion trapping alarm	~40%	Hypothetical; PGC-1 α as a placeholder; MMP upregulation supported [13]
4. Senescence + Tissue Decay	Nuclear DNA breaks → PARP → p53 → senescence → SASP + collagen production stop → irreversible loss of structure	~25%	Mechanistically plausible; FQAD-specific not proven

The percentages are estimates based on mechanistic plausibility, not clinical measurements. This model is a working hypothesis intended as a research roadmap. It may not be used to make treatment decisions or promises of healing; What is crucial is that each chain has yet to be proven in human studies.

1.10 Metabolic emergency power ([YELLOW] Plausible)

Metabolic emergency power: Why standard labs hide the damage

In severe mitochondrial dysfunction, the cell switches from aerobic energy production (36 ATP per glucose molecule via the respiratory chain) to anaerobic glycolysis (2 ATP per molecule, end product: lactate) [4], [35]. This “emergency power generator” explains several clinical phenomena:

Why standard blood counts look normal: At rest, the liver eliminates lactate produced via the Cori cycle quickly enough to keep blood counts within the norm. Standard laboratory parameters (cholesterol, liver enzymes, blood sugar) measure homeostasis outcomes, not the cellular price thereof [35].

Why minimal stress leads to collapse (PEM biochemistry): During stress, energy requirements increase; Glycolysis, which is 18 times more inefficient, suddenly produces more lactate than the liver can eliminate. Local tissue acidification causes muscle heaviness, pain, and cognitive decline—the precise biochemistry of post-exertion malaise [23], [35]. **Consequence for diagnostics:** An isolated lactate value at rest is not sufficient; The lactate-pyruvate quotient is crucial (see stage 2, phase B), as it shows the backlog in front of the blocked mitochondria. In addition, blood collection must be carried out in strict protocol (see biomarker protocol below) [35].

1.11 Chronification — five hypotheses ([YELLOW]/[RED])

Chronification — five hypotheses

Level of evidence:[YELLOW] Plausible for hypotheses 1-3 (mtDNA persistence, ion trapping, epigenetics). [RED] Hypothetical for Hypothesis 4 (cumulative accumulation) and Hypothesis 5 (pre-existing comorbidity / “double hit”). Longitudinal clinical data are completely lacking.

The real core question: Why symptoms persist for years is only half the question. The clinically — and for patients — more pressing question is: **Why does the majority of FQ-injured people recover (often over months to years) to a usable functional level, while a minority remains permanently severely affected?** The first four hypotheses explain *persistence* but not the *outcome dichotomy*. Hypothesis 5 addresses this directly.

1. Prolonged oxidative stress destroys mtDNA; faulty cytochrome complexes create permanent electron leakage. (*Describes a consequence of Hypothesis 2 rather than an independent cause.*)
2. FQ-protein and FQ-cation (Fe-FQ) complexes persist in cells for years — the ion trapping mechanism (see 1.5) explains intracellular accumulation. The INUSpherosis observation (temporary improvement after blood washing, see 3.4) is consistent with this hypothesis.
3. Epigenetic changes in gene regulation become permanent after FQ exposure (see retrograde signaling chain, collagen switch)
4. **Cumulative damage from repeated exposure** — each FQ dose leaves a microscopic footprint. *Limitation:* this hypothesis does not explain cases where a single FQ course causes permanent severe disability.
5. **Pre-existing comorbidity / “double hit”** — perhaps the strongest candidate to explain the outcome dichotomy (see below).

Hypothesis 5: The “double hit” — pre-existing, clinically silent burden ([RED] Hypothesis, high explanatory power)

This hypothesis combines the “double hit” idea from the research literature [77], [79] with clinical observations from the patient community. It is speculative but is the only one that directly explains *why the same drug at the same dose leaves one person unharmed and permanently disables another*.

The core idea: before FQ exposure, a chronic subclinical cellular burden already exists — a pre-damage precisely in the protein/metabolic cluster that fluoroquinolones attack (mitochondria,

pyridine-dependent enzymes, detoxification, connective tissue). This comorbidity was clinically *silent* because the body could compensate for it adequately. The FQ hit is then not the sole cause but the “second hit” that tips an already pre-burdened system over the compensation threshold. In a system without pre-burden, the same hit remains inconsequential.

Category of silent pre-burden	Examples	Why it hits the FQ target cluster
Environmental / toxic load	Mercury from long-standing amalgam fillings, chronic heavy metal or mold exposure	Heavy metals block sulfhydryl enzymes and mitochondria → reduced reserve capacity exactly where FQ additionally attacks
Chronic severe psychological stress	Long-term cortisol overload	Cortisol depletes Mg^{2+} , weakens connective tissue and immune regulation (Treg) → pre-damage of the FQ-vulnerable axes
Genetically determined pre-burden	Metabolic syndrome, diabetes, Ehlers-Danlos syndrome (EDS), Gilbert's syndrome (UGT1A1 deficiency)	Diabetes / metabolic syndrome: mitochondrial pre-damage. EDS: already unstable collagen. Gilbert: reduced hepatic glucuronidation → slower FQ clearance
Age-related	Reduced telomere reserve, lower PGC-1 α capacity, sarcopenic tendency	The age-related already-shortened telomere / mito reserve (see telomere-p53-PGC-1 α axis) is pushed over the critical threshold by FQ

Clinical consequence: Hypothesis 5 shifts the diagnostic focus. It suggests that in severe chronic courses one should specifically search for **pre-existing, previously unrecognized burdens** (heavy metal status, genetic markers such as EDS/Gilbert, metabolic pre-damage) — not because FQ would be “innocent,” but because treating these co-factors may be a more realistic lever than trying to reverse the FQ damage itself. It also explains the INUSpheres observation and the detox approaches (see 3.4): those carrying a heavy-metal co-load might benefit from its reduction, whereas it would be ineffective in FQ damage without such a co-load.

Why symptoms persist for years remains a central open question [4]:

1. Prolonged oxidative stress destroys mtDNA; faulty cytochrome complexes create permanent electron leakage
 2. FQ protein and FQ cation complexes persist in cells for years - the ion trapping mechanism (see D2) explains the intracellular accumulation and the gradual “bit by bit” ion withdrawal over weeks to months after discontinuation
 3. Epigenetic changes in gene regulation become permanent after FQ exposure
- 4. Cumulative Damage from Repeated Exposure**— each FQ dose leaves a microscopic footprint; Repeated use escalates the risk exponentially

Cumulative damage: Why the second or third dose often triggers the crash The fact that not every first tablet leads to an immediate collapse, but repeated intake (months or years later) triggers the severe FQAD syndrome, is explained by four converging accumulation mechanisms:

Mechanism	What happens	Why the next dose is more dangerous
Cumulative Ion Trapping	FQ accumulation in tendons / cartilage is cumulative with repeated administration [13], [51]	Each new intake replenishes the cellular depots more deeply; the chronic Mg^{2+} withdrawal becomes more massive with each round
Irreversible mtDNA scars	Unlike nuclear DNA, mitochondrial DNA has hardly any repair mechanisms. Oxidative damage (Fenton reaction) from the first dose often remains permanent [3], [4]	During re-exposure, the FQ encounters already damaged mitochondria without a compensation reserve → immediate crash into anaerobic emergency power
Used regeneration reserve	Intracellular Mg^{2+} and Zn^{2+} stores (especially in tendons / cartilage) take months to years to replenish	If re-exposure occurs before complete regeneration, the chelation effect continues unabated on enzymes and collagen

Mechanism	What happens	Why the next dose is more dangerous
Senescence accumulation	Each FQ administration drives further healthy cells into the irreversible “senescent state” (SASP secretion, production stop) via DNA breaks → p53 → senescence [34]	At some point the ratio tips over: too many SASP-secreting senescent cells, too few functional fibroblasts → chronic tissue breakdown without replenishment

The body does not forget an FQ dose. The first dose is often compensated for by the healthy cells working overtime. The repeated dose is the straw that breaks the camel’s back, already full and weakened.

The telomere-p53-PGC-1α axis: Why FQAD accelerates biological aging ([YELLOW] Plausible)

Evidence level: [YELLOW] The telomere-p53-PGC-1α axis is well-established in aging research (DePinho et al., Dana-Farber Cancer Institute). The application to FQAD is mechanistically compelling but not investigated in FQAD-specific longitudinal studies.

At the cellular level, FQAD acts as a massively accelerated aging process. The mechanism operates through an axis well-characterized in aging research — the coupling between the cell’s “biological clock” (telomeres) and its “power plant” (mitochondria):

Step	Mechanism	Consequence in FQAD
1. Telomere shortening	FQ-induced oxidative stress (Fenton reaction, see 1.7) directly damages telomeres — telomeres are particularly vulnerable to oxidative damage (guanine-rich sequences)	Telomeres prematurely reach their critical length
2. p53 activation	Critically short telomeres are interpreted as massive DNA damage → guardian protein p53 is activated (protection against malignant transformation)	Cell ceases division → senescence
3. PGC-1α suppression	Activated p53 directly binds to the promoters of PGC-1α and PGC-1β and shuts them down	Mitochondrial biogenesis stops — the cell can no longer build new power plants
4. Downward spiral	Fewer mitochondria → less ATP → more electron leakage → more ROS → further telomere damage	Self-reinforcing vicious cycle: aging accelerates aging

The triple PGC-1α blockade in FQAD: Ciprofloxacin blocks PGC-1α via three converging pathways simultaneously — a constellation unprecedented in aging research:

- 1. **AMPK suppression** (Zeng et al. 2024 [78]) → the build command is directly suppressed
- 2. **NAD⁺ depletion** through B3 mimicry [77] → SIRT1 cannot activate PGC-1α
- 3. **Telomere-p53 axis** → p53 shuts down PGC-1α promoters

Consequence: Simple telomere extension (just upregulating telomerase) falls short in FQAD. A cell with long telomeres but blocked mitochondria can theoretically live forever but has no energy to function. Regeneration requires simultaneous mitochondrial stabilization, reduction of the oxidative fire, and epigenetic reprogramming.

FQAD as chemically induced premature aging — three levels of life shortening ([GREEN]/[YELLOW])

Level	Mechanism	Data
1. Direct excess mortality	Acute, potentially lethal complications: aortic aneurysms, toxic skin reactions (TEN), multi-organ failure	WIdO: ~140 additional deaths / year (Germany) [52]. EMA: Ciprofloxacin 1,161, Levofloxacin 1,532 deaths (2000–2017) [54]

Level	Mechanism	Data
2. Chemically induced premature aging	mtDNA damage → senescence → SASP secretion → enzymatic tissue decay without regeneration = pathobiochemical definition of aging. Telomere-p53-PGC-1α axis cements the aging process	Senescence accumulation, collagen switch (Type I→III), chronic inflammation
3. Indirect life shortening through chronic decline	Catabolic sustained state: muscle protein burning → sarcopenia, premature osteoarthritis, prolonged bedrest → drastically reduced physical resilience	Multisystemic weakness lowers overall resilience capacity

1.12 Why courses are so different - individual vulnerability

Lack of positions in science

A. Lack of disease model. FQAD is clinically recognized but not as a deciphered mechanism. Probably an umbrella of endophenotypes: tendon-dominant, neuro-dominant, dysautonomous, nephrologic, mitochondrial-metabolic, GI/MCAS-dominant, ME/CFS-like [1], [6].

B. Missing biomarkers. No test today says who is actively disease-driven or would respond to which therapy [6], [32].

C. Missing intervention studies. For elementary questions: Which stress helps? Which diet? Which supportive pharmacology?

D. Missing regeneration endpoints. The bridge between biomarker and clinical benefit is not built.

E. Missing failure data. Publication bias — e.g. B. the failure of EPI-743 in mitochondrial diseases is rarely discussed [33].

F. Lack of explanation for course variability. Why some sufferers recover and others become chronic is perhaps the most pressing unanswered question.

Why courses are so different - known and suspected factors

The fact that two people receive the same dose of the same fluoroquinolone and experience completely different outcomes is medically frustrating, but not illogical. There is no gold standard biomarker that predicts who will become chronic [6]. The basic biochemical fact: The drug chemically does the same thing in all people — it blocks topoisomerase, binds to AIFM1/IDH2 and performs chelation. The difference is not due to a different pill, but rather due to individual armor and repair capacity.

Seven Levels of Individual Vulnerability — Affected vs. Non-Affected:

level	For those not affected	For those affected	Testability
1. AIFM1 / IDH2 fit	FQ slips into the binding pocket, binding is loose → enzyme throws off the molecule, respiratory chain quickly normalizes [2]	Genetic variants (SNPs) could create a perfect 3D fit → FQ hooks, blocks ETC extremely long → massive electron jam → ROS explosion	[RED] Research: Targeted NGS + DiffDock Simulation (Level 4-5)
2. Cellular Fire Brigade (SOD2 / GST)	SOD2 and GST enzymes work at full capacity → intercept superoxide and hydroxyl radicals in milliseconds → DNA remains protected [34]	SOD2 polymorphism (e.g. Ala16Val) and / or GSTM1 / GSTT1 null genotype → Fire department can't keep up → Radicals react with free iron (Fenton) → OH• shrapnel destroys mtDNA and nuclear DNA	[YELLOW] Standard Pharmacogenetics Panel

level	For those not affected	For those affected	Testability
3. Magnesium reserve + chelation tolerance	Intracellular Mg^{2+} stores full to the brim → FQ binds some Mg^{2+} , but enough remains for DNA repair, TIMPs and NMDA blockade [4], [51]	Pre-existing intracellular Mg^{2+} deficiency (due to PPI, stress, age, cortisone) → FQ removes the last residue → TIMPs fail (MMP escalation), NMDA receptors overheat (neurotoxicity), collagen loses stability	[GREEN] Mg whole blood analysis; Medical history (PPI, cortisone)
4. Liver clearance / ion trapping	CYP1A2 “Fast Metabolizer” → liver breaks down FQ into harmless metabolites in hours → low tissue accumulation [4], [48]	CYP1A2 “Slow Metabolizer” (often coffee sensitivity) → intact FQ circulates much longer, penetrates deeper into tendons / nerves / CNS, is trapped in acidic lysosomes (ion trapping) → toxic tissue dose potentiates over weeks	[YELLOW] CYP1A2 genotyping; clinical note: coffee sensitivity
5. Renal function (especially levofloxacin)	Normal eGFR + good hydration → FQ is eliminated rapidly renally → short tissue exposure [9], [13]	Even slightly impaired eGFR, dehydration or age → Levofloxacin (almost 100% renal) accumulates massively → prolonged exposure + deeper ion trapping. With low muscle mass: creatinine underestimates the limitation → no dose adjustment → toxic overload (see D3, cystatin C)	[GREEN] eGFR (cystatin C for underweight); Hydration status
6. Pharmacological preload	No pre-medication that depletes Mg^{2+} , damages mitochondria or breaks down GABA receptors → System meets FQ	at full capacity Years of taking PPIs (Mg^{2+} depletion + dysbiosis), statins (Q10 blockade + myotoxicity), bisphosphonates (enthesopathy), NSAIDs (GABA escalation), benzodiazepines (GABA receptor degradation) → System is biochemically weakened before the FQ impact occurs (see Phase A, Pharmacological Pioneers)	[GREEN] Medication history (mandatory before any FQ prescription)
7. Simultaneous corticosteroid use	No cortisone → Collagen formation and immune regulation intact → Tendons can compensate for the chelation stress	Corticosteroids weaken collagen formation independently of FQ (inhibition of procollagen synthesis, reduction of tenocyte proliferation). In combination with the FQ-induced MMP attack, the risk of tendon rupture increases massively. Officially recognized high-risk factor [9], [10], [13]. Additionally: Cortisone depletes Mg^{2+}/Ca^{2+} and suppresses the immune system (Treg function). [GREEN] Medical history; officially recognized - absolute contraindication for simultaneous administration of FQ	

In those who are not affected, there is a short-term cellular stress that the system repairs effortlessly. In those affected, the drug encounters a combination of slower clearance, precisely fitting enzyme pockets, overwhelmed repair mechanisms and depleted mineral stores. The initial spark triggers a cumulative domino effect (DNA breaks → senescence → Treg collapse → MCAS).

The following table summarizes the other factors:

factor	Level of evidence	Mechanism
Dominant endophenotype	Plausible	Not everyone has the same damage type. Tendinous, neuropathic, dysautonomic and nephrological damage have very different chances of regeneration.
Known risk factors	Well supported for Tendon	Older age, renal insufficiency, concurrent corticosteroids, organ transplantation, diabetes increase the risk of tendinous and neuropathic damage [9], [10], [13].
Kidney Function / Clearance	Well supported	Same tablet dose $\neq$ same internal exposure. Differences in eGFR, hydration and concomitant medication massively alter tissue accumulation. Dosage is clinically based on creatinine clearance.
Dose to Weight Ratio / Body Fat	Plausible, not proven as main factor	Body composition influences volume of distribution; Low muscle mass can “whitewash” creatinine and overestimate kidney function. Important Clarification: Lysosomotropic FQ storage involves intracellular acidic compartments (lysosomes), not fat depots. More body fat does not automatically give a protective effect.
Functional vs. structural	Clinically plausible	Predominantly functional dysregulation (metabolic stress, temporary autonomic dysfunction) probably has better chances of regeneration than structural damage (SFN, tendon rupture, podocytic kidney damage).
Secondary Chronification Drivers	Clinically known	After the initial biological damage, sleep loss, deconditioning, PEM / crash patterns, deficiency states, protective posture and fear of stress intensify the progression - not because the illness is “psychological”, but because multiple amplifiers accumulate.
Regeneration reserve	Clinically plausible	Initial fitness, sleep quality, metabolic health, previous illnesses, cortisone exposure and type of tissue affected determine whether recovery or chronification occurs.

Important: If an acquaintance got better with high-dose vitamins, that does not prove causality. The course can be due to milder damage, other organs affected or greater regeneration reserve - not to a specific supplement. There is no reliable evidence of causal FQAD healing for high-dose vitamins.

The cleanest medical formulation: “Some affected people recover because their damage is less, more reversible or limited to better-regenerating systems; others become chronic due to less favorable risk factors, greater organ involvement or secondary chronification mechanisms. This is multifactorial and cannot be explained by a single trick.”

Research priorities

High priority: 1. Validate FQAD biomarkers (mtDNA copy number, PBMC respirometry, lactate, acylcarnitines, cell-free mtDNA) [6], [32] 2. Define subtypes and record them in an international FQAD registry (e.g. via EU rare disease networks) 3. First controlled pilot studies with existing mito-active ingredients in defined FQAD subgroups [7], [24] 4. Set up an N-of-1 study infrastructure: standardized baseline, defined endpoints, termination criteria, washout phases (see Section XV) 5. Develop AIFM1/IDH2-based functional diagnostics: ex vivo challenge protocol (Seahorse + ciprofloxacin exposure of isolated PBMCs), IDH2 metabolite profiling, circulating AIFM1 as apoptosis marker — building on [2]

Secondary: 4. Document spontaneous courses in cohort studies with a standardized phenotyping set 5. Stress control studies: which form helps, which harms, PEM risk [23] 6. Systematically collect microbiome/mast cell/autoimmune parameters in FQAD [18]–[20] 7. Define clinically meaningful endpoints, including patient-reported outcomes (SF-36) 8. Pharmacogenetic risk screening: evaluate polymorphisms in AIFM1, IDH2, CYP enzymes and NAT1/NAT2 as potential

susceptibility factors [2], [48] 9. Annual update, create patient version with infographics

PART 2: DIAGNOSIS — How do you make the damage visible?

From symptom description to objective evidence: step-by-step diagnostics and research vision

Phase B — functional diagnostics and organ phenotyping

aspect	Description
Target	Objectively clarify dominant damage axes
Clinical Status	Diagnostics available, no FQAD-specific protocol
Major Uncertainty	No accepted FQAD biomarker [6]
Endpoint	Organ profile with quantifiable initial values
Risk	Cave: CPET at PEM (see below)

FQAD should remain a diagnosis of exclusion. The search for competing explanations is a medical duty.

Level 1 - Basic diagnostics (standard laboratory, available everywhere) Blood count, CRP/BSG, B12/folate, HbA1c, TSH/ft4, ferritin/transferrin saturation, CK, creatinine/eGFR (if underweight/low muscle mass: cystatin C instead of creatinine for muscle mass-independent eGFR determination), liver values, urine status with albumin/creatinine ratio, vitamin D, magnesium, zinc, selenium, ECG. Orthostasis test for dysautonomic symptoms. ##### Level 2 — Advanced diagnostics (phenotype-guided, partly IGeL/special centers)

Only if clinically indicated by the leading phenotype:

-Neurological: Nerve conduction velocity, EMG, QST, skin punch biopsy (small fiber neuropathy) **-Nephrological:** SPEP/UPEP for proteinuria, nephrological co-care for eGFR decline **-Endocrine:** Cortisol daily profile (saliva), DHEA-S, testosterone/estradiol — only if there is clinical suspicion of HPA dysfunction or treatment-refractory fatigue [17]. IGF-1 as a marker for anabolic reserve in mainly catabolic states **-Autoimmune:** ANA, ENA, possibly GPCR autoantibodies (anti-β-adrenergic, anti-muscarinic receptors) in severe dysautonomia or therapy-refractory SFN [15], [16] **-Dysautonomous:** Tilt table, sweat test, HRV analysis, catecholamines (adrenaline, norepinephrine) in 24-hour urine or plasma levels in the case of chronic sympathetic excess [35] **-GI/Immunological:** Calprotectin, Zonulin in the case of barrier disorders; Tryptase only if there is clinical suspicion of mast cell activation [18]-[20]. For Treg mast cell collapse: flow cytometry (FACS) with CD4+/CD25+/FoxP3+ (Tregs) + IL-10 ([YELLOW] special diagnostics immunology) **-Musculoskeletal:** High-resolution ultrasound with elastography **-Sleep diagnostics**for leading fatigue

Advanced emergency power diagnostics(for severe mitochondrial dysfunction in particular -phenotype-guided, not as a screening):

Emergency power system	Markers	What he shows	Protocol note
Metabolic (glycolysis)	Lactate-Pyruvate Ratio	Backlog of blocked mitochondria; Gold standard for mitochondrial energy shift [4], [35]	See protocol below
Metabolic	M2-PK (Tumor M2 Pyruvate Kinase)	Upregulation in the event of a complete glycolysis switch (Warburg effect) [35]	Research-related

Emergency power system	Markers	What he shows	Protocol note
Autonomous	Catecholamines (24-hour urine)	Chronic increased norepinephrine during long-term sympathetic stimulation [35]	Standard protocol
Catabolic	Ammonia (NH ₃)	Toxic exhaust from muscle protein combustion; explains Brain Fog [35]	Immediately on ice
Catabolic	Amino acid profile (especially BCAA)	Decrease in valine / leucine / isoleucine during severe catabolic stress [35]	Sober
Catabolic	IGF-1	Anabolic reserve; decreased in chronic catabolism [35]	Default
Oxidative stress	GSH / GSSG ratio	reduced / oxidized glutathione ratio; shows excessive demands on the cellular antioxidant system [4], [34]	Special acceptance
DNA damage	8-OHdG (urine)	Markers of oxidative DNA destruction, particularly mtDNA [34]	24-hour urine collection
Protein damage	Nitrotyrosine	Occurs when peroxynitrite damages proteins; shows structural tissue damage [34]	Research-related

Note: The majority of these markers are located at the border between specialized clinical diagnostics and research. Lactate-pyruvate ratio, ammonia and catecholamines are available in specialized laboratories; GSH/GSSG, 8-OHdG, nitrotyrosine and M2-PK are more research-related.

Lactate-pyruvate protocol (critical for validity):

An isolated lactate value at rest is often inconspicuous in mitochondrial dysfunction because the liver eliminates lactate quickly enough at rest via the Cori cycle [35]. Therefore:

1.Uncongested blood collection— no tourniquet or minimal congestion (tourniquet creates local ischemia → lactate artifact) **2.Immediately on ice**— Centrifuge tube or separate plasma within 15 minutes (lactate is extremely volatile; room temperature will distort the value) **3.Measure lactate AND pyruvate**- only the quotient (L/P ratio) shows the backlog in front of the mitochondria; isolated lactate often obscures the defect **4.Ideal case: two-point measurement**- once at rest, once after a defined submaximal load (e.g. 3 minutes of climbing stairs, only if freedom from PEM is documented). The stress measurement shows the collapse of hepatic lactate clearance, which is not visible in the resting value

Note: Many general practitioners' practices and standard laboratories do not adhere to this protocol (lack of immediate cooling, lack of pyruvate determination). Results from non-protocol-compliant testing are of no diagnostic value.

In terms of statutory health insurance: cortisol daily profile, zonulin, skin punch biopsy, GPCR autoantibodies, M2-PK, GSH/GSSG, 8-OHdG, nitrotyrosine and comparable parameters are often not reimbursed or only reimbursed to a limited extent in everyday statutory health insurance (GKV, Germany/Austria). Patients should be transparently informed about possible out-of-pocket costs.

Urine diagnostics: What standard urine tests show and hide Urine offers three diagnostic levels:

Basic coverage (Level 1): Urine status with albumin/creatinine ratio excludes primary kidney disease and significant proteinuria. Glucose and ketone tests rule out metabolic imbalance due to diabetes. Exclusion of active urinary tract infections (since FQAD symptoms often occur after infection treatment). **Special marker (level 2-3):** 8-OHdG in 24-hour urine collection shows oxidative mtDNA destruction [34]. Amino acid profile and nitrogen excretion can objectify catabolic tissue degradation. Methylhistamine in urine in particular due to mast cell activation.

Practical tip - Morning urine pH profile: A morning urine pH profile over 3-5 days (with test strips, self-measurement) can be additionally informative: If the pH remains firmly below 6.0 even after an alkaline diet, this indicates an overload of the body's own buffer systems. A healthy fluctuating pH (acid in the morning, alkaline after meals) speaks against severe systemic acid-base dysregulation. Evidence status: clinical-pragmatic, not FQAD-validated.

Diagnostic Chain of Evidence: Causality Approach There is no single test that “proves” with certainty that the fluoroquinolone alone is the cause. FQAD remains a diagnosis of exclusion [9], [10]. However, the following combination can objectify the typical damage fingerprint:

stage	test	Shows	Evidence class
1. Exposure	Anamnesis: FQ intake, temporal context	Plausible chain of causality	Clinical
2. Loss of function	Lactate-pyruvate quotient (stress protocol)	Shift to anaerobic emergency power	Special diagnostics
3. Structural damage	Skin punch biopsy (IENF density)	Small fiber neuropathy	Special Center
4. Molecular fingerprint	8-OHdG, possibly AIFM1 / IDH2 functional analysis [2]	Oxidative mtDNA damage, ETC blockade	Research-related
5. Exclusion	Standard Laboratory (Level 1)	No competing explanation	Routine

The combination of exposure + loss of function + structural evidence + exclusion constitutes the strongest clinically available causality approximation. The molecular fingerprint (AIFM1/IDH2) is currently only available in research contexts [2].

Level 3 — Research markers (only for studies/centers, not clinical routine) 31P magnetic resonance spectroscopy (ATP in muscle), PBMC respirometry, circulating cell-free mtDNA, acylcarnitine profiles. These markers are important to the research agenda but are not currently part of clinical care.

Calculation of remaining mitochondrial capacity from PBMC laboratory values ([YELLOW] special diagnostics)

Level of evidence:[YELLOW] The individual parameters are established in laboratory chemistry; merging them into a capacity score is a quantitative working hypothesis, not a validated diagnostic index. The “mitochondria baseline profile” (PBMC, available through specialized laboratories such as MMD Magdeburg) provides six parameters that can be put into a mathematical relationship in FQAD to estimate the remaining cellular energy.**Cave:** Standard laboratory findings often interpret these values for healthy populations — in the case of toxic FQ damage, identical numbers can have a completely different meaning.

Parameters	What he shows	FQAD Interpretation
mtDNA:ntDNA ratio (Reference: ~400 ± 230)	Mitochondrial “factory size” — number of mitochondria per cell	High value (e.g. >630) = compensatory mitochondrial proliferation. The body has accumulated masses of power plants to compensate for the failure of any one — a sign of chronic dysfunction, not health
Mitochondrial ATP (pmol / 10 ⁵ PBMC)	Actual energy production at rest	Must be interpreted in relation to the mtDNA:ntDNA ratio. “Normal” ATP with double the number of mitochondria = every single power plant runs at half power
PGC-1α expression (rel. to GAPDH)	Master switch for mitochondrial regeneration (biogenesis)	Hardly detectable (e.g. 0.00031) = lack of new formation. Defective mitochondria accumulate in the cell and are not replaced by fresh ones - exactly the PGC-1α biogenesis trap of the cascade model (1.9, phase 3)

Parameters	What he shows	FQAD Interpretation
Nrf-2 expression (rel. to GAPDH)	Master switch of antioxidant defense	Slightly increased = oxidative stress is present, but the immune system is overwhelmed. Low / normal with high mitochondrial number = mitochondria hardly run (emergency power), therefore little electron leakage - not a sign of health, but of a blocked respiratory chain
Rhodanase expression (rel. to GAPDH)	Formation of iron-sulfur clusters for ETC complexes I, II, III	Very low (e.g. 0.0072) = almost no Fe-S cluster formation. These clusters are the essential “power cables” of the respiratory chain. Without it, aerobic energy production (36 ATP / glucose) cannot physically take place → cell is forced into anaerobic emergency power when stressed (2 ATP / glucose)
mt4977 deletion mutant	Most common “aging deletion” of mtDNA	Undetectable = no gross structural mtDNA destruction. Caveat: More subtle damage (point mutations, oxidative base modifications such as 8-oxo-dG) is not recorded

Three-step capacity calculation (hypothetical, illustrative): *Step 1 — Efficiency Quotient (ATP per Mitochondrion):* If mtDNA:ntDNA is ~210% of the reference value (twice the factory size) but ATP is considered “low” relative to this → the individual mitochondrion produces **<50% of normal capacity** at rest. The body compensates through mass, not efficiency.

Step 2 — Load Capacity (Rhodanase Bottleneck): Almost no Fe-S cluster formation → Respiratory chain physically blocked during exercise → Cell has to switch to glycolysis (2 instead of 36 ATP) → **Energy yield under exercise: ~5-6% of maximum aerobic capacity.** This explains why minimal exertion (sitting up in bed, digesting food) immediately leads to PEM crashes.

Step 3 — Regenerative Ability (PGC-1 α Block): Hardly detectable PGC-1 α = no new formation of mitochondria → the system cannot repair itself → chronic accumulation of dysfunctional power plants. This is molecular evidence for the “biogenesis trap” of the cascade model.

Clinical consequence: The standard laboratory finding (“low ATP production indicates high efficiency”) is a misinterpretation in the case of toxic FQ damage. The combination of increased mitochondrial mass + low ATP/mitochondrion + missing rhodanase + blocked PGC-1 α shows a system that survives on low power in the resting state, but immediately enters energetic collapse under any stress. This constellation of findings should be presented to the treating physician as objective, laboratory evidence of the mitochondrial basis of exercise intolerance.

Stage 4 — Personalized bioinformatic risk analysis ([RED] Research vision)

Note: The following concepts are relevant only to research groups, not to clinical decisions or personal risk assessment. They cannot diagnose FQAD or guide treatment.

Evidence status: Purely experimental. No validated clinical use. The following concepts are intended for research structuring and future pharmacogenetic risk assessment, not for diagnostics.

Pharmacogenetic screening before FQ prescription (future concept): The Reinhardt study [2] identifies AIFM1 and IDH2 as molecular off-targets of fluoroquinolones. Genetic polymorphisms in these proteins may explain why some patients are particularly susceptible to FQ toxicity. Future pharmacogenetic screening before FQ prescription could theoretically identify high-risk patients — analogous to HLA-B*5701 testing before abacavir or CYP2D6 testing before codeine.

Additionally relevant: polymorphisms in clearance enzymes (CYP family, NAT1/NAT2 acetylator status), which could slow hepatic FQ elimination and prolong tissue accumulation [4], [48].

Antioxidant defense genes — the cellular “fire department”: In addition to the direct FQ

binding partners (AIFM1, IDH2) and the clearance enzymes (CYP), polymorphisms in the antioxidant defense deserve special attention:

Gene	Function	Relevance to FQAD vulnerability
SOD2 (superoxide dismutase 2, mitochondrial)	Converts superoxide ($O_2^{\bullet-}$) to H_2O_2 — the first and most critical step in mitochondrial ROS detoxification	In SOD2 polymorphisms (e.g. Ala16Val, rs4880), mitochondrial superoxide detoxification is slowed down. When FQ trigger the “oxidative fire” (Front 1, D5), SOD2 genetics decide whether the cell can extinguish the fire or slip into the FQAD cascade [34]
GST genes (Glutathione S-transferase: GSTM1, GSTT1, GSTP1)	Phase II detoxification: conjugate oxidative damage (lipid peroxides, protein carbonyls) to glutathione for excretion	GSTM1 / GSTT1 null genotypes (homozygous deletions, prevalence 20–50% depending on ethnicity) represent complete loss of function. In FQ-induced oxidative stress, these patients lack the second line of defense after SOD2

These genes do not determine FQ exposure or clearance, but rather the cell’s ability to cope with the resulting oxidative damage. A patient with SOD2-Ala16Val + GSTM1 null genotype has a biochemically doubly weakened “fire department” - the oxidative fire (front 1) escalates faster and more severely with the same FQ dose than in a patient with intact defense genes. SOD2 and GST polymorphisms are testable via standard pharmacogenetics panels (no whole-exome sequencing required).

Personalized binding simulation (in silico pipeline): For patients with available genomic data (e.g. from whole-exome sequencing or consumer genotyping), the following bioinformatics pipeline is theoretically feasible:

step	Method	Input → Output	hurdle
1. Variant identification	Search genomic data for SNPs in AIFM1, IDH2, CYP enzymes, NAT1 / NAT2	Raw data → List of relevant variants	Consumer arrays (23andMe and similar) do not read the entire gene; only preselected SNPs. Rare variants are often not recorded. Complete coverage only by whole-exome sequencing [48]
2. DNA→Protein Translation	Determine the amino acid sequence of each variant via UniProt / dbSNP	DNA letter → amino acid exchange in protein	Not every DNA variant changes the protein (silent mutations)
3. Protein folding	ColabFold (AlphaFold2-based): calculate mutated AIFM1 protein as a 3D structure [49]	FASTA sequence → .pdb file (3D model)	~15–30 min computing time; free in Google Colab
4. Molecular Docking	DiffDock (MIT): Dock ciprofloxacin / levofloxacin into the personalized binding pocket [50]	.pdb + SMILES code → binding affinity (confidence score)	Multiple runs necessary for reproducibility
5. Comparison with wild type	Perform identical pipeline with standard AIFM1 protein (reference sequence)	Control score vs. patient score	Only the difference is meaningful; an isolated score cannot be interpreted

Result Interpretation: If DiffDock shows that ciprofloxacin binds significantly more strongly in the individual AIFM1 variant than in the wild type, this would be a bioinformatic indication of increased personal susceptibility to FQ-induced ETC blockade. This is not evidence of causality, but rather a hypothesis generator for further research.

Critical limitations: - Consumer genotyping (23andMe, MyHeritage) only captures pre-selected SNPs. A negative result (no error found) does not rule out a variant — the chip may

never have read the location [48]. - AlphaFold models are predictions, not experimentally solved structures. Binding simulations are approximations, not exact measurements [49], [50]. - The entire pipeline is a research tool, not a diagnostic test. It cannot diagnose FQAD or justify a personal treatment decision. ##### Level 5 — AIFM1/IDH2-based functional diagnostics ([RED] research vision, building on [2])

Note: None of these tests are currently clinically validated or available. The approaches serve to structure research.

approach	Method	What he shows	Current status
1. IDH2 metabolite profiling	Mass spectrometry in blood: measuring the isocitrate / α -ketoglutarate ratio	Specific metabolic pattern proving that IDH2 does not work efficiently — the Reinhardt study [2] shows that FQ blockade of IDH2 forces reverse carboxylation	Research; Mass spectrometric metabolomics available in specialized laboratories but not standardized for FQAD
2. AIFM1-induced mitochondrial stress test (ex-vivo)	Isolate PBMCs from patient blood → put them under maximum stress in the laboratory with a Seahorse analyzer → measure “spare respiratory capacity”	If AIFM1 no longer stabilizes complexes I / IV, the cells collapse under stress much faster than in healthy people [2]	Research; Seahorse technology available in university mitochondrial centers
3. Ex vivo challenge analysis (ciprofloxacin exposure of isolated cells)	Collect PBMCs or lymphocytes → Sample A (untreated) vs. Sample B (with ciprofloxacin at therapeutic concentration) → Comparison of: mitochondrial membrane potential, oxygen consumption (Seahorse), IDH2 metabolomics	Objective, functional evidence of individual sensitivity: If the patient’s cells collapse at a dose that other cells tolerate, the pathological intolerance is laboratory-proven - causal, not just genetic Research; no patient risk (damage only in Petri dish). Potential future diagnostic gold standard	
4. Detection of circulating AIFM1 in serum	Development of a highly sensitive ELISA test for free AIFM1 or specific degradation products in the blood	AIFM1 is released from mitochondria into the cytoplasm during mitochondrial stress and initiates apoptosis [2]. Circulating AIFM1 would be a direct marker for FQ-induced cell death	Conceptual; ELISA development required. Specificity for FQ damage vs. other mitochondrial stressors would need to be validated
5. Targeted NGS of binding pocket exons	Next-generation sequencing exclusively of the exons encoding the binding pockets of AIFM1 and IDH2	Identification of SNPs that change the pocket in such a way that FQ docks there particularly easily or stays there for a particularly long time - explains individual susceptibility [2]	Research; technically feasible immediately, but interpretation requires functional validation (see point 3)
6. FLIM — Fluorescence Lifetime Microscopy	Examination of skin biopsies or blood cells using FLIM. IDH2 directly influences the NADH / NAD ⁺ ratio; NADH has natural fluorescence properties	Make the redox status of the mitochondria directly visible without cell destruction. Shift in NADH fluorescence lifetime would be visual evidence of ETC disrupted by AIFM1 / IDH2 blockade [2]	Research; FLIM devices available in specialized centers. FQAD-specific standard values would first have to be established

Ex vivo challenge (approach 3) as a potential key innovation — detailed protocol:

Setup: Blood is taken from the affected person; Live PBMCs (lymphocytes, monocytes — high

mitochondrial density) are isolated. **-Sample A (Control):** Cells untreated **-Sample B (provocation):** Cells were treated with ciprofloxacin in a therapeutic concentration ($\sim 7.5 \mu\text{mol/L}$, corresponding to plasma C_{max}).

What is measured (the “AIFM1/IDH2 fingerprint”): **-Mitochondrial membrane potential:** Fluorescent dyes (e.g. JC-1, TMRM) show whether the “battery” of the mitochondria breaks down under ciprofloxacin. In those affected with a vulnerable AIFM1 binding pocket: an above-average drop in voltage **-Real-time respirometry (Seahorse):** Live measurement of oxygen consumption. In genetically vulnerable patients: Oxygen consumption drops immediately under ciprofloxacin, while healthy cells compensate for the stress **-IDH2 stress metabolomics:** Testing whether patient cells on ciprofloxacin immediately divert citrate via the rescue pathway (reverse carboxylation) — direct evidence that IDH2 is blocked in the cells of this specific individual [2]

Why more meaningful than conventional tests: **- Individual sensitivity:** A genetic test (approach 5) only says that a variant exists. This functional test shows what the variant does — supporting the hypothesis of a causal connection **- No patient risk:** The damage only occurs in the Petri dish **- Objective evidence:** If a patient’s cells collapse at a dose that other cells can easily tolerate, this is objective, laboratory evidence of pathological intolerance

Legal and diagnostic relevance: A standardized ex vivo challenge protocol could serve as a long-term functional confirmatory FQAD test — objective, reproducible, without risk to the patient. This would be the operationalization of the missing “diagnostic gold standard” that the model declares as non-existent in Section II. In today’s statutory health insurance standard, FQAD is purely a diagnosis of exclusion. The aim of the translational research (AIFM1/IDH2 fingerprint, lactate-pyruvate ratio) is to make FQAD an objectively positive diagnosis in the medium term.

Critical limitation - time window of ex-vivo challenge: If the individual vulnerability is purely genetic (shape of the AIFM1/IDH2 binding pocket), the test will be effective even years after the damage. However, if the vulnerability at the time of exposure was epigenetic or driven by a temporary cellular deficiency state (e.g. magnesium low during ingestion), the reproduced PBMCs taken today could survive the test in the test tube unscathed - a false-negative finding. This distinction (genetic-permanent vs. situational-temporary) is essential for future research interpretation.

CPET warning Spiroergometry (CPET) is potentially contraindicated in suspected post-exertional malaise (PEM), as maximal exercise can lead to severe worsening of symptoms that lasts for months [23]. Instead, submaximal testing (6-minute walk test, grip strength) or a 2-day CPET exclusively in highly specialized ME/CFS centers.

2.2 Recovery framework: Measurable endpoints per organ axis

Recovery framework: Measurable regeneration endpoints per organ axis

Without measurable endpoints, “regeneration” remains too abstract. For each organ axis, target parameters are defined here, based on which partial regeneration could be objectively recognized. These endpoints define clinical partial and complete remission; Whether one calls a stable complete remission a “cure” is a semantic question and should be operationalized in study protocols. | Organ axis | Measurable endpoints | Partial regeneration can be recognized by | |—|—|—| | Tendon / connective tissue / joint | Ultrasound elastography (tendon), MRI / X-ray (joint), grip strength, pain scores (VAS), joint mobility (ROM) | Decrease in tendon inhomogeneity, increase in strength, stable or improved joint space, reduction in pain | | Neuropathy / Autonomic | Skin punch biopsy (IENF density), QST, NLG, tilt table, HRV, nocturnal pulse oximetry, if necessary polysomnography | Increase in IENF density, improved autonomic reflexes, stable nocturnal SpO_2 | | Nephrological | eGFR, albumin / creatinine ratio, serum albumin | Stable or declining proteinuria, GFR stabilization | | CNS / sleep / stress | CPET (PEM-free only), 6-minute walk test, SF-36, crash frequency | Increase in exercise tolerance, reduction in crash frequency, sleep quality | | Neuropsychiatric | Cognitive Tests (MoCA), Depression / Anxiety Scores (PHQ-9, GAD-7) | Improving cognitive scores, reducing symptom burden | | Endocrine | Cortisol daily profile, DHEA-S, IGF-1 | Normalization of the HPA axis, anabolic reserve |

2.3 Subgroup logic: endophenotypes

Subgroup logic: endophenotypes as the basis of every intervention hypothesis

Any hypothesis regarding mitophagy, senescence, re-conditioning, or curative approaches must be considered subgroup-specific, not for “FQAD as a whole.” Suggested endophenotypes:

Endophenotype	Leading clinic	Specific research questions
Tendon / joint dominant	Tendinopathy, risk of rupture, MSK pain, accelerated osteoarthritis, joint instability	Collagen / matrix regeneration, eccentric loading, cartilage preservation, passive mobilization
Neuropathic-dysautonomous	SFN, POTS, hypersensitivity	GPCR autoantibodies? Immunoadsorption? Nerve fiber regeneration?
PEM / ME / CFS-like	Stress intolerance, crashes, anaerobic emergency power	Metabolic bypass? Ketosis? Pacing algorithms?
Nephrological	Proteinuria, GFR decline	Podocytic stabilization, nephroprotective strategies
Neuropsychiatric-sensitive	Brain fog, anxiety, cognitive dysfunction	GABA / NMDA rebalancing, CNS-directed neuroprotection
Multisystemic heavy	Several axes affected simultaneously	Systemic network failure? Prioritization of the master axis

PART 3: HEALING & REGENERATION — What can you do?

From symptom management to cellular repair: stabilization, reconstruction, curative horizon

3.1 Damage Stabilization ([GREEN] Routine / [YELLOW] Individual)

Phase A — Damage Stabilization ([GREEN] Routine / [YELLOW] Individual)

Level of evidence:[GREEN] Close to guidelines (avoiding further FQ, correcting defects). [YELLOW] Individualized (dysautonomia management, crash protocol — transfer from ME/CFS).

aspect	Description
Target	Prevent further decompensation
Clinical Status	Close to guidelines; Transfer from mitochondrial medicine [21]
Major Uncertainty	No validated data on which measures causally change the course of FQAD
Endpoint	Symptom stability, avoidance of acute deterioration
Risk	Progressive decompensation → advanced diagnostics

Specific measures:

- Strict avoidance of other fluoroquinolones and potentially mitotoxic substances [9], [10]
- Systematic search for aggravating factors: deficiencies, sleep disorder, dysautonomia, malnutrition, endocrine problems, additional neuropathy, renal insufficiency
- Avoidance of catabolic stress: long fasting, dehydration, infection decompensation [21]

Prioritized Risk Factor Ranking: What Makes the Damage Worse A deficiency does not cause the tendon to rupture, but it does deprive the body of the only way to absorb or repair the chemical impact. The following ranking list sorts according to immediate danger:

Rank	Condition / defect	Category	Pathobiochemical escalation mechanism
1	Intracellular magnesium deficiency	Mineral (critical)	FQ actively rob Mg^{2+} (chelation, see D2). If there is a lack of free Mg^{2+} , ions are ripped directly from enzymes (topoisomerase) and collagen. Mg^{2+} is missing as an antagonist against toxic Ca^{2+} influx into cells [4], [51]
2	renal insufficiency + dehydration	Organ (critical)	Blocked waste disposal: Levofloxacin in particular (almost 100% renal) accumulates massively with weak eGFR → prolonged tissue exposure and ion trapping (see D3) [9], [13]
3	Zinc deficiency	Mineral (high)	Missing MMP brake: TIMPs absolutely require Zn^{2+} as a cofactor. FQ also chelate zinc → enzymatic tissue destruction continues unabated (see D2, TIMPs) [13], [34]
4	Simultaneous corticosteroid use	Medicated (high)	Double structural attack: Corticosteroids weaken collagen formation independently of FQ. In combination, the risk of tendon rupture increases. Officially recognized high-risk factor [9], [10], [13]
5	B12 and folic acid deficiency	Vitamin (high)	Defenseless nerve cell: B12 is essential for myelin sheath and cellular energy. If it is missing, if FQ disrupts the ion channels, the nerve cell dies instead of regenerating → SFN risk increases [5]
6	Low muscle mass / underweight	Metabolism (moderate-high)	Dose trap: Little muscle mass → little creatinine → falsely “perfect” eGFR → no dose reduction → toxic load for remaining tissue massively increased. Use cystatin C instead of creatinine (see D3)
7	Vitamin D deficiency	Vitamin (moderate)	Lack of immune / bone control: Vitamin D modulates inflammation and Ca^{2+} incorporation. Deficiency worsens the baseline for tissue repair and fuels SASP chronicity
8	Catabolic Stress / Long Fasting	Metabolism (moderate)	Self-consumption: Cells in emergency power (glycolysis) constantly need energy. Fasting forces muscle protein burning → ammonia → brain fog, sarcopenia, accelerated tissue breakdown [4], [35]
9	Continuous stress + social invalidation	Neurological (moderate)	Mast cell radical amplifier: Chronic sympathetic nerve excess (also through invalidation) fires up unstable mast cells and artificially keeps the inflammatory loop alive [19], [22]

The absolute worst defects are those that leave the FQ’s direct point of attack (Mg^{2+}) defenseless and rob the body of its chemical brakes (Zn^{2+}). The vitamin deficiencies (B12, D) are not the cause of the damage in the acute phase, but are the main reason why the body then slips into chronic disease because there is a lack of building materials for reconstruction.

Pharmacologic Enablers: Pre-medications that escalate FQAD risk Medications taken in the months or years before FQ administration can unnoticed rob the body of the armor it needs to absorb the toxic impact. They don’t cause FQAD directly, but they precondition the system in a toxic way.

Pre-medication	Toxic overlap with FQ mechanics	Clinical consequence
Proton pump inhibitors (PPIs: pantoprazole, omeprazole)	PPIs lower stomach acid → intestines have difficulty absorbing Mg^{2+} → officially recognized side effect: hypomagnesemia after long-term use. Additionally: Dysbiosis preparation (Leaky Gut, MCAS soil)	Patient goes into FQ therapy with chronically empty Mg^{2+} stores → chelation effect hits unprotected tissue. One of the most common and dangerous enablers.
Statins (Simvastatin, Atorvastatin)	Statins block the body's own Q10 production → mitochondria are already energetically damaged. Known myotoxicity (muscle pain to rhabdomyolysis)	FQ-ETC blockade hits muscle cells that are already running at their limit due to Q10 deficiency → immediate energy collapse, deep weakness, increased risk of rhabdomyolysis
Bisphosphonates (Alendronate / Fosamax)	Have a massive impact on bone metabolism and Ca^{2+} balance; are deposited in the bone for years; known long-term NW: chronic bone / joint / muscle pain	FQ-MMP attack hits a bone-tendon framework (enthesis) that has already been structurally altered and mechanically stiffened by bisphosphonates → potentiated therapy-resistant joint / tendon damage
NSAIDs (Ibuprofen, Diclofenac)	NSAIDs displace FQ from plasma proteins and potentiate FQ-induced GABA receptor blockade in the CNS. Explicit warning in the package insert.	Massively increased risk of neurotoxic attacks, seizures, panic attacks, permanent CNS damage if taken promptly
Benzodiazepines (diazepam, lorazepam) / centrally acting psychotropic drugs	Chronic benzo use breaks down GABA receptors (tolerance). FQ blocks exactly these receptors in an additional toxic way "Cold withdrawal at the cellular level": The nervous system loses all self-regulation ability → severe neuropsychiatric FQAD courses (extreme restlessness, insomnia, dysautonomia)	

Clinical consequence: The medication history (particularly PPI long-term therapy, statins, bisphosphonates, NSAIDs, benzodiazepines) is mandatory before every FQ prescription and for phenotyping if FQAD already exists. A patient who has taken pantoprazole + simvastatin for years enters FQ therapy with depleted magnesium and previously damaged mitochondria - the risk of a severe, chronic course is massively increased. **-Dysautonomia management:** In some of those affected, dysautonomous patterns and exercise intolerance may be clinically relevant. Non-pharmacological procedures such as pacing, HRV biofeedback or autonomously oriented rehabilitation are plausible but not validated support approaches for FQAD [22]. The evidence comes from small studies of dysautonomia, anxiety disorders and pain; FQAD-specific data is missing.

Acute relapse management (crash protocol) If a patient presents with severe acute stress decompensation (massive increase in pain, cognitive decline, weakness):

- Absolute protection from stimuli (sensory, physical, cognitive)
- No further diagnostic stress tests in this phase
- Short-term H1/H2 blockade in particular with mast cell involvement (cetirizine + famotidine), electrolyte balancing
- No graded exercise, no physiotherapy during acute decompensation
- Monitoring until restabilization, then reevaluation of the load limit

This crash protocol is clinically pragmatic but not FQAD-specific validated; it is based on ME/CFS acute management [23].

Stress management: Why "training through pain" is dangerous In the case of complaints associated with fluoroquinolone, forced exercise to the point of pain is more risky than sensible.

In the case of tendon pain, authorities expressly recommend that you protect the affected structure and avoid unnecessary strain because the risk of rupture can be increased [9], [10], [13].

Differentiated loading rules according to endophenotype:

axis	Load rule	Evidence
tendons / connective tissue	No forced strain, no “biting through” when in pain. Protection especially in the acute phase. Eccentric therapy only after ultrasound clearance (see organ axis 4)	Best documented; Product information and authorities explicitly warn [9], [13]
Neuropathy / Dysautonomia	Overload plausibly harmful; symptom-guided dosage	Plausible, but less hard evidence than with tendons
PEM / stress intolerance	Pacing is absolute priority; Graded exercise is contraindicated in PEM [23]	Transfer from ME / CFS literature
Neuropsychiatric	Mental stress can increase pain, fatigue and dysautonomia	Biopsychosocial pain model; no FQAD-specific evidence

Rest ≠ complete immobility. Total inactivity has its own disadvantages (deconditioning, circulatory breakdown, muscle breakdown, pain chronification). The golden mean: symptom-guided activity without provocation, without forced increase, without “it has to hurt”.

Warning: Risk of rhabdomyolysis when re-exercising after severe deconditioning In those affected with severe fluoroquinolone-associated muscle weakness, bed rest or severely limited exercise capacity, even slight re-exertion - short standing, sitting up or a few minutes of activity - can in rare cases lead to severe muscle damage (rhabdomyolysis). This is particularly true in cases of severe deconditioning, dehydration or pre-existing muscle vulnerability.

Warning signs (emergency - immediate medical evaluation on the same day):

- Rapidly increasing muscle pain over hours to days
- Unusually severe weakness or sudden loss of function -**Dark urine**(brown to cola colored — myoglobin in urine)
- Increased temperature/feeling feverish
- Swelling, tenderness or feeling of tension in the muscles
- Significant deterioration after relatively minor stress

Immediate diagnosis: CK (creatine kinase - massively increased in rhabdomyolysis), creatinine (kidney function - myoglobin can cause acute kidney failure), electrolytes (potassium - life-threatening hyperkalemia possible), urine status (myoglobinuria).

Disclaimer: This is not normal muscle soreness and not the typical FQAD crash/PEM (which results from excess lactate and mitochondrial emergency flow). Rhabdomyolysis is an acute muscle breakdown with potentially life-threatening complications (kidney failure, cardiac arrhythmia) and requires immediate inpatient treatment.

Practical Reloading Precautions:

- No forced increase in mobilization after prolonged bed rest
- Exercise only very carefully, symptom-guided and in minimal steps
- For previous episodes with dark urine or extreme aggravation of pain: special restraint
- Ensure adequate fluid intake (unless there are medical reasons against it)
- Arrange a low-threshold CK check if muscle damage is suspected This warning does not mean that every activity is dangerous. It points out that in severely affected patients, forced or too rapid re-exertion can in individual cases lead to serious, possibly permanently deteriorating muscle damage.

3.2 Organ axes — parallel treatment paths ([YELLOW] Individualized)

Organ axes — parallel treatment paths

A realistic regeneration plan is primarily tissue specific. The following axes require their own, parallel treatment paths.

1) Nephrological axis Proteinuria monitoring, eGFR progression. Kidney-adapted diet for protein loss. Nephrological co-care for creatinine increases [1].

2) Neurological-autonomic axis Small fiber neuropathy, peripheral neuropathy, dysautonomia. Symptom-oriented therapy: Gabapentinoids for neuropathic pain - taking into account the tolerability restrictions mentioned at the beginning (see Section I), starting with fractions of the standard dose [5]. Beta-blockers for POTS, fluid and salt management for orthostasis [22]. If GPCR autoantibodies are detected: if necessary, immunoadsorption in a university setting as an experimental option [15], [16].

Sleep-disordered breathing — a commonly overlooked problem:

Those affected report nightly drops in SpO₂ (measurable with a pulse oximeter). In organically healthy lungs, the FQAD model explains two mechanisms:

Diaphragm exhaustion (hypoventilation): The diaphragm requires large amounts of ATP to breathe for hours at night. When the respiratory chain is blocked and anaerobic emergency flow occurs, the respiratory muscle becomes fatigued - breathing becomes increasingly shallow during sleep (hypoventilation) and blood oxygen measurably drops [35].

Central sleep apnea (dysautonomia): The automatic breathing drive during deep sleep is controlled by the autonomic nervous system. In severe dysautonomia, the brain temporarily “forgets” to send the breathing impulse - breathing pauses (central apneas) occur in which the remaining oxygen is used up without replenishment [22].

Diagnostics: Pulse oximetry screening at home (over several nights); in the case of documented declines: polysomnography in the sleep laboratory to differentiate between obstructive, central and mixed sleep apnea as well as hypoventilation. *Therapy:* The solution is primarily mechanical, not chemical: CPAP or BiPAP breathing support relieves the exhausted diaphragm of work and prevents breathing pauses. The sole administration of high-dose oxygen via a nasal cannula without supporting the breathing mechanics does not solve the problem of diaphragm exhaustion and remains an incalculable risk in the case of blocked mitochondria due to the risk of oxidative stress (Fenton reaction) [34].

3) Gastrointestinal-immunological axis Fluoroquinolones are primarily broad-spectrum antibiotics - the resulting dysbiosis can act as an independent, chronically relevant axis of damage. This axis is hypothetically classified in FQAD (biologically plausible, FQAD-specific has hardly been investigated); the prevalence of clearly defined gut-driven courses in FQAD is not known [18]-[20].

Mechanism: Antibiotic-induced dysbiosis → loss of protective bacteria (bifidobacteria, lactobacilli) → barrier disruption (“leaky gut”) → bacterial translocation (toxins in the bloodstream) → systemic inflammation → mast cell activation. Fluoroquinolones can also activate mast cells directly via MRGPRX2 [19]. This could explain why intestinal problems in FQAD are often accompanied by allergy-like reactions, cognitive decline or joint pain.

Regulatory T cells (Tregs) as microbiome victims — the Treg mast cell collapse:

In healthy people, mast cells are kept in check by regulatory T cells (Tregs). Tregs act as an immunological “brake” by releasing anti-inflammatory messenger substances (especially IL-10) that signal to the mast cells: stay calm. Fluoroquinolones could attack this brake via two converging pathways:

1. Microbiome loss → Butyrate deficiency: The broad-spectrum antibiotic destroys commensal intestinal bacteria that produce short-chain fatty acids (especially butyrate). Butyrate is essential for the differentiation and survival of Tregs. Without butyrate, the Treg population could collapse [18], [46]. **2. Oxidative stress → direct T cell damage:** The FQ-induced mitochondrial energy deficiency and the mass ROS production (see D5, front 1) also directly damage

the sensitive T lymphocytes [4], [34]. **Consequence:** If the Treg brake breaks down, mast cells degranulate uncontrollably at the slightest stimulus. The nervous system is flooded with histamine and cytokines, the stimulus threshold drops - the development of MCAS and MCS is immunologically disinhibited.

Causal chain: FQ toxicity → butyrate deficiency + ROS → Treg collapse → mast cell disinhibition (MCAS) → MCS.

Diagnostics of Treg deficiency ([YELLOW] Special diagnostics, immunology):

Parameters	Method	What he shows	Interpretation
CD4⁺/CD25⁺/FoxP3⁺ (Tregs)	Flow cytometry (FACS) of lymphocyte subpopulations	Quantification of functional regulatory T cells. FoxP3 is the transcription factor that identifies true, suppressive Tregs	Significantly reduced fraction → immunological disinhibition proven in laboratory chemistry → supports Treg mast cell collapse as the cause of MCAS
Interleukin-10 (IL-10)	Serum / plasma	Primary anti-inflammatory messenger Tregs	With Treg deficiency often reduced → lack of immunological brake objectified
Tryptase / Methylhistamine	Serum (tryptase) / 24h urine (methylhistamine)	Proof that the mast cells actually degranulate in an uncontrolled manner	In combination with reduced Tregs: confirmation of the Treg-mast cell axis

The CD4⁺/CD25⁺/FoxP3⁺ determination is available from specialist immunological laboratories (not routine blood work). The combination of decreased Tregs + increased tryptase/methylhistamine + clinical MCAS symptoms results in a complete biological causal chain that goes beyond a pure description of symptoms.

Treg recovery — five biochemical levers ([YELLOW] individualized / [RED] research):

Level of evidence: The individual mechanisms are immunologically established; the specific use in FQAD is not validated. Each approach is to be introduced “low and slow” — an FQAD nervous system is sensitive to any sudden change. Basic principle: only one variable at a time, lowest dose, discontinuation if deterioration occurs.

#	Strategy	Mechanism	Implementation at FQAD	Status
1	Postbiotics (Butyrate)	Butyrate is a potent HDAC inhibitor → epigenetically unblocks the FoxP3 gene → naive T cells differentiate into functional Tregs. Additionally: Butyrate is the main energy supplier of the intestinal epithelial cells → stabilizes tight junctions → repairs leaky gut. Avoids the “histamine trap” of normal probiotics (no live bacteria, no histamine risk) [46]	Enteric-coated sodium butyrate or tributyrin (so that butyrate reaches the colon without damage). Start with microdose (e.g. half a capsule every other day), only increase if tolerated	[YELLOW] Biologically based; FQAD-specific not examined

#	Strategy	Mechanism	Implementation at FQAD	Status
2	Vitamin D3 + Retinoic Acid (Vitamin A)	T cells have vitamin D receptors; activated 1,25-(OH) ₂ -D3 accelerates Treg differentiation (CD25 ⁺ /FoxP3 ⁺). Retinoic acid (vitamin A) works synergistically and enhances Treg induction in the intestine. Cave in FQAD: Renal activation of 25-OH-D3 to 1,25-(OH) ₂ -D3 could be limited in nephrological involvement	Laboratory-guided vitamin D level (target: 40–60 ng / mL), always with K2 and Mg as cofactors (see vitamin D+K2+Mg synergy, organ axis 3). Retinoic acid only through natural food sources (liver, egg yolk) or in low doses	[YELLOW] Vitamin D is well proven as a Treg inducer; FQAD-specific not examined
3	Low-Dose Naltrexone (LDN)	In normal doses, an opiate blocker; in microdoses (0.5-4.5 mg) it only blocks opioid receptors for a short time → rebound effect: increased endorphin release → measurable Th17→Treg shift + TLR4 dampening on mast cells. Increasingly used in MCAS and autoimmune therapy	Only under medical supervision. Start with 0.5 mg, gradually increase. Caveat: can initially cause sleep disorders or a short-term increase in symptoms. Contraindicated with concurrent opioid therapy	[YELLOW] Off label; RCTs in fibromyalgia and Crohn's disease; FQAD-specific not examined
4	Vagus nerve activation (cholinergic anti-inflammatory pathway)	Vagus nerve releases acetylcholine → immune cells with acetylcholine receptors switch to regulatory (Treg) mode → histamine release is slowed down. Chronic dysautonomia (long-term sympathetic stress) suppresses this pathway Non-invasive methods: 4-7-8 breathing techniques, HRV biofeedback, gentle humming, if necessary transcutaneous vagus nerve stimulation (tvNS) on the ear [22], [44]. No extreme cold / stress training	[YELLOW] Physiologically based; plausible in dysautonomia / MCAS; FQAD-specific not validated	
5	SPMs (Specialized Pro-resolving Mediators): Resolvine / Maresine	SPMs are formed from omega-3 fatty acids (EPA / DHA) and are the body's own "inflammatory stop switches" → they actively recruit Tregs to the site of inflammation → tissue is cleaned up and repaired. Pure anti-inflammatory drugs (such as NSAIDs) only suppress; SPMs dissolve the inflammation (Resolution)	Laboratory-guided omega-3 supply (high-quality algae oil preferred as it is free of heavy metals). Measure the omega-3 index in the blood (target: >8%). Caveat: In severe MCAS, fish oil can also act as a trigger → Test algae oil	[YELLOW] SPM research active; FQAD-specific not examined

Important: These strategies are aimed at restoring basic immunological regulation, not symptom suppression. They do not replace MCAS acute therapy (H1/H2 blockers, mast cell stabilizers), but complement it as a long-term rehabilitation approach for the Treg-mast cell axis.

Molecular Mechanics of Mast Cell Activation — DAMPs and Calcium Influx:

Mast cell activation in FQAD likely occurs through at least three converging pathways: 1. **DAMPs (Damage-Associated Molecular Patterns)**: Damaged mitochondria release molecular “alarm waste” (mtDNA fragments, cardiolipin, ATP) - so-called DAMPs, which act as endogenous danger signals. These DAMPs activate pattern recognition receptors (TLR9, NLRP3) on mast cells and signal: “Tissue dies here” → inflammatory cascade [4], [34]. 2. **Direct MRGPRX2 pathway**: Ciprofloxacin directly activates the MRGPRX2 receptor on mast cells, triggering immediate, non-IgE-mediated degranulation (histamine release) — without classic allergy sensitization [19]. 3. **Intracellular calcium influx**: Ciprofloxacin induces a rapid intracellular Ca^{2+} increase in mast cells, which acts as a direct physical switch for exocytosis (granule release). In combination with the FQ-induced NMDA overactivation and GABA blockade (see section C, axis 4), a systemic Ca^{2+} excess is created, which permanently lowers the mast cell degranulation threshold.

Important distinction: Proven dysbiosis is not evidence of mtDNA damage or collagen breakdown — it is a separate, parallel axis of damage. A commercial stool microbiome test shows who lives in the gut, but not what damage the imbalance has done to the barrier.

Diagnostics — functional markers before bacterial counting:

Markers	What he shows	Availability
Calprotectin (stool)	Structural inflammation of the intestinal mucosa	Available from statutory health insurance providers
Zonulin (blood / stool)	Clinically relevant barrier dysfunction (leaky gut)	IGeL / special laboratory
Tryptase (blood)	Mast cell activation (only if clinically suspected)	Statutory health insurance provider if indicated
Methylhistamine (24-hour urine)	Histamine overproduction	Special laboratory
Microbiome Stool Profile	Bacterial composition, diversity index	IGeL / commercial providers

Recommended sequence: First functional markers (calprotectin, zonulin, possibly tryptase), then - only if the result is positive - extended microbiome profiling to guide therapy.

Probiotics for FQAD — The Histamine Trap:

Many commercially available probiotics (particularly favorable Lactobacillus strains) produce their own histamine in the intestine. If mast cell activation is present, the wrong probiotic can trigger severe inflammatory attacks, cognitive deterioration or pain [18], [20]. Low-histamine bacterial strains (e.g. Bifidobacterium longum, Lactobacillus rhamnosus GG - selection based on individual tolerance) are the safer option if MCAS is suspected. The document does not name any specific preparations as recommendations. **Postbiotics as an alternative to probiotics**: Instead of swallowing live bacteria and hoping for their survival, the curative horizon (Section XIV) refers to synthetic postbiotics: the pure metabolites of healthy bacteria (short-chain fatty acids such as butyrate) that directly stabilize the intestinal barrier and act as anti-inflammatory signaling molecules [46]. This is research, not a therapeutic available today.

Increase butyrate production through diet — MCAS-safe strategies ([YELLOW] individualized):

Level of evidence: [YELLOW] Butyrate production by prebiotics is microbiologically established. The specific use in FQAD/MCAS is not validated. Basic rule: “Low and Slow”, introduce one at a time, stop if it worsens.

The histamine trap extends not only to probiotic capsules, but also to foods: fermented foods (yoghurt, kefir, sauerkraut, kombucha) are created through fermentation, during which a lot of histamine is naturally produced - often a direct trigger for flare-ups in the case of leaky gut and unstable mast cells.

Strategy	Mechanism	Implementation	MCAS Security
Resistant Starch (Prebiotics)	Is not digested in the small intestine → migrates intact into the large intestine → fermented there by commensal bacteria directly to butyrate — histamine-free	Cooled potatoes or rice (12-24 hours in the fridge → starch becomes “resistant”; slight reheating allowed, not boiling hot). Green (unripe) bananas or banana flour. Oat flakes (beta-glucans → excellent butyrate substrate, if gluten-free / oat-compatible)	High — no histamine during fermentation; introduce very slowly (flatulence possible with dysbiosis)
Soluble fiber	Feed the butyrate-producing bacteria and protect the irritated mucous membrane	Apple pectin (cooked apples / sugar-free applesauce - gently coats the intestinal mucosa). Psyllium husks (small amounts + lots of water; bind toxins + SCFA substrate)	High - histamine neutral; Increase quantities carefully
Butter / Ghee (direct source of butyrate)	“Butyrate” is derived from the Latin butyrum (butter) - grass-fed butter and ghee (clarified butter, milk protein removed) contain natural butyric acid Ghee is usually excellently tolerated in low-histamine cuisine; small amounts as cooking fat	High — Ghee is low in histamine and casein	
Low histamine probiotic strains	Bifidobacteria do not produce butyrate themselves, but engage in “cross-feeding”: prepare substrate so that butyrate bacteria utilize it optimally	Safe bifidobacteria: B. infantis, B. longum. Safe lactobacilli: L. rhamnosus GG, L. plantarum (histamine-lowering or -neutral). To avoid: L. casei, L. bulgaricus, L. delbrueckii (histamine producing) [18], [20]	Means — test individually; only after GI diagnosis and introduce individually

Practical Summary: Avoid fermented and standard probiotics. Instead: cooled potatoes/rice, applesauce, ghee and, if necessary, low-histamine bifido strains - so that the mast cells do not get fuel while the Tregs are rebuilt via butyrate production.

Therapy if clinically suspected: Mast cell stabilizers (cromoglicic acid, ketotifen; caveat: GI irritation as a side effect), low-histamine diet (avoidance of aged cheese, wine, tomatoes, fermented foods), targeted microbiome build-up only with low-histamine strains and after diagnostics [18]-[20].

Severe mast cell activation — when even water cannot be tolerated:

In extreme cases, those affected report reactions to almost all foods and even still water. This is often clinically dismissed as psychosomatic, but it has an organic basis: in severe MCAS, mast cells no longer only fire in response to chemical allergens, but also in response to physical stimuli - mechanical gastric distension (even from a sip of water), temperature differences in the drink or osmotic differences are sufficient to cause unstable mast cells to degranulate [19]. The root causes of chronicity: senescent cells in the intestinal tissue constantly release SASP inflammatory substances, which keep mast cells in a state of alert 24 hours a day; At the same time, a barrier damaged by dysbiosis (leaky gut) keeps the immune system in constant stress. Mast cell stabilizers (H1/H2 blockers, cromoglicic acid, ketotifen) do not repair mitochondrial damage or the intestinal barrier, but rather create a “chemical armor” around the fragile mast cells so that they do not degranulate immediately with minimal stimuli. For many sufferers, this symptom control is the only way to maintain food or fluids at all - an essential measure of damage stabilization (Phase A), which may remain necessary in the long term or permanently as long as the underlying cell damage is not repaired.

Multiple Chemical Sensitivity (MCS) as a result of the MCAS/GABA double blockade:

Level of evidence:[YELLOW] Plausible. Mechanistically justifiable from the combination of MCAS literature and GABA neurotoxicity data; FQAD-specific not systematically examined.

MCS (hypersensitivity to chemicals, odors, fumes in concentrations that healthy people can easily tolerate) is frequently reported by those affected by FQAD, but has not yet been discussed as an independent complication in the document. The pathogenesis could result from the convergence of two damage mechanisms:

1.Immunological component (MCAS): Chronically unstable mast cells with a lowered degranulation threshold (due to DAMPs, Treg loss and MRGPRX2 sensitization) respond to minimal chemical stimuli with histamine release. **2.Neurological component (GABA filter failure):** FQ competitively block GABA-A receptors in the CNS [5]. The resulting neuronal hyperexcitability - especially in the limbic system (amygdala, olfactory cortex) - could lead to the central nervous system no longer adequately filtering chemical environmental stimuli and misinterpreting them as life-threatening danger signals.

The combination of immunological overreaction (MCAS) and neuronal misinterpretation (GABA deficit) could explain why MCS occurs so frequently and so severely in those affected by FQAD - and why purely immunological therapies (mast cell stabilizers) alone are often not enough: the neurological filter disorder persists independently of mast cell stabilization. Therapeutically, a dual approach (MCAS stabilization + neural desensitization/GABA support) should be considered, although there is no FQAD-specific evidence for any of these approaches.

The three key markers - practical use by the doctor:

Markers	What he proves	Crucial practice details
Tryptase (blood)	Systemic mast cell activation (MCAS). Tryptase is produced almost exclusively by mast cells; an increase shows their degranulation.	Timing is critical: Baseline may be completely normal at rest. The blood sample should ideally be taken 1-4 hours after an acute reaction in order to document the increase. A normal resting value does not rule out MCAS.
Zonulin (blood / stool)	Barrier disruption (leaky gut). Zonulin controls the tight junctions of the intestinal mucosa; an increased value shows that the protective layer is defective and irritants are entering the blood in an uncontrolled manner.	Available as IGeL or through special laboratories. Not eligible for insurance everywhere - inform patients about costs.
Calprotectin (stool)	Structural intestinal inflammation. Objectively differentiates between a real inflammatory process and functional "irritable bowel syndrome". A high value medically rules out a purely psychosomatic cause.	Available to statutory health insurance providers for gastroenterological indications. Standard in gastroenterology - not an exotic test.

These three parameters are established in conventional medicine (gastroenterology, immunology). The advantage: They provide hard, objective findings that can be used to argue to skeptical doctors that the symptoms represent a physical and not primarily a psychological problem.

MCAS Stabilization Protocol (10-step approach):

There is no single solution to toxically induced MCAS — it requires a multi-pronged approach: shield with medication, reduce histamine supply, calm an overstimulated nervous system. Each change must be tested individually. Evidence status: transferred from general MCAS literature; FQAD-specific not validated [18]-[20].

Medicinal basis (medical support): **1.H1 + H2 blocker combination:** Cetirizine/desloratadine (H1, skin/respiratory) plus famotidine (H2, gastrointestinal). Only the combination adequately protects both types of receptors. **2.Oral mast cell stabilizers:** Cromoglicic acid or ketotifen coats the mast cell and prevents degranulation. Slow dosing is mandatory - even these medications can initially cause irritation. **3.Leukotriene antagonists:** Montelukast cuts off the second axis of inflammation (leukotrienes) when H1/H2 blockers alone are not enough.

Nutrition and intestinal axis: **4.Consistent low-histamine elimination diet:** Avoiding fermented, matured and warmed foods (cheese, red wine, tomatoes, soy sauce). Doesn't cure mast

cells, but takes the constant pressure out of the system. **5.DAO enzyme substitution before meals:** Diamine oxidase (DAO) breaks down histamine in the intestines. In antibiotic-induced dysbiosis, the body's own DAO production is often reduced. DAO capsules (e.g. Daosin) shortly before eating act as enzymatic histamine disposal. **6.Gentle barrier stabilization:** Postbiotics (butyrate) or L-glutamine (test extremely carefully and in low doses in FQAD) can stabilize tight junctions [46]. No irritating standard probiotics (see histamine trap above).

Nervous system and environment: **7.Trigger identification via symptom diary:** Hidden triggers: strong smells (perfume, cleaning products), temperature fluctuations, additives in medications (titanium dioxide, dyes). **8.Soothe autonomic axis:** Vagus nerve activation (4-7-8 breathing, gentle humming, non-invasive vagal stimulation if necessary). No extreme cold/stress training. Sympathetic excess directly fuels mast cells [22]. **9.Strictly avoid PEM:** Metabolic crashes (lactate buildup, cellular stress) are one of the strongest internal mast cell triggers. Pacing is also active MCAS therapy [23].

Gentle cofactors: **10.Buffered Vitamin C:** Most important natural cofactor for DAO production and histamine breakdown. Regular ascorbic acid can irritate the stomach and mast cells — hence calcium ascorbate or liposomal vitamin C in small doses. No high-dose infusions.

Calcium and magnesium — a common misconception:

A common misconception is the assumption that if you have FQAD you have to reduce calcium in your diet because intracellular calcium influx triggers cell death (apoptosis). The pathobiochemical reality is more nuanced:

The problem is intracellular, not dietary: Cell death occurs due to calcium buildup inside the cell and in the mitochondria. The calcium from the diet ends up in the blood, the level of which the body keeps exactly constant through homeostasis - if necessary by breaking down bones. Eating less calcium does not lower intracellular calcium but increases the risk of osteoporosis — particularly dangerous in immobile FQAD patients. *Why cells “drown” in calcium (the ATP problem):* Healthy cells constantly pump excess calcium via calcium ATPases — these pumps require ATP. When the respiratory chain is blocked and there is anaerobic emergency power, ATP is missing, the pumps stop, and calcium floods the cell uncontrollably [4], [35].

The true antagonist - Magnesium: Magnesium and calcium are antagonists on the ion channels (especially NMDA receptors). Adequate magnesium levels block uncontrolled calcium influx, even in the case of ATP deficiency. Since fluoroquinolones bind magnesium through chelation [4], many of those affected suffer from intracellular magnesium deficiency, which further escalates the calcium problem.

Consequence: Do not reduce calcium in the diet. Check magnesium status instead — serum magnesium alone is not enough (only shows extracellular Mg); a whole blood mineral analysis (intracellular magnesium) is more informative. Magnesium is already part of the basic diagnosis of statutory health insurance (level 1); the whole blood analysis is IGeL/special laboratory.

4) Musculoskeletal-tendinous axis Pathomechanism: Fluoroquinolones upregulate matrix metalloproteinases (MMPs), which actively degrade tendon collagen [13]. At the same time, damaged fibroblasts go into senescence and stop producing collagen. The result: enzymatic decay with a simultaneous stop in production - a gap that cannot be closed by adding building materials alone.

The retrograde signaling chain: Why damaged mitochondria rewrite the collagen blueprint in the cell nucleus ([YELLOW] Plausible)

The clinically observed phenomenon - after FQ damage, tendon tissue no longer grows parallel and elastic, but rather chaotic, thickened and scarred (chronic tendinopathy/fibrosis) - is caused by a chain reaction between mitochondrial and nuclear DNA:

step	Location	What happens
1. Origin: mtDNA damage	Mitochondria	FQ inhibit mitochondrial topoisomerase II → mtDNA damage [2], [3]. The mtDNA does not contain any collagen blueprints, only the blueprints of the respiratory chain complexes. Consequence: Cell hardly produces ATP, but massive ROS (oxidative stress)

step	Location	What happens
2. Retrograde signaling	Cytoplasm → cell nucleus	The oxidative stress from the broken mitochondria floods the entire cell and penetrates into the nucleus. In addition, released DAMPs (mtDNA fragments, cardiolipin) signal to the cell nucleus: “Energy crisis, tissue damage, activate emergency program”
3. Epigenetic shift in nDNA	Cell nucleus (nuclear DNA)	ROS + energy deficiency alters the epigenetic switches (DNA methylation, histone acetylation) on the collagen genes in nuclear DNA. Epigenetics determines which genes can be read - the oxidative stress “pastes over” the correct blueprints
4. Collagen Type Switch	Tenocytes	The epigenetically misprogrammed tenocytes switch off genes for healthy, parallel type I collagen (elastic, load-transmitting) and instead activate type III collagen (thin, rigid, unstructured - “emergency collagen” for rapid wound healing) + MMP overproduction [13]
5. Chronic tendinopathy	tendon tissue	Result: Instead of elastic, parallel fibers, chaotic, knotted scar tissue (fibrosis) is created. The tendon is thickened but mechanically weaker — it can no longer transmit tensile forces along one axis, which increases the risk of rupture

Clinical consequence: The collagen blueprint defect is located in the nuclear DNA (nDNA), but is almost always the direct result of oxidative fire from the damaged mitochondrial DNA (mtDNA). This explains why isolated collagen supplementation (gelatin, hydrolyzate) does not solve the problem: the building material is supplied, but the blueprint in the cell nucleus is incorrectly programmed. The therapy must therefore primarily reduce oxidative stress (Mg^{2+} , antioxidants) and restore mitochondrial function (PGC-1 α activation) so that the epigenetic switch in the cell nucleus can switch back to type I collagen. **The MMP problem — why diet alone is not enough:** There are currently no approved agents or dietary supplements that specifically and safely stop MMPs in FQAD. Herbal “MMP inhibitors” (curcumin, OPC, quercetin, green tea extract) from forums have not been proven for this level of damage and are not listed as therapy in the model. The Curative Horizon (Section XIV) outlines siRNA hydrogels as a future concept: locally injected gels that genetically silence MMP blueprints in tissue [45]. Indirectly, reducing redox stress (diet, avoiding blood sugar peaks) can reduce the long-term inflammatory stress on the cells and thus weaken MMP upregulation - an indirect, not a direct lever [34].

Collagen hydrolyzate - realistic assessment: Useful as an easily digestible protein source (glycine, proline) to support and prevent catabolic tissue breakdown. However, it will not stop or reverse the enzymatic tendon breakdown alone: The problem is not primarily a lack of building blocks, but rather a stop in cell production (senescence) with simultaneous active degradation (MMPs). Visually: truckloads of new bricks don't help when the bricklayers go on strike and an excavator tears down the wall. The absorbed protein is metabolized without being specifically incorporated into the damaged tendon - unless the fibroblasts are given a mechanical remodeling command at the same time.

Eccentric stress therapy — the central tool for today:

Normal rest allows pain to subside, but does not build new tissue. Normal strength training can overload the damaged tendon. Eccentric loading is the mechanical middle ground:

Principle: The tendon is lengthened slowly and in a controlled manner under tension (example: from standing on your toes, lower your heels extremely slowly). The uniform mechanical pull signals the remaining healthy fibroblasts precisely in which direction they need to weave new collagen fibers. The unstructured scar tissue is realigned — a structural remodeling process that takes months.

Three hard FQAD conditions:

Condition	Requirement	Consequence of non-fulfillment
1. Acute phase must be over	No active inflammation, no acute pain attack	In case of acute irritation: strict protection, no training

Condition	Requirement	Consequence of non-fulfillment
2. Ultrasound monitoring before start	High-resolution ultrasound with elastography: assess collagen thickness, microcracks, inhomogeneity [13]	If collagen is dangerously thinned or microtears: risk of rupture → training contraindicated
3. No PEM	Documented stress tolerance without post-exertional malaise	If you have a tendency to PEM: Pacing has absolute priority; Eccentric training only after PEM stabilization

Protocol (example Achilles tendon): Start with minimal body weight, extremely slow eccentrics (5-10 seconds per repetition), only 1-2 sets of 5-10 repetitions. Increase only if tolerability is documented for at least 48 hours (no flare-ups, no PEM, no swelling). Physiotherapeutic support recommended. Ultrasound check after 4-6 weeks.

Differentiation between training pain and FQAD crash: A slight pulling sensation in the tendon during or shortly after exercise that subsides within hours is to be expected. However, if the exercise results in days of full-body fatigue, cognitive decline, searing nerve pain, or radiating swelling, the metabolic or structural limit has been exceeded → immediate stop, crash protocol (Phase A).

PRP (Platelet-Rich Plasma) is case-dependent and has not been validated for FQAD [13].

Accelerated Osteoarthritis in Young FQAD Sufferers — Levels of Evidence:

It is often reported in the community that those affected develop severe osteoarthritis in their 30s or 40s. However, the evidence is tiered:

Level of evidence	statement
Occupied	Fluoroquinolones can cause long-lasting damage to tendons, muscles, joints and nerves. Joint pain and swelling are recognized as possible long-lasting side effects [9], [10], [13]
Plausible	Accelerated development of osteoarthritis can occur secondarily when joint mechanics deteriorate due to tendon, muscle and possibly cartilage / matrix damage and chronic incorrect loading occurs. Older toxicological works describe FQ-induced chondrocyte apoptosis, reduced matrix formation and altered MMP expression in cartilage cells [13]
Not documented	That fluoroquinolones generally primarily trigger rapidly progressive osteoarthritis as an independent direct effect in humans. The evidence for this is significantly weaker than for tendinopathy

Three converging mechanisms (hypothetical): **1.Chondrocyte crash:** FQ attack the mitochondria of the cartilage cells → apoptosis or senescence → repair stops in the cartilage. MMP-3 and MMP-13 specifically break down collagen type II (main component of cartilage) from the inside [13].

2.Mechanical instability: Softened ligaments and degenerated tendons no longer keep the joint on track → micro-instability → asymmetrical abrasion of the remaining cartilage (comparable to a crooked car tire).

3.Chronic joint inflammation: SASP factors from senescent cells maintain smoldering inflammation in the joint, which further accelerates degradation.

The cleanest formulation: “The antibiotic could have damaged tendons, muscles, matrix and possibly cartilage cells; this can lead to altered joint mechanics and thus accelerated wear.”

Cartilage preservation during bed rest - passive micro-mobilization:

Cartilage is avascular; it is nourished exclusively by diffusion from the synovial fluid. This diffusion process requires rhythmic compression and release — a “pump” through movement. When you are forced to stay in bed, this pump stops working and the cartilage also degenerates due to a lack of nutrients.

Passive micro-movements (also possible with severe PEM): - Slow, passive bending/stretching of the joints (knees, ankles, fingers) while lying down - no use of muscle strength, but gravity or hand guidance - Heel rocking: Allow the heel to slide minimally forward/back on the mattress (a

few centimeters) - Toe circling: Allow the ankle joint to rotate passively, without muscle tension
 - Goal: 2-3 times a day for 2-5 minutes per joint, pain-free, without any effort - Discontinue if there is pain, swelling or subsequent deterioration

This puts no strain on tendons or the cardiovascular system and does not provoke PEM, but maintains minimal synovial fluid circulation. Physiotherapeutic guidance is recommended.

5) Endocrine axis With long-term stress, secondary HPA axis dysfunction can become clinically relevant [17]. Evaluate cortisol, DHEA-S and sex hormones in a phenotype-guided manner. Endocrine balancing should, if necessary, be checked before intensive re-conditioning.

6) Neuropsychiatric axis Anxiety, depression, cognitive dysfunction, sleep disorder. Psychological support, if necessary cognitive behavioral therapy. SSRI/SNRI only with extreme caution (see Section I). The integration of neuropsychiatric care serves to treat real symptoms and coping stress, not to question organic damage [5], [14].

Social invalidation as a chronification amplifier:

Devaluation, pressure and disbelief from relatives, friends or doctors (“don’t act like that”, “that can hurt”, “you’re imagining it”) are extremely common among those affected by FQAD. This is not medically harmless: in chronic pain and multisystem diseases, it is well documented that social support is more protective, while invalidation is associated with worse symptom burden, more depression and poorer outcome.

This does not mean that the illness is “just psychological” - on the contrary: authorities recognize that fluoroquinolones can cause long-lasting, disabling and sometimes irreversible damage to nerves, tendons, muscles and the CNS [9], [10]. If someone is constantly invalidated in this situation, social pressure meets real damage. The resulting vicious circle of sleep disturbance, tension, incorrect stress behavior and more pain can measurably increase chronicity.

Short information for relatives:

- Fluoroquinolones can rarely but cause serious long-term damage to multiple organ systems. This is recognized by FDA, EMA and BfArM [9], [10].
- “Try harder, it may hurt” is medically counterproductive if there is possible tendon/nerve involvement. If you have tendon problems, rest is explicitly recommended [13].
- “You’re imagining it” is a form of invalidation that can worsen the outcome.
- More helpful: “I believe you,” “We’ll see what you can tolerate today,” “We’ll take a symptom-guided approach.”
- The goal is not forced normality, but gradual, measurable stabilization and functional gain.

3.3 Re-conditioning ([YELLOW] Individualized)

Phase C — Re-conditioning

C1 — Clinical re-conditioning (acceptable today)

aspect	Description
Target	Improve functional reserves
Clinical Status	transfer from mitochondrial medicine; no FQAD-specific evidence [21], [24]
Major Uncertainty	Which form helps, which harms; Subgroups unknown
Endpoint	Walking distance, grip strength, quality of life (SF-36), symptom-free duration of exercise
Risk	Severe load decompensation → immediate stop, crash protocol (see phase A)

Specifically:

-Nutrition: Individually dosed, needs-based nutrition. Sufficient protein for muscle maintenance, kidney-adapted for proteinuria. This does not follow any general nutritional instructions;

The biologically plausible but not FQAD-validated hypothesis of protein cycling (mTOR unloading, see Section XIV) should not be misunderstood as self-therapy [17], [25].

Nutritional strategy based on the pathobiochemical model (plausible, not FQAD validated):

The diet is aimed at three axes: relieving the load on mitochondria, reducing anaerobic emergency power (excess lactate) and stopping catabolic tissue breakdown. Any changeover should be done carefully ("low and slow") [47].

1. Mitochondria support and anti-catabolism: High-quality protein (eggs, fish, legumes) in sufficient quantities to counteract the breakdown of muscle protein caused by the lack of energy. Omega-3 fatty acids (linseed oil, walnuts, algae oil) support cell membranes and counteract oxidative stress. B vitamins (whole grains if tolerated, green leafy vegetables) as cofactors for energy metabolism. Foods rich in vitamin C (peppers, citrus fruits) for collagen synthesis in damaged tendons and fascia [13], [47]. Collagen hydrolyzate (glycine, proline) can be useful as an easily digestible supplementary protein source, but will not alone stop enzymatic tendon breakdown (see organ axis 4: MSK).

2. Lactate management (reduction of acid load): Avoiding isolated sugars and white flour — these fuel glycolysis (the emergency mode), which leads to rapid lactate increases and can worsen post-exertional malaise [35]. Instead: complex carbohydrates (oats, quinoa, potatoes) that provide stable energy without acute lactate peaks. Green vegetables (spinach, broccoli) provide magnesium and potassium as acid buffers. Berries (blueberries, raspberries) contain antioxidants that can counteract the oxidative cascade (Fenton reaction) [34], [47].

3. Organ axis-specific nutrition: Kidney-adapted when eGFR is compromised: moderate, high-quality protein intake, sparing use of salt. Low-histamine diet if there are signs of mast cell activation (avoidance of aged cheese, wine, tomatoes, fermented foods) [18]–[20]. Digestive relief: light evening meals save energy for nightly repair processes.

4. Protein cycling (only under medical supervision): Alternation of building phases (sufficient protein for muscle maintenance, if necessary with BCAA supplementation) and relief phases (moderately reduced protein intake to relieve mTOR and enable autophagy) [25]. Contraindicated in cases of underweight, severe sarcopenia or unstable adrenal status. Any form of protein or calorie reduction must ensure the preservation of muscle mass; Unintentional sarcopenia would further impair the ability to regenerate [17], [25].

Disclaimer: These nutritional principles are biologically plausible but have not been validated in FQAD studies. They do not replace individual nutritional advice.

Weight loss in FQAD: warning signal, not treatment goal ([GREEN] clinically relevant)

Unintentional or forced weight loss is a massive red flag with FQAD. If the mitochondria are blocked (emergency power/glycolysis) and too few calories are consumed, the body immediately begins to burn its own muscle tissue (gluconeogenesis → catabolic state). The consequences: (1) Ammonia as a toxic waste product → brain fog; (2) muscle wasting (sarcopenia) → impaired ability to regenerate; (3) low muscle mass distorts creatinine-based kidney values → toxic overdose in future medications (see cystatin C case, 1.6). The absolute priority is to maintain muscle mass and avoid catabolic stress.

Calorie source prioritization for FQAD:

Since fast carbohydrates (sugar, white flour) lead to lactate peaks and PEM crashes (see food triggers), calories (goal: ≥2000 kcal) must be achieved via three safe pillars:

pillar	Why preferred at FQAD	Examples
Safe Fats (#1 Calorie Carrier)	Highest calorie density / g; do not produce lactate; do not trigger mast cells; are metabolized differently than carbohydrates	Ghee (direct source of butyrate), olive oil, coconut oil, algae oil (omega-3 / SPMs), macadamia nuts (oxalate-free, low histamine)
Complex carbohydrates	Slow release → no glycolysis peaks; resistant starch → histamine-free butyrate production	Cooled potatoes / rice (12-24 hour refrigerator), quinoa, millet (gluten-free), sweet potatoes, oat flakes (beta-glucans)

pillar	Why preferred at FQAD	Examples
High Quality Protein	Preservation of muscle mass against catabolic breakdown; essential for collagen repair	Organic poultry / salmon (organic → no FQ residues), eggs, legumes if tolerated (test individually)

MCAS and Treg compliant sample nutrition plan ([YELLOW] individualized):

All information is an example, not a therapy prescription. Individual tolerance determines food selection; introduce each new food individually.

Day 1 — Focus: Magnesium, Zinc and Collagen Protection (Vitamin C)

meal	Ingredients (MCAS Compliant)	Nutrient Booster	Principle	~kcal
Breakfast	100g gluten-free oat flakes (B vitamins, beta-glucans) + 2 tbsp ghee (butyrate, fat-soluble vitamins A / D/E / K) + 150g applesauce (pectin) + blueberries	2 tbsp peeled pumpkin seeds — MCAS compatible, low in oxalates; one of the best plant sources of Mg, Zn and Fe at the same time	Butyrate Substrate + Chelate Repair Minerals	~650
Lunch	100g quinoa (vegetable Fe, Mg, all essential AS) + 150g organic wild salmon (Vit. D3, B12, Omega-3) + 150g steamed broccoli (Ca, K, folic acid) + 3 tbsp olive oil	½ raw yellow pepper — low in histamine, vitamin C bomb. Vit. C essential for collagen synthesis in tendons + histamine breakdown (DAO enzyme). Eat raw (heat destroys vitamin C)	Omega-3→SPMs + Collagen Repair	~800
Evening	350g pre-cooked, cooled potatoes (resistant starch, K-buffer) + 3 organic eggs (protein, B vitamins, choline, biotin) + 2 tbsp ghee + zucchini	Large handful of fresh parsley - massive vitamin K1 (bone) + folic acid, without triggering oxalates or histamine	Resistant starch→Tregs + Protein→Anti-catabolism + K1 for bones	~700

Day 2 — Focus: B vitamins, iron and cell membrane protection

meal	Ingredients (MCAS Compliant)	Nutrient Booster	Principle	~kcal
Breakfast	80g millet (gluten-free, rich in vegetable Fe + silicon for connective tissue) + 150ml full-fat coconut milk (safe quick energy) + green banana (resistant starch, K, B6)	2 tbsp peeled hemp seeds — perfect Omega-3:6 ratio, high-quality protein, cover a large part of the daily Mg and Fe requirements, gentle on the intestines	MCT energy + mineral build-up	~700
Lunch	150g organic chicken breast (Vit. B3, B6, Zn, clean protein) + 100g basmati rice from the previous day (cooled → resistant starch) + ghee / olive oil	200g steamed pak choi - spinach substitute without oxalates: massive Ca, folic acid, vitamin A; mild on the intestines, no risk of kidney stones	Protein preservation + Ca without oxalate risk	~750
Snack	40g macadamia nuts - highest calorie density of all nuts, oxalate-free, low in histamine	Provides Manganese — “ignition key” for SOD2 (mitochondrial fire extinguisher)	Safe calorie compression + SOD2 cofactor	~300
Evening	300g sweet potatoes (beta-carotene→Vit. A, K, copper) + 100ml coconut milk + mild herbs → pureed soup	Shot cold-pressed pumpkin seed oil over the finished soup - Vitamin E (cell membrane antioxidant) + plant sterols	Easily digestible + membrane protection	~500

Micronutrient summary of the two days:

Micronutrient	Primary Sources in the Plan	FQAD Relevance
Magnesium	Pumpkin seeds, hemp seeds, quinoa, oatmeal, broccoli	Chelation Repair (#1 Priority)
Zinc	Pumpkin seeds, quinoa, organic poultry, eggs	TIMP cofactor (MMP brake) + SOD cofactor
Iron (vegetable)	Pumpkin seeds, hemp seeds, millet, quinoa	Low dose via food = safe; Avoid high-dose supplements (Fenton!)
Vitamin C	Yellow peppers (raw!), broccoli, blueberries	Collagen synthesis + DAO enzyme (histamine breakdown)
Vitamin K1	Parsley, broccoli, bok choy	Bone protection in the event of immobility
Vitamin D3	Organic wild salmon, ghee, eggs	Treg induction (see Treg restoration)
B vitamins	Salmon (B12), chicken (B3 / B6), eggs (biotin / choline), millet, oats	Myelin sheath + cellular energy
Omega-3 (EPA / DHA)	Organic wild salmon, hemp seeds, algae oil	SPM production → inflammation resolution
Manganese	Macadamia, millet, quinoa	SOD2 cofactor (mitochondrial fire extinguisher)
Calcium (without oxalate)	Pak choy, broccoli, eggs	Bone protection without NMDA escalation

The Realistic Limit of Nutrition: This plan maximizes the nutrients possible from natural foods under FQAD/MCAS conditions. However, if chelation has massively depleted deep intracellular Mg^{2+}/Zn^{2+} depots, even the best nutrition is not sufficient to resolve this pathological deficiency quickly enough. Food provides the maintenance dose; A therapeutic balance requires laboratory-managed monopreparations - see supplementation paradox (1.5). **Cave glycine paradox:** Magnesium glycinate can cause paradoxical fatigue in FQ-damaged nervous systems (glycine = NMDA co-agonist). MCAS-safe alternatives: Mg malate, Mg taurate or transdermal $MgCl_2$ (see 1.5).

Compact FQAD/MCAS food classification (traffic light system):

This table is a guide, not a diet prescription. Individual tolerance varies; introduce each new food individually.

[GREEN] GREEN — Safe Base: Maximum nutrients, safe calories, no mast cell triggers.

Food	FQAD justification
Ghee and Coconut Oil	Heat stable energy. Ghee directly delivers butyrate (Treg induction), coconut oil provides quick MCT energy without lactate build-up
Potatoes and rice (cooled, 12-24 hours)	Resistant starch → histamine-free butyrate production in the colon for Treg buildup
Broccoli and Pak Choi	Safe spinach alternative: low in oxalate, provides Ca, K (acid buffer), folic acid, vitamin A
Hemp seeds and pumpkin seeds	Low Oxalate Nutrient Booster: Mg + Zn + Fe (Chelate Repair Trio). Hemp: perfect Omega 3:6 ratio
Organic wild salmon (frozen / freshly slaughtered)	Active EPA / DHA → SPM production (inflammatory resolution) + Vit. D3 + B12. Never from a can (histamine!)
Organic beef (freshly slaughtered, not aged)	Top for zinc and protein. Condition: not “dependent” — maturation produces histamine
Sweet Potatoes	Safe complex KH, potassium, massive beta-carotene (→ Vit. A)

Food	FQAD justification
Blueberries and applesauce (cooked)	Apple: Pectin (intestinal mucosa protection). Blueberries: safe antioxidants without histamine
Algae oil (high quality)	Safest, purest, histamine-free source of therapeutic EPA / DHA (no fish oil risk for MCAS)
Mild herbs (dill, coriander)	Safe micronutrients and digestive support, no mast cell irritation
Macadamia Nuts	Highest calorie density of all nuts, oxalate-free, low in histamine. Supply manganese (SOD2 cofactor)
Buckwheat	Gluten-free, easy to digest, low in histamine. Best natural source of quercetin (23 mg / 100g) compatible with MCAS / gastroparesis. Alternative to rice noodles
Olive oil (extra virgin)	Calories + polyphenols (anti-inflammatory), histamine neutral
MCT oil (pure C8 / caprylic acid)	Gastroparesis bypass: Does not require bile acid, hardly triggers CCK gastric brake → goes directly to the liver → ketone bodies as mito-energy. Increase dropwise (initially laxative)
Slippery Elm bark	Mechanical mucous membrane protection: Forms a gel-like film over the stomach / esophageal wall → physically shields mast cells from food stimuli. Histamine-free, no acid blockage
Pure rock salt (without iodine / fluoride / free-flowing aid)	Electrolyte balancing for dysautonomia; no chemical additives
Short meat broth (organic, max. 60-90 min.)	Safe alternative to bone broth: briefly boil freshly slaughtered organic beef (no bones) → amino acids + minerals without massive histamine / glutamate formation. Warming, gentle on the intestines, suitable as a soup base
Blueberries (blueberries)	Antioxidant leaders: extremely rich in anthocyanins → directly intercept ROS and protect mtDNA. Very low in histamine and low in sugar
Apples (cooked as puree, with peel)	Intestinal protector: Releases massive amounts of pectin → coats permeable intestinal mucosa (leaky gut) like a protective film + feeds butyrate bacteria. Prepare sugar-free
Green, unripe bananas	Treg booster: Hardly any sugar when immature, but massively resistant starch → histamine-free butyrate feeding. Cave: Once yellow / brown → [RED] RED (histamine + sugar)
Pomegranate seeds (fresh)	Highly potent polyphenols → systemic inflammation reduction + mast cell stabilizing effect. Dose moderately
Cranberries / Cranberries (fresh / unsweetened)	Proanthocyanidins: extremely strong antioxidant + traditional kidney / bladder protection. No dried fruits (sugar + sulfites)

[YELLOW] YELLOW — Test depending on dose: Basically healthy, but can irritate the unstable system.

Food	FQAD justification
Organic eggs	Important choline + biotin + B vitamins + protein. Caveat: egg white can be triggering at first → possibly only egg yolk for now
parsley, basil	Top for vitamin C / K1, but contain oxalic acid → only as a spice, never as a smoothie ingredient in large quantities

Food	FQAD justification
Quinoa, millet	Good plant nutrients (Fe, Mg, essential AS), but dose moderately (avoid glycolysis overload)
Linseed (crushed)	Soothe intestinal mucosa + mucilage; but plant-based ALA-Omega-3 is not therapeutically sufficient for SPMs
Lactose-free quark, sheep's cheese	Good source of protein, hardly any histamine. Caveat: casein can irritate leaky gut → sheep / goats are usually better tolerated
Mozzarella, young Gouda	Low histamine (unfermented), but keep an eye on milk proteins if there is an intestinal barrier disorder
Psyllium husks	Feed butyrate bacteria + bind toxins, but with impaired intestinal motor function there is a risk of constipation → microdoses + lots of water
Sweet cherries, peaches	Low histamine but high fructose + FODMAPs → rapid intestinal fermentation → flatulence (mechanical MCAS trigger). Test only half a handful
Watermelon	Very low in histamine and hydrating, but high glycemic index → blood sugar spike. Eat with fat (e.g. macadamia) to slow absorption
Blackberries, Raspberries	Good vitamin C, but moderate histamine and salicylate levels. Some tolerate it, others react with nerve pain → test individually
Spelt (original wheat)	Contains gluten + lectins; Usually tolerated if the intestine is intact, but with leaky gut gluten fragments + lectins can get into the blood → inflammatory mediators → brain fog / fatigue. Test individually; Gluten-free alternatives (quinoa, millet, rice) are safer with MCAS
Peppers (raw)	Provides excellent vitamin C (collagen + DAO), but: nightshade → contains capsaicin + solanine + salicylates. In cases of salicylate intolerance (common in FQAD / MCAS), peppers can cause paradoxical fatigue → test individually, if reaction occurs, replace with broccoli / parsley as a source of vitamin C
Onions (cooked, small quantities)	Provide quercetin (mast cell stabilizer!), but: high sulfur content + FODMAPs. In case of sulfur metabolism weakness (CBS / SUOX) or SIBO → fermentation, flatulence, mechanical MCAS trigger. Strictly avoid raw (inulin trigger); cooked and tested individually in small quantities

[RED] RED - Strictly avoid: Mineral robbers, neuro-triggers and inflammation enhancers.

Food	FQAD justification
Spinach, almonds, beetroot	Oxalate robbers: bind urgently needed Mg^{2+}/Ca^{2+} in the intestines and blood, putting a strain on the kidneys
Pork (also organic)	Histamine liberator: directly forces the body's own mast cells to degranulate
Avocado	Histamine rich (~20-30 mg / kg) and histamine liberator. Often recommended as "healthy," but in MCAS it is a common trigger for fatigue, brain fog, and mast cell flare-ups
Aged cheese (Parmesan, Cheddar, etc.)	Histamine Bombs: Fermentation / ripening produces massive amounts of biogenic amines
Yogurt, kefir, sauerkraut, kombucha	Fermentation products: Classic cultures (<i>L. bulgaricus</i> , <i>L. casei</i>) produce histamine in the intestine
Walnuts, Tomatoes, Citrus, Strawberries	Classic histamine liberators: maintain the chronic mast cell inflammatory loop

Food	FQAD justification
Conventional meat and bone broth (organic too!)	Triple trap: (1) <i>Residues</i> : Enrofloxacin (veterinary FQ) → Ciprofloxacin metabolite in bone; Ion trapping in animal bones → Boiling extracts FQ as a highly concentrated eluate (in conventional husbandry). (2) <i>Histamine bomb</i> : Organic bone broth is also boiled for 12-24 hours → extreme protein breakdown → massive histamine and biogenic amine formation → immediate mast cell surges. (3) <i>Glutamate trap</i> : Long boiling of bones / cartilage releases massive glutamine → becomes free glutamate (= natural flavor enhancer; identical to yeast extract) → if GABA receptors are blocked: neuronal overheating (excitotoxicity), brain fog, restlessness, nerve burning. Safe alternative: "Short meat broth" - freshly slaughtered organic beef (no bones), cook for max. 60-90 minutes → Amino acids and minerals are lost, but there is not enough time for massive histamine / glutamate formation
Jerusalem artichokes, onions (large quantities)	Mechanical trigger: Inulin ferments extremely quickly, bloating the intestines → mechanical mast cell degranulation
Sugar, white flour	Danger of crash: forces massive glycolysis throughput → lactate flood → PEM
Glutamate, yeast extract, aspartame	Neuro-Triggers: Excitotoxicity with FQ-blocked GABA receptors → panic, insomnia, nerve burning
Black tea, green tea (large quantities)	Fluoride + aluminum from old tea leaves; compete with Mg^{2+} and increase oxidative stress
Citrus fruits (lemon, orange, grapefruit)	Classic histamine liberators + irritate damaged stomach lining. Grapefruit additionally blocks CYP3A4 (liver detoxification enzyme)
Strawberries	One of the strongest histamine liberators known → pseudoallergic reactions, MCAS flare-ups (skin, brain fog, joints)
Yellow / brown (ripe) bananas	Histamine trap: The riper, the more biogenic amines + sugar; Resistant starch has completely disappeared. Just green / immature = [GREEN]
Kiwi, Pineapple, Papaya	Aggressive protein-splitting enzymes (bromelain etc.): useful in healthy people, but irritate extremely sensitive / permeable mucous membranes + trigger mast cells
Dried fruit (raisins, dates, figs)	Triple bomb: gigantic amounts of sugar (lactate backlog) + dehydration + often contaminated with mold spores / sulfites → immediate immunological crashes
Commercial drinking formula (Fresubin, Fortimel, Ensure)	Maltodextrin / sugar = lactate crash; Casein / Soy = MCAS trigger; Carrageenan = intestinal inflammation inducer; synthetic vitamins = hardly usable in the event of oxidative stress. Designed for tumor patients with intact mitos, not for FQAD
HGH Injections, Growth Factor Extracts (YTE, Laminine, FGF)	Violent stimulation of a blocked system: energy crash (ATP missing for build-up command) + senescence overdrive (risk of cancer) + MMP feeding + MCAS hormonal shock. Avian egg extracts additionally: foreign protein MCAS triggers for leaky gut

Drinks: Mineral water as a therapeutic tool ([GREEN]/[RED])

Mineral water is an elegant, trigger-free method of mineral absorption: the ions are dissolved and are directly absorbed in the upper GI tract - without the risk of oxalate, histamine or FODMAP. However, the bottle label determines benefit or harm:

Parameters	Target value	FQAD justification
Hydrogen carbonate (HCO₃⁻)	>1000 mg / L (ideal 1000-1500)	Strongest natural lactate buffer system: cells in emergency flow (glycolysis) produce lactic acid → HCO ₃ ⁻ buffers continuously without neutralizing the stomach acid in a shock
Magnesium (Mg²⁺)	>100 mg / L	Chelation Repair Foundation; Bioavailability from water almost like from dairy products - without casein / histamine risk
Calcium (Ca²⁺)	250-500 mg / L	Bone protection during immobility + nerve signals; without oxalate robbers in the way
Sulphate (SO₄²⁻)	<50 mg / L (as low as possible)	CAUTION: Sulfur trap : FQ-damaged microbiome often contains overgrown sulfate-reducing bacteria → convert sulfate directly into H ₂ S (hydrogen sulfide) → this gas blocks complex IV of the respiratory chain (identical target as cyanide!) and massively worsens the mitochondrial emergency flow. Organic sulfur for detoxification / joints is better obtained from broccoli, pak choi or, if necessary, MSM
Sodium (Na⁺)	<20 mg / L	Renal protection: Keep low to avoid osmotic stress on glomeruli. Recognizable by the imprint "Suitable for infant formula"
Carbonation	None (still / natural)	CAUTION: Mechanical MCAS trigger : CO ₂ bubbles physically stretch the stomach / intestinal wall → enough for unstable mast cells for degranulation, even without histamine in the drink

Bottle label summary: Still water with lots of HCO₃⁻ (>1000) + Mg (>100) + Ca (250-500), but minimal sulfate (<50) and sodium (<20). Drink throughout the day → continuous lactate buffering without stomach acid shock.

Warning: Avoid base powder with FQAD ([RED])

Alkaline powders seem to be the logical "fire brigade" against lactate hyperacidity - but in FQAD/MCAS they are counterproductive in most cases:

trap	Mechanism
Stomach acid trap	Carbonates first neutralize stomach acid → identical effect to PPIs. Without stomach acid: Mg ²⁺ absorption massively reduced (exactly the mineral that is needed most) + proteins are not pre-digested → fermentation in the intestine → dysbiosis → mast cell trigger
MCAS Citrate Trap	High-quality base powders use citrates (Mg citrate, K citrate). Problem: Citrates are often produced industrially by fermentation with <i>Aspergillus niger</i> (mold) → unstable mast cells often cause immediate attacks (burning nerves, brain fog). Safer forms: magnesium glycinate, malate, threonate
Kidney Sodium Trap	Cheap base powders consist largely of sodium hydrogen carbonate (baking soda) → sudden Na ⁺ flood → osmotic stress on kidney filters. Contraindicated in cases of nephrological involvement

Safe alternatives to lactate buffering:(1) Still mineral water with lots of HCO₃⁻ throughout the day (gentlest option). (2) Natural K⁺/Mg²⁺ buffers over broccoli, bok choy, sweet potatoes. (3) Pure monoprparations based on laboratory values (whole blood), no imprecise base mixtures. **Transdermal magnesium absorption: The gastrointestinal bypass ([YELLOW] individualized)**

Level of Evidence:[YELLOW] Biologically plausible. Transdermal Mg absorption has been described pharmacokinetically but has not been studied in FQAD studies.

Oral magnesium supplementation often reaches a mechanical limit in FQAD: the affected GI tract reacts to high doses of Mg with diarrhea → loss of electrolytes → irritation of the mucous membranes → counterproductive. Transdermal absorption (via the skin) bypasses the entire

digestive system: Magnesium reaches the tissue and bloodstream directly without triggering mast cells in the intestines or stressing the kidneys through oral overdose.

Parameters	Recommendation	FQAD justification
Magnesium form	Prefer magnesium chloride (e.g. Zechstein flakes)	Penetrates cells easily, no sulfate side effects. Magnesium sulfate (Epsom / Epsom salt) also supplies sulfate → potentially problematic in dysbiosis (sulfate-reducing bacteria → H ₂ S, see sulfur trap). MgCl ₂ safer for very sensitive patients
Water temperature	Max. body temperature (36-37°C), never hot	CAUTION: MCAS temperature trap: Heat → sudden vasodilation → mast cells in the skin massively degranulate → palpitations, flushing, itching, drop in blood pressure. Identical principle as with the sauna warning (see evaluation matrix)
Introduction	Foot baths instead of full bath (2-3 times / week, 20-30 min.)	Feet have large pores → excellent Mg absorption. No circulatory stress due to full bath immersion. Systematic replenishment of cellular stores without systemic heat triggers
Increase	Only after 2-4 weeks of problem-free foot baths: test a body-warm half bath (lower body only)	Low and slow principle: one variable at a time. With any deterioration (flushing, racing heart, exhaustion) → lower temperature or return to foot bath
Timing	In the evening before sleep	Mg acts as a natural NMDA antagonist → calms the overexcited nervous system → can improve sleep quality. Dual function: chelation repair + sleep aid

Practical procedure: Dissolve 3-5 tablespoons of magnesium chloride flakes in a foot bath with water at body temperature (36-37°C) → immerse your feet for 20-30 minutes → then rinse, do not rub off → go straight to bed. Complements, but does not replace, oral supplementation (magnesium glycinate according to laboratory values).

Paradoxical weakness reaction to transdermal magnesium - three mechanisms: > Some FQAD sufferers report massive, leaden muscle weakness instead of relief after rubbing with “magnesium oil” (highly concentrated MgCl₂ brine, ~31%). This is not a sign of intolerance, but can be explained by three converging physiological effects:

Mechanism	What happens	Why dangerous with FQAD
1. Osmotic mast cell shock	Highly concentrated brine creates massive osmotic pressure on the skin → draws water from the tissue → mast cells in the skin react to osmotic / mechanical stress with degranulation	Local histamine release → vasodilation → drop in blood pressure → blood flow to the muscles collapses → leaden weakness + exhaustion
2. Calcium blockade (loss of contraction)	Mg ²⁺ and Ca ²⁺ are direct opponents in the muscle: Ca ²⁺ = contraction, Mg ²⁺ = relaxation. Highly concentrated Mg directly on large muscle areas → floods cells with relaxation signal	Mg ²⁺ blocks the remaining Ca ²⁺ channels completely → muscle loses basic tension (tone) → massive, soft weakness. Often amplified in FQAD because ionic equilibrium is already shifted
3. Vasodilation in dysautonomia	Mg ²⁺ is a powerful vasodilator. Rubbing large areas (calves, thighs) → local vasodilation → blood pools in the legs	In dysautonomia / POTS, the ANS cannot narrow the vessels reflexively → brain + central muscles lose oxygen + glucose → systemic weakness + brain fog

Safe use despite weakness reaction — three adjustments:

Customization	Why	Practical
Dilute brine (1:1 or 1:2)	Lowers osmotic pressure below the mast cell degranulation threshold	Dilute the MgCl ₂ solution in a spray bottle at least 1:1 with distilled water
Only the soles of the feet, not large muscle areas	Soles of feet: thicker skin, less reactive mast cells, area too small for systemic drop in blood pressure. Pores still absorb Mg excellently → slow, systemic distribution	Never on calves / thighs / arms → only soles of feet or foot bath
Wash-off after 20 minutes	After ~20 minutes, the skin's maximum absorption capacity is reached; After that, salt only creates osmotic stress without additional absorption	Apply → lie down for 20 minutes (protect your blood pressure) → wash off completely with lukewarm water

The combination of dilution + soles + time limit bypasses the three mechanisms of weakness while maintaining therapeutic GI bypass.

311 nm narrowband UVB phototherapy: vitamin D synthesis without oral Mg depletion ([YELLOW]/[RED] with reservation)

Level of evidence:[YELLOW] Vitamin D synthesis through UVB is physiologically proven; 311 nm narrowband UVB is the dermatological standard (psoriasis, vitiligo). The specific use in FQAD/MCAS has not been studied. **Cave:** Medical UVB lamps are not wellness devices — improper use causes severe burns and MCAS flare-ups within minutes. Only after medical consultation.

Rationale: Oral vitamin D supplementation (drops/capsules) deprives the body of magnesium during activation - in FQAD with already depleted Mg²⁺ stores this can trigger paradoxical crashes. Cutaneous vitamin D synthesis by UVB bypasses this oral Mg withdrawal and supplies vitamin D directly via the physiological production chain in the skin.

Why 311 nm could be an advantage in FQAD: Vitamin D production peak is ~297 nm; 311 nm is less efficient by a factor of 2-3. In standard medicine this is a “disadvantage”. For MCAS patients, this inefficiency acts as a natural dampener — not an uncontrollable flood of vitamin D that overwhelms the liver with Mg deprivation.

Risk of phototoxicity after long-term FQAD: During acute FQ intake and in the months afterwards, UV phototoxicity is extremely high (see D5, front 2: photolytic C-F splitting). After a decade, the amount of intact, reactive FQ molecules in the upper layer of the skin is vanishingly small. The remaining risk is not the acute fluorine release, but the defenseless skin: blocked mitochondria → missing antioxidants (glutathione) → every UV light creates oxidative stress that the skin cells cannot buffer.

Rule	Implementation	FQAD justification
Eye protection (absolute)	Put on UV protection glasses before the device switches on and only take them off after switching it off. Eyelids alone are too thin	UVB burns cornea (photokeratitis) and promotes cataracts
Micro-dosing (time, not distance!)	Start: max. 30-45 seconds / area. Increase: +10-15 seconds every 2 days, only if there is zero redness. Limit: rarely >2-3 min./area. At minimal pink → immediately time step back	Burnt skin releases massive amounts of histamine → MCAS boost. Goal: Vitamin D pulse below the mast cell stimulation threshold
Spacers firmly on the skin	Never increase the distance to make it “softer”. Quadratic distance law: double the distance = only ¼ intensity → 10-15 minutes necessary → scattered light risk for eyes + massive oxidative surface stress	Fixed distance + short time = scalpel approach: controlled energy on the smallest area, rest of the body unaffected
Rotation of areas	Mon: stomach, Wed: right thigh, Fri: left thigh. Inside forearms as an alternative. Never the same place every day	Skin needs days to regenerate; Vitamin D is synthesized across the entire skin surface

Rule	Implementation	FQAD justification
Skin Prep	Naked, dry skin without creams / oils / soap residue. Never apply transdermal MgCl_2 to the same area before UVB (salt reflects / irritates) → foot bath in the evening, lamp on the stomach at midday	Avoid chemical reactions with foreign substances
Photodegradation limit	Longer irradiation $\neq$ more vitamin D. From a local maximum, pre-vitamin D is destroyed by further UVB into inactive substances (suprasterol) → zero yield + only oxidative stress	Biological safety valve of the skin; confirms micro-dosing strategy

MED titration (determine minimum erythema dose): 1. Test spot: Lamp with spacer directly on stomach → exactly 15 seconds 2. Wait 24 hours 3. Result: Minimally pink/itchy → dose was too high (reduce to 10 seconds). Completely pale → next time test for 20 seconds elsewhere 4. Goal: Find the maximum time at which zero redness occurs. Maintain this and rotate 2-3 different areas every day

Realistic Expectation: A 7-watt handheld lamp can only produce ~50-100 IU of vitamin D per 30-second area — far less than a full daily dose. This is not a disadvantage with FQAD, but rather intentional: crash-free microdoses that accumulate over weeks. Combination with natural sources of vitamin D (egg yolk, ghee, organic wild salmon) for additive effect. **This method is risky and requires absolute discipline.** If there is a lack of compliance (forgot safety glasses, inaccurate timing, increased distance), the risks outweigh the benefits. Only after consultation with a doctor and with laboratory-based vitamin D monitoring (25-OH-D3 every 8-12 weeks).

The calorie trick with FQAD: If the weight continues to fall: Never reduce protein or complex carbohydrates, but rather increase safe fats (ghee, coconut oil, olive oil, macadamia nut butter). Fats do not produce lactate, do not overtax the liver and provide 9 kcal/g (vs. 4 kcal/g for carbohydrate/protein).

Gastroparesis management in FQAD/MCAS ([YELLOW] individualized):

Background: Dysautonomia (vagal nerve block) → Stomach does not receive an electrical signal to contract → Food sits for hours, ferments → Reflux + MCAS trigger. The reflux is almost never “too much acid”, but rather a mechanical backlog + MMP-weakened sphincter.

Problem	Solution	Justification
Solid food remains lying down	Mince everything to the maximum: meat through a meat grinder (double minced), fish into pulp with a fork, all meals can be pureed. Chew 30-40 times → shift the mechanical work of the stomach into the mouth	Paralyzed stomach cannot knead fibers; Pre-digested food immediately offers a target for stomach acid and slips into the small intestine more quickly
Fat further slows down the stomach (CCK reflex)	MCT oil (C8 caprylic acid) instead of LCT fats: MCT is biochemically so short that the body does not recognize it as fat → hardly triggers CCK gastric brake, does not need bile acid, goes directly into the portal vein of the liver → ketone bodies as alternative energy	For gastroparesis: MCT oil as the primary source of calories; Dose LCT fats (ghee, nut butter) more carefully. Increase MCT dropwise (initially laxative)
Reflux when eating lying down	Never lie flat <3 hours after eating; wedge pillow at night; consistently lying on the left side (stomach entrance then above). No PPIs (see alkaline powder warning: Mg absorption blocked, dysbiosis fueled)	Gravity as a physical reflux brake. PPIs make the underlying problem worse in FQAD

Problem	Solution	Justification
Burning mucous membrane	Slippery Elm bark: Allow 1 teaspoon of powder to swell in 200 ml of still, lukewarm water → gel-like mucus that forms a fireproof protective film over the esophagus and stomach wall. Drink before eating	Histamine-free, mechanical mucous membrane protection: Mast cells are physically shielded from food stimuli. No chemical acid blockade necessary
Vagus Nerve Hack Before Meal	2-3 min. 4-7-8 breathing (4s in, 7s hold, 8s out) right before the first bite	Long exhalation physically stimulates the vagus nerve → briefly switches the system from stress to digestion mode
Loss of appetite	Calorie compression: liquid emulsions instead of solid meals. No appetite stimulants (bitter substances irritate leaky gut; sedating antihistamines increase brain fog)	Stretch receptors in the stomach are less triggered by liquids than by solids

Low histamine meat/fish preparation - four iron rules:

1.**Mechanical pre-digestion:** Beef from the organic butcher directly through the meat grinder (double minced); Chop the fish into mm pieces or mash it with a fork after cooking 2.**Flash cooking (2-4 minutes):** Fry minced/chopped protein only extremely briefly and hot → kills germs, leaves no time for enzymatic histamine to build up. No braising, no goulash, no bone broth 3.**Zero tolerance for leftovers:** Prepare exactly the portion that can be eaten immediately. No meal prep - refrigerator only slows down histamine formation. If you don't eat it straight away, throw it away or freeze it steaming hot 4.**Never packaged meat** from the plastic tray - histamine clock starts from slaughter

Warning: Commercial Drinking Food at FQAD ([RED] Avoid)

Standard drinking food (Fresubin, Fortimel, Ensure) is designed for tumor patients with intact mitochondria and is usually counterproductive in FQAD/MCAS: -**Maltodextrin/Sugar**(first ingredient): Massive glycolysis overload → lactate flood → PEM. Feeds misinhabitation (SIBO) -**Casein/Soy Protein:** Strong mast cell triggers for leaky gut -**Emulsifiers (carrageenan):** Used in research to specifically trigger intestinal inflammation in laboratory mice -**Synthetic vitamins:** Inactive forms (e.g. folic acid instead of methylfolate) that can hardly be converted during oxidative stress

Safe Alternative — Homemade Gastroparesis Bypass Drink (~800 kcal): 200 ml coconut milk (only coconut + water, no carrageenan) + 2 tbsp macadamia nut butter + 1 tsp MCT oil (C8, can be increased) + 1 tbsp wild frozen blueberries + pinch of rock salt → mix vigorously. **Don't drink all at once:** small sips every 15-20 minutes throughout the morning (gastroparesis hack). Optional: slippery elm bark as a base instead of water.

Warning: Growth Hormone (HGH) and Growth Factor Extracts in FQAD ([RED] Strictly Contraindicated)

Forcible stimulation of a blocked system is one of the most dangerous interventions in FQAD.

Risk	Mechanism
Energy Trap	HGH commands all cells to "Build and Grow!" → Cell division / protein biosynthesis devours gigantic amounts of ATP → blocked mitochondria cannot deliver → catastrophic PEM crash
Senescence Override	FQ causes DNA breaks → p53 sends damaged cells into senescence (cancer protection). HGH forces these DNA-damaged cells to divide → cancer brake overridden → apoptosis or uncontrolled proliferation
MMP feeding	Even if new collagen were produced → is immediately destroyed by upregulated MMPs as long as oxidative burn + ion deficiency persists
MCAS hormonal shock	Sudden HGH / IGF-1 flooding changes receptor density → in severe MCAS: massive, uncontrollable degranulations

The same applies to **egg extracts (YTE, laminins, FGF concentrates)**: Highly concentrated avian peptides = MCAS grenade (foreign protein trigger for leaky gut) + identical growth signal as HGH (cell forced to crash without ATP). Cave: Whole, cooked organic eggs are safe — stomach acid destroys the peptide structure (denaturation), and the egg then only provides amino acid building blocks, not growth instructions.

Warning: Therapeutic Peptides (BPC-157, TB-500, CJC-1295/Ipamorelin, Epithalon) in FQAD ([RED] Not recommended — no evidence, high risk)

Evidence level: [RED] No randomized controlled trials (RCTs) exist for any of these peptides in humans. All evidence is limited to in vitro and animal studies. FQAD forums nonetheless regularly share experience reports — frequently following the pattern: initial euphoric improvement, then reverse effect and forced discontinuation.

In FQAD communities, peptides are discussed as a “secret weapon” for tendon regeneration and mitochondrial repair. This model warns for five converging reasons:

Peptide	Claimed effect	FQAD-specific risk
BPC-157 (Body Protection Compound)	Tendon regeneration, angiogenesis, anti-inflammatory	(1) VEGF upregulation → in FQ-damaged tissue: neoangiogenesis into brittle, MMP-degraded collagen = structureless vessel formation in a decayed scaffold. (2) Immune modulation unclear → in MCAS: mast cell trigger from foreign peptide cannot be excluded. (3) Zero human long-term safety data
TB-500 (Thymosin Beta-4)	Wound healing, cell migration, anti-inflammatory	(1) Stimulates cell migration and proliferation → in DNA-damaged, senescent cells: division command to cells with defective mtDNA = copying the broken blueprint (identical problem as with HGH / PQQ). (2) Senescence override : p53 brake is bypassed → cancer risk with pre-damaged DNA
CJC-1295 / Ipamorelin (GH secretagogues)	Stimulate endogenous HGH release → muscle building, regeneration	Identical risk profile to direct HGH administration (see above), but delayed and less dose-controllable: energy trap (ATP lacking for build command), senescence override, MMP feeding, MCAS hormonal shock. The indirect pathway via the pituitary makes dosing harder to control than exogenous HGH
Epithalon (Epitalon, Ala-Glu-Asp-Gly)	Telomerase activation → telomere extension → “rejuvenation”	(1) Telomere extension without mito stabilization falls short (see telomere-p53-PGC-1 α axis, 1.11): cell with long telomeres + blocked mitochondria = can theoretically live forever but cannot function. (2) Epigenetic false start : with FQ-induced collagen switch (Type I→III), a cell forced to divide would read the wrong blueprint → more scar collagen instead of healthy tissue. (3) Cancer risk : telomerase activation in cells with FQ-induced DNA breaks = removing the cancer brake while DNA damage is ongoing

The typical forum pattern: “Initially euphoric, then reverse effect”

This pattern frequently reported in FQAD forums is mechanistically explained: (1) **Initial effect**: The peptide temporarily stimulates repair processes — the few remaining functional cells respond to the build command → subjective improvement. (2) **Reverse effect after days/weeks**: The systemic blockades (AMPK suppression, PGC-1 α shutdown, MMP overactivity, Fe-FQ₃ crystals in tissue) catch up with the system → ATP reserves exhausted → the stimulated repair processes collapse, but the metabolic cost (energy expenditure + cellular debris) has already been paid → net deterioration compared to baseline.

Fundamental rule for all anabolic/regenerative interventions in FQAD: Every “build command” (whether HGH, peptides, forced training, or senolytics) requires that (a) mitochondria can supply enough ATP to execute the command, (b) MMPs are under control (otherwise every new collagen is immediately destroyed again), and (c) the epigenetic blueprint is read correctly (otherwise scar rather than tissue is produced). Without these three prerequisites, any anabolic stimulus in FQAD is a net loss.

Hidden food triggers in FQAD — four risk groups:

When mitochondria are blocked and mineral stores are depleted, the body does not forgive nutritional errors. Certain everyday foods act like chemical radical amplifiers because they hit the pathobiochemical wounds:

Trigger group	Examples	Pathobiochemical mechanism	Typical symptoms
Excitotoxins	Monosodium glutamate (yeast extract, ready meals), aspartame	Glutamate is the direct opponent of GABA. With FQ-blocked GABA receptors → neuronal overheating (excitotoxicity) [5]	Panic attacks, insomnia, burning nerves, muscle twitches
Quick Sugars	Table sugar, white flour, high glycemic fruit in large quantities	Forces massive glycolysis throughput in emergency power → sudden flood of lactate that overwhelms liver clearance [35]	Lead fatigue (PEM), muscle heaviness, brain fog within 1-2 hours
Mineral Robbers	Foods rich in oxalates (spinach, beets, almonds, dark chocolate); Fluoride sources (black / green tea, fluoridated salt)	Oxalic acid binds Mg^{2+}/Ca^{2+} in the intestines and blood → deprives the system of its last reserves. Free fluoride competes with Mg^{2+} and disrupts enzymatic processes [51]	Muscle cramps, tendon pain, worsening after “healthy” smoothies
Histamine liberators	Aged cheese, red wine, fermented foods, tomatoes, strawberries, citrus fruits, conventional pork	For leaky gut + unstable mast cells: histamine supply or mast cell degranulation → MCAS surge [18]-[20]	Cardiac arrhythmia, drop in blood pressure, joint swelling, burning skin

Fluoroquinolone residues in conventional meat:

In factory farming, enrofloxacin (a veterinary fluoroquinolone) is used prophylactically in drinking water. The primary metabolite of enrofloxacin in animals is exactly ciprofloxacin. Measurable residues of intact ciprofloxacin can be found in slaughter tissue. Legal limit values (MRLs) guarantee safety for healthy people, but do not guarantee 100% freedom from the molecule.

For those affected by FQAD, these microgram amounts can be critical: the system balances on the energetic precipice; Even traces of the intact ciprofloxacin molecule can create a new oxidative spark in previously damaged cells.

Special danger - bone broth from conventional farming: Ion trapping (intracellular FQ storage in bones/cartilage) occurs in animals just as it does in humans. Boiling these bones extracts the deep-seated, heat-stable fluoroquinolone as a highly concentrated “highly concentrated FQ eluate”. Despite the best intentions (collagen buildup), bone broth from conventional animal husbandry can trigger the most severe attacks. Alternative: bones from organic farming (no FQ use) or plant-based collagen building blocks. Evidence status: FQ residues in meat have been documented in regulatory and analytical terms. The causal relationship between microdose residues and FQAD relapses is biologically plausible (cumulative damage + loss of tolerance) but has not been examined in controlled studies. - Sleep optimization as a basic pillar - Very careful progressive activation: only if tolerability is documented, symptom-guided - Standardized graded exercise protocols are potentially contraindicated in FQAD with PEM propensity [23] - With ME/CFS/PEM-like pattern: Pacing has absolute priority over training

C2 — Mitochondrial modulation (study context only)

aspect	Description
Target	Improve mitochondrial quality control
Clinical Status	Research field without clinically established significance in FQAD [1], [24]
Major Uncertainty	Which patients benefit; what dose; which tissues can be reached
Endpoint	Stage 3 research markers, clinical symptoms
Risk	Any deterioration → immediate stop

Mitophagy/stress axis modulation is currently a research field with no clinically established role in FQAD. Biologically conceivable approaches (none validated): careful mitophagy modulation, redox stress reduction, indirect promotion of mitochondrial biogenesis [24], [26].

Safety note - Risk of apoptosis: Based on general cell biology, the following theoretical risk assessment results: An isolated pharmacological increase in mitochondrial turnover without an understanding of the parallel biogenesis could be detrimental; Direct data on this are lacking for FQAD [24], [27].

3.4 Evaluation matrix: Popular interventions vs. FQAD mechanics

Evaluation Matrix: Popular Interventions vs. FQAD Mechanics

FQAD is an acquired, toxic blockade — not analogous to congenital mitochondrial diseases. What can help with a genetic defect can be dangerous with a toxically blocked cell [7], [8]. The idea of repairing a fragile cell system blocked by toxins by “pressure refueling” with high-dose infusions is neither scientifically proven nor safe. Community reports of severe deterioration and death following unmonitored off-label therapies underscore this warning.

Infusions and oxidative therapies

Intervention	Target (forums / alternative medicine)	FQAD mechanics according to model	Risk assessment
Glutathione infusions (high dose)	Replenish Cellular Antioxidant, Scavenge Radicals	GSH / GSSG quotient often shows massive overtaxing of the antioxidant system [4], [34]	Mechanistically plausible, but as a high-dose infusion with blocked emergency power metabolism there is a risk of derailment. Serious crashes and deaths reported in the community. Can only be discussed with close monitoring and biomarker support - never as self-therapy.
Ozone therapy (autologous blood)	Oxidative stimulus for immune activation	FQAD cells are already destroyed by uncontrolled oxidative stress (Fenton reaction) [4], [34]	Potentially dangerous. Adding another strong oxidant to a system whose main damage comes from reactive oxygen species contradicts the goal of stabilization.

Intervention	Target (forums / alternative medicine)	FQAD mechanics according to model	Risk assessment
Hyperbaric Oxygen Therapy (HBOT)	Drastically increase oxygen saturation in the tissue	FQs block complexes I and IV of the ETC [2]. Oxygen that cannot be processed in the cell generates hydroxyl radicals via free iron	[YELLOW]/[RED] Double-edged risk. Theoretical risk: more O ₂ in a system with blocked ETC + free iron (Fe-FQ) could amplify the oxidative hit and mtDNA destruction via the Fenton reaction. Theoretical benefit: HBOT can promote angiogenesis, stem-cell mobilization and inflammation modulation; some patients report anecdotal improvement. Caution protocol if attempting: very slow pressure approach instead of immediate target pressure — first session as a short trial (e.g. 15 min, low pressure), then symptom-guided increase over many sessions. Any deterioration (crash, nerve burning) → immediate stop. Optimize antioxidant status (glutathione) beforehand. No HBOT in the highly active acute phase with ongoing oxidative fire.
High-dose vitamin C infusions	Antioxidant effect, collagen cofactor	Collagen damage through MMP upregulation is a well-supported axis of damage [13]	Plausible as moderate support; However, mega doses via vein can overwhelm the fragile system. Oral, moderate intake via diet is safer (see phase C1).

Vitamins and supplements — basic supplies

Intervention	Target	FQAD mechanics	Risk assessment
B12, B complex (infusion / oral)	Nerve regeneration, energy cofactors	B12 / folate are part of the basic diagnosis level 1; Deficiencies are aggravating factors [1]	Very useful for laboratory-confirmed defects. Blind high-dose injections without blood control are incalculable.
Vitamin D + K2 + Magnesium	Immune modulation, bone protection, calcium control	Vitamin D is standard parameter in stage 1; Mg is part of the basic diagnostics	Only refill in a laboratory manner. Vitamin D does not drive calcium into diseased cells, but rather promotes absorption from the intestines into the blood. K2 directs calcium to the bones instead of soft tissue. Mg blocks uncontrolled calcium influx into cells (see calcium / magnesium section, organ axis 3). Mega doses without monitoring (e.g. Coimbra Protocol) are not supported in the model.
Molecular Hydrogen (H₂)	Selective neutralization of hydroxyl radicals	Hydroxyl radicals (Fenton reaction) are the main cause of mtDNA destruction [4], [34]	Mechanistically plausible, but no FQAD data whatsoever.

“Mito-Boosters” from forums — the dangerous trinity

What works preventatively in a healthy cell can cause massive damage in a toxically blocked system. The document warns against “forcible stimulation” of a blocked system.

Intervention	Forum theory	FQAD-specific risk	Risk assessment
Coenzyme Q10 (Ubiquinol)	Transports electrons in the respiratory chain; is said to increase ATP production	FQs block complexes I and IV [2]. More Q10 = more electrons on a closed road → electron buildup → radical formation (superoxide) instead of ATP	Can theoretically accelerate oxidative cell death instead of stopping it. Only consider if there is a laboratory-confirmed Q10 deficiency and under close monitoring; no blind high-dose supplementation.
PQQ (pyrrolo-quinoline quinone)	Stimulates mitochondrial biogenesis; is intended to force cells to build new mitochondria	The nuclear damage in FQAD is often a defective blueprint (damaged mtDNA). In the worst case, PQQ-forced proliferation copies the broken blueprint → even more dysfunctional power plants	Danger of energetic exhaustion without functional gain. The model prioritizes first cleanup (mitophagy), then careful rebuilding — not blind growth.
Alpha Lipoic Acid (ALA)	super antioxidant; recycles used glutathione	Two FQAD-specific pitfalls: (1) ALA lowers blood sugar — in cells that rely 100% on glycolysis emergency power, this can trigger an immediate metabolic crash (PEM) [35]. (2) ALA contains a lot of sulfur (thiol) - with MCAS / dysbiosis, many sufferers react to isolated sulfur with inflammatory attacks and nerve burning.	Double risk: hypoglycemia + MCAS trigger. Don't supplement blindly; Before Consideration: Check blood glucose stability and MCAS status.

Basic rule: A toxically blocked system does not need “coffee” (Q10/PQQ/ALA), but rather stabilization, diagnostics and bottleneck elimination first. Before taking Mito-Booster, the model requires objective laboratory evidence of a deficiency (e.g. Q10 in whole blood, ATP in leukocytes - both special diagnostics/IGeL).

Polyphenol concentrates — the triple trap

High-dose polyphenol capsules (quercetin, resveratrol, curcumin, green tea extract, fisetin) are celebrated in forums as “cell rejuvenators”. With FQAD, the concentrated form poses three specific risks:

Risk	Mechanism	Consequence
Senolytic effect	As senolytics, quercetin / fisetin kill masses of senescent cells [28]	Gigantic amounts of cell waste in a body whose liver is already overwhelmed with lactate clearance → week-long toxic crash
MCAS / DAO inhibition	Curcumin, green tea extract and berry extracts contain salicylates or inhibit the DAO enzyme in the intestine	Histamine breakdown is blocked → violent mast cell surges (brain fog, skin burning, dizziness) in MCAS sufferers
Pro-oxidative poultice	In the highly concentrated doses of NEM capsules, polyphenols switch from antioxidant to pro-oxidant Increase in oxidative stress instead of reduction; In tissue damaged by the Fenton reaction, this can accelerate collagen breakdown	

Alternative: Absorb polyphenols through the natural food matrix (blueberries, dark grapes, olive oil), where the body processes them slowly and safely - no concentrated capsules.

Metabolic and physical interventions

Intervention	Target	FQAD mechanics	Risk assessment
Medicinal fasting / long interval fasting	Enforce autophagy / mitophagy	FQAD patients are running on backup power (glycolysis); When there is a lack of energy, the body burns muscle tissue (ammonia production) [4], [35]	High risk. The model strictly warns against catabolic stress [21]. Accidental sarcopenia impairs regeneration.
Sport / Graded Exercise (GET)	Building muscle, increasing stress	Stress creates a massive backlog of lactate in front of blocked mitochondria → local hyperacidification → PEM [23], [35]	Contraindicated if you have a tendency to PEM. Standardized training protocols can trigger setbacks that last for months. Strict pacing takes priority.
Therapeutic Ketosis	Glycolysis bypass, ketones as alternative energy	BHB bypasses blocked PDH, acts as an HDAC inhibitor and PGC-1α inducer [36]	Important research hypothesis for deep remission (see Section XIV). However, self-experimentation is risky (weight loss, dyslipidemia, kidney strain); only responsible in a monitored study context.
Red / Near Infrared Light (Photobiomodulation)	Stimulation of cytochrome c oxidase (complex IV)	FQs disrupt exactly this complex IV [2]	Risk unclear. Forced stimulation of toxically blocked machinery could increase electron leakage. Not rated as safe in the model.

Thermal and environmental stressors

Intervention	Physiological effect	FQAD-specific risk	Assessment
Finnish sauna (80-100°C)	Massive vasodilation, sweating, circulatory strain, high ATP consumption for thermoregulation	Dysautonomia: blood pooling, tachycardia, risk of collapse [22]. Dehydration: strictly contradicts the stabilization phase [21]. SFN: defective sweat gland innervation → overheating. PEM triggers due to energy consumption [23], [35]	High risk with unstable course. Mild alternative: local heat (hot water bottle) or low-temperature infrared cabin (~35°C).
Ice bathing / Wim Hof method	Massive sympathetic shock (adrenaline / norepinephrine), extreme energy requirements for thermogenesis, cortisol release	Energy collapse: sudden lactate production when the respiratory chain is blocked → PEM crash [35]. Sympathetic shock: if the ANS is miscontrolled, extreme cold stimuli can trigger POTS attacks, rapid heartbeat, and nerve pain [22]. HPA stress: avoid catabolic stress [17], [21]	Very risky. Light Kneipp waterings (lower leg only, short) are the safer alternative if there is no PEM and stable autonomy.
Kneipp showers / contrast showers (mild)	Moderate vascular training stimulus	More tolerable than extreme stimuli; with stable autonomies and no tendency to PEM, plausibly for gentle improvement of vascular tone	Only if tolerated is documented. Discontinue if you have a racing heart, dizziness or subsequent exhaustion.

Basic rule for all environmental stimuli: As soon as a thermal stimulus (heat or cold) leads to days of exhaustion, muscle pain, palpitations or aggravation of neuropathy, the autonomic or metabolic stress limit has been exceeded. The principle of “low and slow” applies to temperature as well as physical stress.

Pharmacological and procedural approaches discussed in forums ([YELLOW]/[RED])

Evidence level: [YELLOW]/[RED] The following approaches are widely discussed in patient forums and partly offered experimentally by physicians. They are listed here with pros and cons — not as a recommendation, but for transparent classification. No FQAD-specific RCT exists for any of them.

Approach	Rationale / possible pro	FQAD-specific con / risk	Assessment
Low-dose doxycycline (e.g. 20 mg 2×/day, ~12 weeks)	Subantimicrobial doxycycline is an established MMP inhibitor (approved for periodontitis / rosacea). Since MMP escalation is a core mechanism of FQAD tissue destruction (see 1.9), targeted MMP inhibition could slow collagen breakdown [81]. Offered experimentally by some physicians	Double problem: (1) Doxycycline inhibits MMPs partly through zinc chelation (Zn^{2+} is an MMP cofactor) → may further lower the already-critical zinc status in FQAD → paradoxically also weakens TIMPs and SOD. (2) It is itself an antibiotic with mitochondrial affinity (tetracyclines inhibit mitochondrial protein synthesis). Not trivial in a mito-damaged system	[RED] Only considerable under medical supervision with zinc monitoring. The MMP-inhibition benefit must be weighed against zinc depletion and mitochondrial load
INUSpheres / therapeutic apheresis (“blood washing”)	Filters lipophilic toxins, cytokines and immune complexes from the blood. Patients report consistent but temporary improvement	The temporariness is diagnostically informative: it is consistent with Hypothesis 2 (tissue-trapped Fe-FQ slowly leaching back into the blood) and Hypothesis 5 (co-load). But: high cost, invasive, effect often lasts only weeks	[YELLOW] Mechanistically interesting as a hint of a circulating component; questionable as long-term therapy due to temporariness and cost. May support Hypothesis 2 / 5
High-dose NAC (+ glycine) (GlyNAC, ~50–100 mg / kg)	N-acetylcysteine is a glutathione precursor; GlyNAC (glycine + NAC) markedly improves mitochondrial markers and glutathione status in studies. Anecdotal + MMD lab-test improvement reported	Sulfur / sulfite sensitivity: in the sulfur-metabolism weakness common in FQAD (CBS / SUOX), high-dose NAC can trigger sulfite load and MCAS flares. Glycine is an NMDA co-agonist (see glycine paradox) → use with caution when the GABA brake is blocked	[YELLOW] Plausible glutathione lever; test only by slow titration (toothpick method). Unsuitable with sulfur sensitivity
Detox (environmental toxins + FQ elimination)	Follows directly from Hypothesis 5: reducing a pre-existing co-load (heavy metals) and promoting excretion pathways could relieve the overall system	Aggressive chelation therapies (DMSA, EDTA) are high-risk in an unstable, mineral-depleted FQAD system (additionally remove Mg^{2+}/Zn^{2+}). “Gentle” detox concepts are often unproven	[RED] Targeted diagnostics (heavy metal status) before any detox measure. Avoid aggressive chelators. Sensible only with proven co-load

Important diagnostic note — whole blood/RBC instead of serum: For the lab-guided mineral and vitamin optimization recommended throughout this model: serum/plasma values measure only the extracellular space and can appear normal despite intracellular deficiency. For magnesium, zinc, selenium and potassium, **whole-blood or erythrocyte analysis (RBC, intracellular)** is considerably more informative and should — where available — be preferred over serum measurement. This is already noted in several places in the model (see 1.12, mineral water section) and applies generally.

Quantifying the damage: Why a percentage is not possible

A common patient question is: “What percentage of my cells are damaged?” The model could explain why this is currently unanswerable:

-No validated overall biomarker exists that reliably measures the proportion of damaged cells in the body [6]. **-Mosaic Pattern:** Fluoroquinolones do not damage every cell equally. Depending on the endophenotype, tendon cells, small nerve fibers, podocytes or muscle cells are primarily affected - to varying degrees [1], [6].

Instead of absolute cell numbers, the model requires the measurement of the **loss of function** per organ axis:

Method	What it shows	Practical significance
PEM threshold (everyday observation)	From what duration / intensity of stress does deterioration follow? Best everyday indicator of the extent of mitochondrial blockage; the lower the threshold, the more pronounced the functional failure	
Skin punch biopsy (IENF density)	Actual thinning of small nerve fibers	Quantitative structural evidence of the neurological damage axis
Lactate-pyruvate quotient (exposure)	Collapse of hepatic lactate clearance under load	Functional evidence of the metabolic emergency power shift
Ultrasonic elastography (tendon)	Inhomogeneity, cracks, thickening of collagen	Structural evidence of tendinous degeneration
6-minute walk test / grip strength	Everyday stress reserve	Comparative value for progress observation and regeneration monitoring

Looking for an absolute percentage is not helpful. Therapy and stabilization are based on the clinical behavior and the measurable endpoints of the affected organs (see Recovery Framework, Section XII).

Basic principle

The fundamental difference between forum trends and the clinical-translational model: Forums often attempt to “force” healing (through high-dose infusions, aggressive fasting or oxygen pressure). The model positions itself in opposite directions: Goal number 1 is to stabilize damage and prevent further decompensation, not aggressive stimulation of diseased cells.

Before any experimental intervention, the following applies: Objective diagnostics and phenotyping (Phase B) → laboratory-guided correction of defects → stabilization → only then, under medical supervision, consider cautious translationally plausible approaches.

Commercial “Regenerative Medicine” Offerings for FQAD — Evaluation Framework

Desperate patients are increasingly encountering commercial clinics offering specific FQAD treatment programs — typically combinations of IV vitamin infusions, diet/supplement protocols, and “stem cell therapy” or “cellular therapy.” Such programs are advertised with success stories, often cost five to six-figure amounts and often require serial or multi-week treatments.

Core Medical Review: Marketing and testimonials are not the same as robust evidence. The fact that a practice takes FQAD seriously and offers a plausible recovery narrative is not evidence of effectiveness, safety, or suitability for specific subgroups. The FDA specifically warns that many regenerative/stem cell-based offerings are not approved and have not been proven safe or effective for numerous conditions [9]. The ISSCR (International Society for Stem Cell Research) emphasizes that with commercial stem cell therapies it is often not even clear which cells are given in what quality, dose and processing. **Typical problem structure:** - Several different elements are offered together (IV nutrients, diet, supplements, cell therapy) → in the end it is unclear which part will work if improvement occurs - Strong success narratives, but no transparent published data with inclusion/exclusion criteria, endpoints, side effects and follow-up - No registered study protocol, no peer-reviewed clinical outcome publication for FQAD - Treatment is advertised for an area in which there is real medical damage, but no established causal therapy exists - this is precisely where patients are particularly vulnerable

Benefit-Risk Assessment:

Dimension	Review
Potential Benefit	Theoretically present (cell / secretome / supportive approaches not biologically absurd), but not reliably proven for FQAD
Medical risk	Real and difficult to calculate due to lack of transparency; for stem cell products: infections, immune reactions, tumor risk cannot be ruled out
Financial risk	High; Serial treatments with uncertain effect
Overall verdict	If there is no study protocol, no publication and no safety documentation: benefit-risk ratio unclear to unfavorable

The fair counterposition: It is possible that individual patients benefit from such programs. But without clean data you don't know who, why, how and at what price. Individual success stories are not enough to classify a therapy as sensibly calculable.

Checklist: 7 questions before making a decision

For security reasons, anyone seriously considering such an offer should ask at least the following questions — and insist on clear, verifiable answers:

1. **What cell source is used?** Autologous/allogeneic? MSCs, secretome or something else? Which quality standards apply? 2. **Is there a standard written protocol?** With defined doses, timing and follow-up? 3. **Which side effects/deteriorations are systematically documented?** 4. **Is there published or properly anonymized outcome data specifically for FQAD?** (Not just testimonials, but data with endpoints and follow-up) 5. **Which patients are rejected** because the risk is too high? (If no one gets rejected, that's a warning sign) 6. **What is the plan if things get worse?** Is there an emergency/abort protocol? 7. **Which parts of the program are evidence-based and which are purely experimental?**

If there are no clear, verifiable answers to these questions, that is a strong warning sign.

3.5 Experimental Technologies ([RED] Research Only)**Long-term experimental technologies with lack of clinical connectivity****A. Senotherapeutics**

Senolytics (dasatinib+quercetin) can reduce senescence markers in initial human studies [28]. Whether FQAD is driven by pathological senescence is unclear. Serious use is only conceivable after FQAD-specific biomarkers and safety data [28].

B. Mitochondrial cell and transfer therapy (MAT)

MAT is in early trials for primary mitochondrial diseases [29], [30]. Target tissue detection, efficacy and safety data are lacking for FQAD [30], [31].

C. Biogenesis and stress networks as hypothetical reprogramming targets

The idea of targeting the regeneration network (PGC-1 α /NRF1/ERR α /AMPK/SIRT1) is conceptually attractive. There are no FQAD data and no tissue-specific active substances [24].

3.6 Mitochondrial medicine as a translational context ([RED] Research only)

Developments in mitochondrial medicine as a translational context

Disclaimer

The following approvals do not justify transferability to FQAD, but merely demonstrate the principal pharmacological addressability of mitochondrial processes. Barth syndrome and TK2 deficiency are monogenetic diseases with a defined structural defect; FQAD is an acquired, multi-system toxicity with different pathogenesis. Elamipretide data are based on very small, selected populations; transferability is even being discussed within the Mito community [7], [8].

Elamipretide(Stealth BioTherapeutics): First FDA approval for a mitochondrial drug (September 2025, Barth syndrome). Stabilizes cardiolipin [7].

Kygevvi(UCB): FDA approval for TK2 deficiency (November 2025). Restores mtDNA copy numbers [8].

USP30 Inhibitors(Vincere Biosciences): \$5 million grant from MJFF, Phase I planned for 2026 [27].

Mayo Clinic: Ongoing study of biomarkers for FQ-induced mitochondrial toxicity [32].

Target hierarchy

Clinically justifiable today (A, B, C1): Stabilization, crash protocol, exclusion diagnostics, organ phenotyping, parallel organ-related therapy, careful re-conditioning with protein cycling and pacing. **Translationally plausible — only study context (C2):** Mitophagy/stress axis modulation, only if deficit is proven, under biomarker support, with apoptosis caution.

Conceivable in the long term (Section VII): Senotherapeutics, MAT, biogenesis reprogramming - each only after FQAD-specific validation.

Not seriously justifiable today: A promise to derive a reliable healing plan from this. Likewise: off-label self-medication.

3.7 The curative horizon ([RED] research vision)

The Curative Horizon: Hypothetical Strategies for Deep Remission (Research Vision)

Disclaimer: The following concepts are speculative research visions — not a clinical suggestion. Any strategy would need to demonstrate safety, targeting, and tissue-specific benefit before conceivable FQAD application. The presentation serves to structure future research programs.

XIV.1 Intracellular regeneration

Strategy	Promise	Biggest security problem	Minimal evidence before human studies
A. Therapeutic ketosis	BHB bypasses blocked PDH, acts as an HDAC inhibitor and PGC-1α inducer [36]	weight loss, dyslipidemia, kidney strain; In severe mitoblockade, ketone oxidation may also be restricted	FQAD-specific pilot study with metabolic monitoring
B. Mito genome editing	DdCBEs / mitoTALENs: heteroplasmy shift in favor of healthy mtDNA [37]	Off-target sections, mosaicism, potential tumorigenesis	Animal model with FQ-induced mito-dysfunction
C. Epigenetic Unlocking	dCas9-TET1: demethylation of the PGC-1α promoter [38]	Off-target demethylation, unintentional gene activation, germline risk	In vitro detection in FQAD-relevant cell types

Strategy	Promise	Biggest security problem	Minimal evidence before human studies
D. Transient OSKM reprogramming	Epigenetic rejuvenation of senescent cells without stem cell conversion [39]	Teratoma / carcinoma development in case of incorrect / overdose; no humane safety profile	Long-term safety data in primate models
E. LNP-Mito transfer	Cell-free, scalable mitochondrial exchange via lipid nanoparticles [40]	Immunogenicity, targeting imprecision, unknown persistence	Tissue-specific targeting demonstrated in vivo

Ketogenic diets and exogenous ketones can only be used in seriously ill patients in the context of prospective studies in nutritional medicine and nephrology; Spontaneous self-experiments should clearly be avoided due to weight, lipid and kidney risks [36]. Any form of protein or calorie reduction (see mTOR hypothesis [25]) must ensure the preservation of muscle mass; unintentional sarcopenia would further impair regenerative ability.

XIV.2 Extracellular and systemic regeneration

Strategy	Promise	Biggest security problem	Minimal evidence before human studies
F. Lysosomotropic Nano-Scavenger	Tailored nanoparticles that penetrate acidic lysosomes, bind persistent FQ complexes and force exocytosis [43]	Non-specific lysosome disorder, toxic degradation products	Detection of FQ persistence in human cells; In vitro clearance
G. Autonomous hardware reset (SGB / bioelectronics)	Ganglion-stellate blockade disrupts chronic sympathetic loops; Closed-loop vagus nerve microstimulators modulate inflammation in real time [44]	SGB: vascular / nerve injury; Bioelectronics: implantation risks, unknown long-term effects	SGB: Case series in FQAD dysautonomia; Bioelectronics: Safety in Chronic Inflammation Models
H. ECM reprogramming (siRNA hydrogels)	Biomimetic hydrogels physically bridge tendon tears; embedded siRNA silences MMP expression [45]	Off-target gene silencing, immune response to hydrogel, local inflammation	Animal model of FQ-induced tendinopathy with proof of function
I. Microbiome editing (phages + postbiotics)	Tailored bacteriophages specifically eliminate pro-inflammatory species; synthetic postbiotics (SCFA in nanoparticles) seal the intestinal barrier [46]	Phage resistance, dysbiosis worsening, immune stimulation	Microbiome Profiling in FQAD; controlled phage study in dysbiosis
J. Endothelial regeneration	EPC cell therapy, VEGF modulation or NO signal rebalancing to normalize microcirculation [41]	Neoangiogenesis risk, tumor promotion, systemic hemodynamic effects	Evidence of capillary rarefaction in FQAD
K. Stem cell niche repair and tissue engineering	Local MSC therapy or tendon-specific scaffold implants with mito-support to restore tissue regeneration competence [41]	Cell transformation, immune rejection, scaffold degradation	Stem cell niche analysis in FQAD tissue

XIV.3 Systemic coordination and in silico planning

Adaptive sequencing: Instead of a fixed phase plan: small biomarker panels per subgroup, serial monitoring, clear termination criteria, n-of-1 designs in study centers. Intervention blocks (turnover → biogenesis → senescence) are dynamically sequenced based on biomarker responses — as a study algorithm, not a clinical plan [42].

Digital twin models: Patient-specific computer models on which curative sequences are first run through in silico to simulate risk/benefit profiles before initial n-of-1 curative trial protocols are considered in strictly controlled settings [42].

Clinical prioritization matrix for research approaches

criterion	Highest priority	Medium priority	distant horizon
Threatens organ function?	Nephroprotection, crash prevention	SFN stabilization, tendon management	Systemic reprogramming
Is everyday life the most limiting?	PEM / exercise intolerance, dysautonomia	Neuropathic pain, sleep	Cognitive optimization
Reversible?	Dysbiosis, endocrine axis	Mitochondrial re-conditioning	Senescence, mtDNA damage
Measurable?	L / P ratio, eGFR, walking distance	HRV, SFN biopsy, ultrasound	Heteroplasmy, senescence markers
Can be approached with little risk?	Pacing, nutrition, sleep	Ketosis (monitored), phages	Genome editing, OSKM

Five-stage model of theoretical complete remission

1.**Trigger-Free / Stabilization**— No further exposure, avoid catabolic stress 2.**Objective calming of the dominant organ axis**— measurable at the end points from Section XII 3.**Regaining exercise reserve**— documented increase in VO2max, walking distance, crash freedom 4.**Regression of the chronification mechanisms**— only measurable if validated biomarkers are available 5.**Long-term stable re-integration without crash**— at least 12 months of stable function under normal everyday stress

3.8 Practical consequences for care today

Practical consequences for care today

- 1.**FQAD remains a clinical probabilistic diagnosis without a gold standard.**Diagnosis is based on exposure history, clinical pattern, and exclusion of competing explanations.
- 2.**Care must be multi-organ and symptom-oriented.**Parallel, tissue-specific treatment pathways depending on the leading endophenotype. Multidisciplinary care is desirable.
- 3.**Experimental approaches do not belong in routine.**All translational and speculative modules (C2, VII, XIV) are research agenda.

Supply classification: What belongs where?

Category	Measures	Setting
Routine (today, family doctor / specialist)	Basic diagnostics level 1, correction of deficiencies (B12, D, Mg), pacing, sleep optimization, low-histamine diet for suspected MCAS, crash protocol, beta-blockers for POTS, physiotherapy (FQAD-adapted), neuropsychiatric co-care	Standard care
Special Center	Skin punch biopsy (SFN), advanced autonomic testing, tilt table, daily cortisol profile, GPCR autoantibodies, polysomnography, eccentric stress therapy with ultrasound monitoring, mast cell stabilizer titration	Neurology, gastroenterology, rheumatology, sleep medicine

Category	Measures	Setting
Only in studies / research	C2 modules (mitophagy modulation), senolytics, MAT, genome editing, OSKM reprogramming, phage therapy, LNP mito transfer, siRNA hydrogels, therapeutic ketosis (monitored), multi-omics profiling	University study centers

N-of-1 study logic: How to seriously test hypotheses on individual cases

In the absence of FQAD RCTs, individualized n-of-1 studies under strict medical supervision can provide an intermediate step to translation. The basic protocol:

step	Content	Safety rule
1. Baseline	2-4 weeks documentation of endpoints (crash frequency, walking distance, VAS pain, SF-36, if necessary L / P ratio) under stable conditions	No changes during baseline
2. Intervention	A single, narrowly defined intervention (e.g., low-histamine diet, DAO supplementation, eccentric training, or — in the study context — urolithin A)	Lowest dose; change only one variable at a time
3. Monitoring	Weekly endpoint recording over a defined period (4-12 weeks depending on the measure)	Clear termination criteria: any significant deterioration → immediate stop
4. Washout	2-4 weeks return to baseline, measure endpoints again	Only if medically justifiable
5. Retest	Repetition of the intervention for reproducibility	Only if the result is positive in round 1

This scheme is much closer to real translation than great promises for the future and allows those affected and their doctors to check under controlled conditions whether a particular measure helps individually - without uncontrolled experimentation.

3.9 Complementary therapies, devices and care requirements ([YELLOW]/[RED] individualized)

All of the following measures are biologically plausible but have not been validated in FQAD studies. They do not replace medical treatment. “Low and Slow” — one variable at a time.

Natural Sun Strategy: UV Index Based Protocol ([YELLOW])

In addition to vitamin D, the natural sun provides three other therapeutic signals: (1) serotonin → melatonin chain (the body’s strongest mitochondrial antioxidant at night), (2) nitric oxide (NO) via UVA → vasodilation → improved microcirculation in tendons/nerves, (3) circadian vagal clock (morning light → hypothalamus →). Vagus nerve starts digestion).

UV index	Safe time (skin type I-II)	FQAD remark
0-2 (Winter / March)	30-60 mins	Biologically inefficient; Cooling down is often more harmful than beneficial → prefer 311 nm lamp indoors
3-5 (Spring)	10-15 mins	“Goldilocks” range: sufficient for vitamin D impulse with minimal oxidative stress
6-7 (Summer)	3-6 mins	Very effective; A tea break on the balcony is enough

UV index	Safe time (skin type I-II)	FQAD remark
8-10 (mid-summer)	1-3 mins	"Flash tanning", then immediately shade

Three rules: (1) Shadow law: If your own shadow is longer than your height → UVB too weak (best time: 11 a.m. – 3 p.m.). (2) Don't wait for redness - "silent" production. (3) Antioxidant flood: blueberry drink 30-60 minutes before sun exposure (anthocyanins scavenge ROS in the skin).

Morning light (no UVB necessary): Within 30 minutes of waking up, go to the window/balcony for 5-10 minutes without sunglasses → Serotonin + vagal timing for gastroparesis.

Photobiomodulation: red light/near infrared without heat ([YELLOW] with reservations)

Parameters	Recommendation	FQAD justification
Wavelengths	660 nm (deep red, skin / superficial inflammation) + 850 nm (NIR, deep tissue / tendons / stomach)	Absorbed by cytochrome c oxidase (complex IV) → displaces blocking NO → oxygen flows → ATP production increases. Cave: FQs disrupt exactly complex IV — forced stimulation during acute blockade could increase electron leakage
Device Type	Cold LED panels / masks, not IR saunas (heat = MCAS trigger)	Saunas: histamine surge + electrolyte depletion (Mg / K/Na) + blood from stomach into skin (worsens gastroparesis)
Dose	Start: 3-5 minutes from 30 cm. 24 hour observation. Pay attention to the biphasic dose curve: too much = oxidative stress	Find the "sweet spot"; more ≠ better
Purchasing criteria	EMF: "Low EMF" at 10-15 cm. Flicker: 0%. Irradiance: ≥100 mW / cm ² at 15 cm (otherwise sessions too long → heat). Cheap masks often only 20 mW / cm ² = unusable	Cheap devices: EMF irritates small fibers, flicker triggers brain fog

Neuromodulation and devices ([YELLOW]/[RED])

Device	Review	FQAD justification
tVNS (transcutaneous vagus nerve stimulation of the ear)	[YELLOW] Very useful	Stimulates ramus auricularis of the vagus → activates parasympathetic nervous system → can directly address gastroparesis. Most logical device addition
PEMF / BEMER (Magnetic Field Microcirculation)	[YELLOW] Test carefully	Patented signal for vasomotion (capillary rhythm) → oxygen + nutrients in tendons / nerve endings with poor blood supply. No heat. Cave: test electrical sensitivity; Herxheimer reaction possible (metabolic waste in a blocked liver)
Combi mats (Vitori etc.)	[YELLOW]/[RED] Caution	PEMF + infrared heat + ions. The heating function is a potential MCAS trigger → only with the heating switched off or lowest setting
Microcurrent (MENS)	[YELLOW] Insider tip	Directly charges mitochondrial cell voltage (μA range). Minimally invasive
alternating pressure mattresses	[RED] Mostly contraindicated	Permanent mechanical stress → mast cell mechanoreceptors in constant fire. Gastroparesis worsening (wave-like pressure → reflux). Only in cases of absolute immobility (pressure sore prophylaxis)
Classic TENS	[YELLOW] Caution	Can put additional stress on overstimulated small fibers

Always request a test phase (14-30 day right of return) - the body decides after 3 applications whether the magnetic field is “food” or “stress”.

Sound and frequency therapy ([YELLOW])

Frequency Type	Target	Application	Cave
Delta Beats (0.5-4 Hz)	Deep sleep / cellular repair	At night using headphones, very quiet	Binaural beats require headphones (frequency separation L / R)
Theta beats (4-8 Hz)	Vagus activation / gastroparesis	15 minutes before lunch	In hyperacusis (common in FQAD): extremely low volume
528 Hz sine	calming nerves	Background music while resting	“DNA repair” scientifically controversial; calming effect on ANS clinically reported
Brown noise	Tinnitus / sensory overload	For stress or to fall asleep	Smoother than white noise; masks overstimulated FQ nerve signals
Vagus Toning (Humming)	Parasympathetic activation	Deep “Vooom” hum: mechanical vibration stimulates vagus nerve via larynx	No device required; Can be combined with 4-7-8 breathing

Mind-Body Interventions ([YELLOW])

Method	Priority	FQAD justification	Cave
Guided Meditation	***	Lowers CRH (main trigger for mast cells). Before eating → Parasympathetic activation for gastroparesis	With Brain Fog: guided body scans instead of “empty” meditation. No “Toxic Positivity”
Qigong / Tai Chi (gentle)	***	Flowing movements lubricate joints, stretch tendons without micro-tears in damaged collagen, upper body rotation massages stomach (peristalsis)	Best form of movement for tendon problems. “Earth Element” Exercises for Stomach
Breathing techniques	***	4-7-8 (vagal activation), box breathing (panic attacks / CNS toxicity), nasal breathing (NO production → vasodilation)	If you feel dizzy or have a racing heart, stop immediately. Max. 2-3 minutes to start. No holotropic breathing (crash risk)
Autogenic training	*	Heavy exercises only (grounds nervous system). Heat exercises contraindicated in MCAS (heat flush)	Paradoxical reactions possible
Aromatherapy [YELLOW] Careful		Lavender (linalool → NMDA calming), peppermint (gastroparesis nausea), lemon (brain fog). Only 100% natural essential oils organic	[RED] Avoid: Incense sticks (fine dust / benzene → MCAS attack), synthetic fragrances, heavy floral scents (ylang-ylang, jasmine). Dry inhalation (1 drop on cloth, 20-30 cm away), no diffuser at the beginning

Warning: Homeopathic/isopathic ciprofloxacin ([RED] strictly avoid) The idea of “getting used to” the body through isopathic preparations (homeopathic Cipro) is dangerous in FQAD: (1) Real FQ molecules are contained in low potencies (D6-D12) → re-exposure of the damaged system. (2) In MCAS, mast cells often react to nanogram quantities with degranulation. (3) No validated mechanism of action for reversing mtDNA damage or chelation loss.

FQAD/MCAS Care Level Guide ([GREEN] Eligibility)

Applying for a nursing degree is not “giving up,” but rather a strategic decision to save energy for healing.

Application: An informal three-line letter to the health insurance provider is sufficient. Entitlement begins from the month of application - don't wait.

The 6 modules — FQAD-relevant focal points: **-Mobility:** Tendon instability (climbing stairs, repositioning) **-Self-care:** Gastroparesis → special food preparation (flash cooking, Wolfen); PEM → Help with personal care **-Illness management:** Mg foot baths, UVB therapy, doctor's visits, special meals = high time expenditure (often underestimated!) **-Everyday organization:** Brain Fog → Organization not possible

Assessment tips: Care diary (2 weeks, document every minute of help). Describe the worst days, not the best. Have all findings reports ready.

Benefits from care level 1: Relief allowance (131 €/month → household help/shopping), care aids (42 €/month), living space adjustments (up to 4,000 € subsidy, e.g. grab bar for tendon instability). From level 2: care allowance (from €347). Without level of care: Household help is possible as a health insurance benefit (§ 38 SGB V) if the family doctor certifies that housekeeping is not possible due to illness.

3.10 Health policy structural reforms: International role models for Germany

The authorities' previous strategy (sending red hand letters and hoping that doctors read them) has obviously failed - the number of prescriptions fell by ~34% [57], but ~17 million DDD/year (2019) and an estimated 8-9 million DDD (2023) are not acceptable given the documented burden of damage. The following international role models do not rely on hope, but rather on hard system limits.

1. No-fault damage fund based on the ONIAM model (France) — the most important point for those affected In France, the state authority ONIAM (Office National d'Indemnisation des Accidents Médicaux) compensates patients with serious medication injuries regardless of fault (no-fault). The patient does not have to prove medical negligence or manufacturer intent - it is sufficient to prove that the damage was most likely caused by the medication.

FQAD relevance for Germany: The year-long, ruinous legal battle against pharmaceutical lawyers (which systematically overburdens those affected by FQAD with PEM, brain fog and disability) is no longer necessary. Fast financial help for care, household and living expenses. The estimated 150,000-450,000 permanent losses justify a dedicated fund. Financing model: Manufacturer levy per FQ pack sold (analogous to the Contergan Foundation).

2. Delayed prescription / "Wait and See" prescription (UK, NICE guidelines)

The doctor writes the AB prescription but gives the instruction: "Don't fill it for 3-5 days, only if it gets worse." 70-80% of uncomplicated infections heal spontaneously.

FQAD relevance: Would immediately eliminate millions of unnecessary FQ prescriptions for minor infections (uUTI, sinusitis, bronchitis - together ~63% of all FQ prescriptions [63]). The patient has psychological safety (prescription in his pocket), but the pharmacy never sees him in the majority of cases. Not yet established as a guideline standard in Germany.

3. Real-time AI-powered prescription review (Netherlands)

When the doctor enters ciprofloxacin into the practice system, software checks the electronic patient file for absolute contraindications in milliseconds.

FQAD relevance: Immediate alarm for the most common high-risk constellations that are overlooked in the stress of practice: "Patient is taking corticosteroids at the same time → officially recognized high-risk factor for tendon rupture" (see vulnerability level 7). "Patient takes PPI (Pantoprazole) → Mg²⁺ depletion → FQ toxicity increased" (see pharmacological preload). "Kidney function eGFR <60 → risk of accumulation, especially with levofloxacin" (see cystatin C case). "Age >65 + diabetes → double the risk of seniors". The prescription is only activated when the doctor actively and justifiably overwrites the warning. The technical infrastructure (ePA, telematics infrastructure) has existed in Germany since 2025 - the algorithms just need to be implemented.

4. Legal protection for non-regulation (Sweden) Many German doctors prescribe FQ for defensive medicine: "What if the cystitis turns into pyelonephritis and the patient sues me because I didn't give an antibiotic?"

FQAD relevance: Sweden legally protects doctors if they strictly adhere to guidelines and do not prescribe AB for minor infections. This explicit liability protection is missing in Germany. A legal clarification that non-prescription in accordance with guidelines is not treatment error would immediately reduce the systemic prescribing pressure for minor infections.

5. Visual risk classes on the packaging + mandatory second opinion

Risk classes: A red warning symbol on the ciprofloxacin package (Class 3: “Reserve antibiotic / Severe, potentially irreversible side effects possible”) would motivate patients to ask: “Is there nothing more harmless?” FQ packs currently look identical to harmless antibiotics.

Mandatory second opinion: For FQ prescriptions outside of inpatient emergency indications (sepsis, febrile neutropenia), digital approval by an infectiologist or ABS representative (Antibiotic Stewardship) would be conceivable - analogous to the second opinion for spinal surgeries (§ 27b SGB V). The organizational effort (hurdle) would immediately reduce trivial prescriptions to zero.

6. National FQAD registry (model: Denmark)

Denmark collects health data centrally and can systematically track the long-term effects of medications using register queries.

FQAD relevance: A German FQAD registry (connected to the ePA and the BfArM) would for the first time answer the questions that have been unanswered for 30 years: How many patients who received ciprofloxacin in 2018 had tendon tears, neuropathies or needed a level of care in 2020? What is the actual chronification rate? Which subgroups (CYP1A2, SOD2, Mg status) are particularly at risk? Without a register, all prevalence estimates (see 150,000–450,000) remain model calculations. A register would make the late damages undeniably visible - and thus provide the basis for the no-fault damage fund (point 1).

Summary: 6-point reform plan for Germany

No.	Measure	role model	Effect
1	No-Fault Damage Fund	ONIAM (France)	Fast compensation without litigation
2	Delayed prescription	NICE (UK)	~70-80% of minor prescriptions eliminated
3	AI regulation check	e-medication (Netherlands)	Real-time warning of contraindications
4	Liability protection for non-regulation	Sweden	Defensive medicine pressure eliminated
5	Risk classes + mandatory second opinion	Analogy § 27b SGB V	Minor prescriptions → zero
6	National FQAD Registry	Denmark	Late damage visible → evidence base for all other points

Synthesis and outlook

Practical take-home messages

1. FQAD remains a diagnosis of exclusion - basic diagnostics are mandatory.
2. Goal 1: Damage stabilization and pacing, not “force healing”.
3. Think of organ axes separately and treat them in parallel.
4. No off-label senolytics, mitophagy substances or mito-agents outside of studies.
5. Re-conditioning only “low and slow”; with PEM no graded exercise.
6. Titrate CNS medications very carefully and individually.
7. Know and use the crash protocol in the event of acute decompensation.

8. Standard blood counts do not rule out severe mitochondrial dysfunction - the lactate-pyruvate ratio can be helpful in standardized special diagnostics, but is not a validated FQAD biomarker.
9. Any hypothesis about mitophagy, senescence, or reconditioning should be considered subgroup-specific, not "FQAD as a whole."
10. Diet: minimize sugar/white flour (lactate peaks), sufficient protein against catabolism, kidney-adapted for proteinuria, low histamine if MCAS is suspected - no radical diets without medical supervision. Probiotics only low in histamine and after GI diagnostics (Caution: histamine trap with standard probiotics).
11. Extreme temperature stimuli (sauna, ice bathing) can trigger severe crashes in cases of dysautonomia, SFN and mitochondrial blockade - "low and slow" also applies to environmental stressors.
12. Nocturnal SpO₂ drops in FQAD: pulse oximetry screening; If there are abnormalities, polysomnography → mechanical respiratory support (CPAP/BiPAP), not high-dose oxygen.
13. MCAS stabilization requires multi-pronged approach: H1+H2 blockers, mast cell stabilizers, low histamine diet, DAO substitution, trigger avoidance, pacing as MCAS therapy.
14. Do not reduce calcium (risk of osteoporosis); instead check magnesium status (whole blood analysis) - Magnesium is the cellular counterpart for uncontrolled calcium influx and protects against the ion-robbing chelation effect of fluoroquinolones on tissue (see III.D2).
15. Do not blindly supplement mito-boosters (Q10, PQQ, ALA) and polyphenol concentrates - they can trigger crashes if the respiratory chain is blocked, MCAS or glycolysis dependency. Only in the case of a laboratory-confirmed defect and under monitoring.
16. "Training through pain" is medically counterproductive when there is tendon/nerve involvement. Rest in the acute phase, then symptom-guided activity - not forced, not provoking pain.
17. Social invalidation ("don't act like that") measurably worsens outcomes. Inform relatives: FQAD is recognized by authorities, protection is medically justified, pressure is counterproductive.
18. When re-exercising after being confined to bed: Be aware of the risk of rhabdomyolysis. Warning signs (dark urine, rapid increase in pain, weakness) require immediate CK/creatinine control—emergency, not muscle soreness.
19. Critically examine commercial "stem cell/regenerative clinics" for FQAD: use the 7-question checklist (see Section VI). Missing study protocols, publications and safety data are warning signs - individual successes are no substitute for evidence.
20. Accelerated osteoarthritis in young FQAD sufferers is plausible (secondary to tendon/matrix damage + mechanical instability), but not proven as a direct FQ effect. Cartilage preservation when confined to bed: passive micro-mobilization (painless, without muscle strength) maintains synovial fluid circulation.
21. TIMPs (body's MMP brakes) require zinc as a cofactor; FQ also chelate zinc → check in the laboratory and compensate if there is a deficiency. Selenium is also relevant as a glutathione peroxidase cofactor.
22. Levofloxacin often heavier than ciprofloxacin: no liver back door, longer HWZ, perfected 3D key. Coffee sensitivity = CYP1A2 warning signal for Cipro risk; with Levo, kidney function (cystatin C, not just creatinine) is the bottleneck.
23. Be aware of cross-vulnerability: Severe reactions to mefloquine/chloroquine, chemotherapy drugs (topoisomerase inhibitors) or extreme codeine/SSRI sensitivity are clinical warning signs of increased risk of FQ toxicity.
24. Prioritized risk factors for FQ administration: (1) Mg²⁺ deficiency → (2) renal insufficiency/dehydration → (3) Zn²⁺ deficiency → (4) corticosteroids → (5) B12 deficiency. This ranking determines the sequence of stabilization measures.
25. Pre-medications as a precursor: PPI long-term therapy (Mg²⁺ depletion), statins (Q10 blockade + myotoxicity), bisphosphonates (enthesopathy), NSAIDs (GABA escalation), benzodiazepines (GABA receptor degradation). Medication history is mandatory before FQ prescription and for FQAD phenotyping.
26. FQAD is not "fluoride poisoning." The fluorine atom on C6 is not split off in the body (<0.1%). It is the architectural enabler: increases lipophilicity (→ tissue penetration), protects against liver degradation (→ intact molecule persists) and sharpens the chelation claw at C3/C4.
27. Repeated FQ administration escalates the risk exponentially: cumulative ion trapping, irreversible mtDNA scars, depleted Mg²⁺/Zn²⁺ reserves, senescence accumulation. The first dose is often compensated; the repeated one is the last straw.

28. FQ residues in conventional meat (enrofloxacin → ciprofloxacin metabolite) can trigger relapses in those affected by FQAD. Particular danger: Bone broth from conventional farming (ion trapping in animal bones). Strictly avoid food triggers (glutamate, sugar, oxalates, histamine liberators) in the acute phase.
29. Complete damage mechanics: Front 0 (chelation disarms SOD/repair enzymes) → Front 1 (ETC blockade → Fenton → OH• radicals) → Front 2 (phototoxicity in skin only; only context of true fluoride release). Consequential damage: ~50% lipid peroxidation, ~45% protein oxidation, ~1-5% DNA damage (most serious long-term consequences).
30. Nemonoxacin (non-fluoroquinolone) as a pharmacological counterevidence: removal of the fluorine atom eliminates phototoxicity, reduces lipophilicity/ion trapping and accelerates liver degradation. Residual chelation risk remains (quinolone backbone), but catastrophic mito chain reaction is almost impossible.
31. Chelate affinity quantitatively proven: Antacids (Al/Mg) reduce FQ absorption by up to 90%, iron by 80-90%, zinc 50-70%, calcium 40-60%. The same force that blocks absorption in the intestine acts as an ion robber of enzymes and collagen after absorption in the tissue.
32. Supplementation paradox: Mg²⁺ and Zn²⁺ are the most important repair building blocks (priority 1), but iron is a radical amplifier in active oxidative stress (Fenton reaction) and calcium can increase excitotoxicity in NMDA overstimulation. Staggered protocol: first Mg/Zn, iron/Ca restrictively until stabilization.
33. AIFM1/IDH2-based functional diagnostics (research vision, 6 approaches): (1) IDH2 metabolite profiling, (2) Seahorse stress test, (3) ex vivo challenge with ciprofloxacin (potential gold standard: sample A/B comparison with membrane potential + respirometry + metabolomics), (4) circulating AIFM1 as Apoptosis markers, (5) Targeted NGS of binding pocket exons, (6) FLIM redox imaging. All approaches purely conceptual [2].
34. MCS (Multiple Chemical Sensitivity) in FQAD could result from the convergence of MCAS (immunological: unstable mast cells + DAMPs + Treg loss) and GABA filter failure (neurological: limbic misinterpretation of chemical stimuli). Dual therapy approach (MCAS stabilization + neural desensitization) could be considered, but without FQAD-specific evidence.
35. SOD2 and GST polymorphisms as additional genetic vulnerability factors: SOD2 (mitochondrial superoxide detoxification) and GSTM1/GSTT1 (phase II detoxification) determine the ability of the cell to cope with FQ-induced oxidative damage. Testable via standard pharmacogenetics panels.
36. Treg-mast cell collapse as an immunological causal chain for MCAS: FQ → butyrate deficiency + ROS → Treg depletion → mast cell disinhibition. Diagnostics: CD4⁺/CD25⁺/FoxP3⁺ by FACS (special diagnostics) + IL-10 + tryptase/methylhistamine. Decreased Tregs + increased mast cell markers = objectified immunological disinhibition.
37. Individual vulnerability is explained by seven converging levels: (1) AIFM1/IDH2 binding pocket (fit genetics), (2) cellular fire department (SOD2/GST polymorphisms), (3) magnesium reserve at the start of therapy, (4) liver clearance (CYP1A2 slow/fast metabolizer), (5) kidney function (eGFR/cystatin C, esp. at Levofloxacin), (6) pharmacological preload (PPI, statins, bisphosphonates, NSAIDs, benzodiazepines), (7) concurrent corticosteroids (officially recognized high-risk factor: double collagen attack + Mg/Ca depletion + immunosuppression).
38. Treg recovery as long-term MCAS rehabilitation: Five levers — (1) butyrate/postbiotics (HDAC inhibitor → FoxP3 → Treg differentiation, bypasses histamine trap), (2) vitamin D3+retinoic acid (Treg induction via VDR), (3) low-dose naltrexone/LDN (Th17→Treg shift), (4) Vagus nerve activation (cholinergic anti-inflammatory pathway), (5) SPMs/resolvins from Omega-3 (inflammatory resolution + Treg recruitment). All [YELLOW] customized / not FQAD validated; “Low and Slow”.
39. MCAS-safe butyrate increase via diet: Resistant starch (cooled potatoes/rice → histamine-free colon fermentation to butyrate), soluble fiber (apple pectin, psyllium), ghee as a direct source of butyrate, low-histamine probiotics (*B. infantis*, *B. longum*, *L. rhamnosus* GG). Strictly avoid: Fermented foods (yoghurt, kefir, sauerkraut) and histamine-producing strains (*L. casei*, *L. bulgaricus*).
40. Weight loss in FQAD is a warning signal, not a therapeutic goal. Catabolism → ammonia (brain fog) + sarcopenia + creatinine adulteration. Calories (≥2000 kcal) across three safe pillars: (1) fats as the main calorie carrier (ghee, coconut oil, olive oil - no lactate, no mast cell triggers), (2) complex carbohydrates (resistant starch), (3) high-quality protein (organic, FQ residue-free). Never reduce protein/carbohydrate - if you lose weight, increase fats.

41. FQAD/MCAS Food Classification (including fruit): [GREEN] Safe (ghee, cooled potatoes/rice, broccoli/bok choy, pumpkin seeds/hemp seeds, organic frozen wild salmon, algae oil, macadamia, blueberries, applesauce, green bananas, cranberries, short meat broth). [YELLOW] Testing (eggs, quinoa/millet, cherries/peaches, watermelon, blackberries, psyllium, young cheese). [RED] Avoid (oxalate robbers, histamine liberators including strawberries/citrus/ripe bananas, bone broth, dried fruit, sugar, glutamate).
42. Mineral water as a therapeutic tool: Still water with $\text{HCO}_3^- > 1000 \text{ mg/L}$ (lactate buffer) + $\text{Mg} > 100$ + $\text{Ca} 250\text{-}500$, but sulfate < 50 (sulfur trap: sulfate-reducing bacteria $\rightarrow \text{H}_2\text{S}$ blocks complex IV like cyanide) and $\text{Na} < 20$ (kidney protection). No carbonation (mechanical MCAS trigger). Avoid base powder with FQAD: carbonates = PPI effect, citrates = MCAS trigger (Aspergillus fermentation), baking soda = kidney Na^+ flood.
43. Transdermal Mg intake (foot baths): Magnesium chloride flakes (no sulfate/Epsom), body temperature max. $36\text{-}37^\circ\text{C}$ (MCAS temperature trap!), foot baths before a full bath, in the evening as a sleep aid ($\text{Mg} = \text{NMDA}$ antagonist). In the case of a paradoxical weakness reaction due to Mg oil: dilute the brine (1:1), only the soles of the feet (not muscle areas), wash-off after 20 minutes. Three mechanisms: osmotic mast cell shock + Ca^{2+} channel blockade + vasodilation in dysautonomia.
44. Glycine paradox: Magnesium glycinate can trigger paradoxical fatigue in FQAD — not because of Mg, but because glycine, as an NMDA co-agonist, triggers excitotoxicity in FQ-blocked GABA receptors. MCAS-safe alternatives: Mg-malate (ATP support), Mg-aurate (GABAerg), transdermal MgCl_2 . The reaction is qualitative (carrier intolerance), not quantitative (deficiency).
45. Metabolic PEM through food: In long-term FQAD/MCAS, histamine liberators (avocado), salicylates (peppers, onions) and gluten/lectins (spelled) trigger a cytokine cascade $\rightarrow$ Immune reaction deprives the weak mitochondria of remaining ATP $\rightarrow$ Exhaustion is not a lack of willpower, but a measurable loss of energy. Third trigger dimension besides histamine and oxalate: salicylate intolerance.
46. Residual mitochondrial capacity can be calculated from PBMC laboratory values: (1) Efficiency quotient = $\text{ATP} / \text{mtDNA}:\text{ntDNA}$ ratio $\rightarrow$ with compensatory mito proliferation + low ATP: single mitochondrion runs at $< 50\%$. (2) Rhodanase bottleneck: missing Fe-S clusters $\rightarrow$ aerobic capacity under load only $\sim 5\text{-}6\%$. (3) PGC-1 α block: no mito-neogenesis $\rightarrow$ biogenesis trap. Standard laboratory findings often misinterpret this constellation as “efficient” — in FQAD it shows a massive energy deficit.
47. Gastroparesis in FQAD = vagus nerve block (dysautonomia) + MMP-weakened sphincter + MCAS mucosa. MCT oil (C8) as a gastroparesis bypass (no CCK reflex), slippery elm bark as mechanical mucous membrane protection, flash cooking (2–4 min.), zero tolerance for leftovers. No PPIs; no food in one piece - chop everything up to the maximum.
48. Growth hormones (HGH), therapeutic peptides (BPC-157, TB-500, CJC-1295/Ipamorelin, Epithalon) and egg extracts (YTE, FGF) not recommended in FQAD: energy crash + senescence override + MMP feeding + MCAS hormonal shock. Zero RCTs for all peptides. Typical forum pattern: initially euphoric $\rightarrow$ reverse effect $\rightarrow$ discontinuation. Epithalon (telomerase activation) with FQ-DNA damage = removing cancer brake while damage is ongoing. Fundamental rule: any anabolic stimulus requires ATP capacity + MMP control + correct epigenetic blueprint — without these three prerequisites it is a net loss. Also avoid commercial drinking foods (Fresubin etc.) (maltodextrin/casein/carrageenan). For testosterone deficiency: bioidentical gel in microdose from an endocrinologist.
49. 311 nm UVB phototherapy as a source of vitamin D without oral Mg withdrawal ([YELLOW]/[RED] with reservations): micro-dosing (30-60 seconds/area), fixed distance (scalpel approach), rotating areas, MED titration, absolute eye protection requirement. Expectation: only $\sim 50\text{-}100 \text{ IU/session}$ — wanted with FQAD (crash-free). 311 nm is a factor of 2-3 more inefficient than 297 nm peak $\rightarrow$ a natural attenuator in MCAS. Risky - only after consultation with a doctor + vitamin D monitoring.
50. Complementary therapies for FQAD: Photobiomodulation (660/850 nm LED, no IR sauna, note biphasic dose), tVNS as the most logical device addition (vagus $\rightarrow$ gastroparesis), sound therapy (theta beats before meals, brown noise for tinnitus), Qigong as the best form of exercise for tendon problems, meditation before eating (CRH reduction $\rightarrow$ Mast cell stabilization). Avoid: alternating pressure mattresses (MCAS mechanostress), incense sticks (fine dust), isopathic Cipro (re-exposure).
51. Apply for nursing care level at FQAD: Informal application to nursing insurance fund. Care diary (2 weeks). FQAD-relevant modules: mobility (tendons), self-care (gastroparesis $\rightarrow$ special cuisine), disease management (therapy time expenditure), everyday life (brain fog).

- From level 1: relief amount 131 €/month. Without level of care: Section 38 SGB V (household help via family doctor certificate).
52. Quercetin as PGC-1 α activator: Stimulates AMPK/SIRT1/PGC-1 α $\rightarrow$ mitochondrial biogenesis (directly addresses PGC-1 α blockade) + mast cell stabilization. Necessarily pure substance (no fillers), creep in microdosing (knife tip $\rightarrow$ 250-500 mg), as it is also a senolytic (risk of crash with full dose). Natural sources (capers, onions) not feasible for MCAS/gastroparesis; Buckwheat as the only compatible food source (23 mg/100g).
 53. Omega-3 supplementation: TOTOX value necessarily <5 (oxidized fish oil = mito-toxin + histamine). Liquid algae oil (darkened glass bottle, cool immediately) instead of fish oil capsules. ALA from flaxseed is not enough for FQAD (conversion to EPA/DHA is massively restricted if mitochondria are damaged).
 54. Regulatory consequence: ~83% of FQ regulations can be replaced by conventional AB + phages [63]. The ~17% previously considered “unreplaceable” (CAP, complicated UTI, prostatitis) can be addressed by nemonoxacin (non-fluoroquinolone, approved in Taiwan/China, phase III in the EU [69]): quinolone tissue permeability without phototoxicity, without ion trapping, with retained residual chelation risk. Triple combination of conventional AB + phage + nemonoxacin $\rightarrow$ almost complete FQ replacement possible in the future.
 55. Retrograde signaling chain mtDNA $\rightarrow$ nDNA $\rightarrow$ collagen switch: FQ-damaged mtDNA $\rightarrow$ ROS + energy deficiency $\rightarrow$ retrograde signaling to cell nucleus $\rightarrow$ epigenetic shift (methylation/acetylation) at collagen genes in nDNA $\rightarrow$ tenocytes switch from elastic type I collagen to rigid type III collagen (scar tissue) around + MMP overproduction. The blueprint error is in the nDNA, but is a result of the mtDNA crisis. Collagen supplementation alone does not solve the problem - therapy must primarily reduce oxidative stress and activate PGC-1 α so that the epigenetic switch resets.
 56. Economic burden (own projection, three calculation paths disclosed): Path 1 bottom-up (damage categories summed) = €1.75 billion; Path 2 top-down (Kuula model [71], 4 million $\times$ €528) = €2.11 billion; Path 3 synthesis + long-term costs of chronic old cases = **3.5-4.5 billion €/year**. Biggest uncertainty factor: cumulative old cases (150,000-450,000) — cannot be reliably limited without an FQAD register. Prevalence in the order of Parkinson’s (~420,000 [72]) and MS (~250,000 [73]).
 57. Global dimension: Estimated ~40-50% of the world’s population (~3.2-4 billion people) have been treated with FQ at least once (model estimate from DDD consumption data [74], [75], accumulated over ~40 years). With a conservative permanent damage rate of 0.1% $\rightarrow$ ~3.6 million potentially permanently injured people worldwide without registry data or ICD code.
 58. 6-point reform plan for Germany: (1) No-fault damage fund based on the ONIAM model (France) — no-fault compensation. (2) Delayed prescribing (UK/NICE) — eliminates ~70-80% of minor prescriptions. (3) Real-time AI prescription check (Netherlands) — automatic cortisone/PPI/renal function alert. (4) Liability protection for non-prescription (Sweden) — eliminates defensive medicine pressure. (5) Risk classes on packaging + mandatory second opinion for outpatient FQ prescription. (6) National FQAD Registry (Denmark) — makes late damage visible and provides the evidence base for all other points.

Conclusion

The responsible goal today is not the planning of a complete cure, but rather the systematic maximization of remission, organ recovery and stress reserve along defined endophenotypes and measurable organ endpoints. Deep remission up to extensive functional recovery remains an open possibility that can only be tested by research. Why some sufferers recover and others become chronic is multifactorial and cannot be explained by a single factor. Dominant endophenotype, renal function, known risk factors (age, corticosteroids, diabetes), genetic vulnerability and secondary drivers of chronicity interact (see Section X.F). There is no reliable evidence of causal FQAD healing for any single supplement, diet or infusion therapy.

In the future, a sequential, biomarker-guided intervention program could theoretically enable extensive restoration of normal function for defined subgroups. The curative horizon (Section XIV) shows that biotechnological tools are fundamentally evolving—from genome editing to microbiome surgery. However, the development of such tools does not result in any proximity to clinical applicability in FQAD; each strategy must demonstrate safety, tissue-specific targeting,

and clinical benefit prior to FQAD use.

The most important advance would not be a single new drug, but: FQAD biomarkers, FQAD subtypes, the first controlled FQAD intervention studies and an n-of-1 study culture that makes individualized hypotheses seriously testable — tied to the measurable endpoints of the recovery framework (Section XII).

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This document describes hypotheses and research directions — not a therapy recommended for use. It does not replace medical advice. The off-label use of the substances mentioned is associated with significant risks; FQAD-specific safety data is missing. Deep remission and extensive functional recovery are long-term research hypotheses, not a plannable therapeutic goal.